

深度学习在神经网络中的概述

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摘要

近年来, 深度人工神经网络 (包括循环网络) 在模式识别和机器学习领域赢得了众多比赛。这篇历史综述简要总结了相关研究工作, 其中大部分来自上一个千年。浅层学习和深度学习通过它们的<样式 id='1'>信用分配路径</样式>来区分, 这些路径是由可能可学习的、因果链接组成的链, 连接着动作和效果。我回顾了深度监督学习 (也回顾了反向传播的历史), 无监督学习、强化学习与进化计算, 以及间接搜索用于编码深度和大型网络的短程序。

LATEX 源文件: <http://www.idsia.ch/~juergen/DeepLearning8Oct2014.tex> 完整
BIBTEX 文件 (888 kB): <http://www.idsia.ch/~juergen/deep.bib>

前言

这是关于深度学习(DL)的预印本。其中一个目标是表彰那些为当前技术现状做出贡献的人。我承认尝试实现这一目标的局限性。DL研究社区本身可以被视为一个不断演化的、深层的科学家网络, 他们以复杂的方式相互影响。从最近的DL成果出发, 我试图通过过去半个世纪甚至更早的时间追溯相关思想的起源, 有时使用“局部搜索”方法沿着被引用的引用回溯时间。由于并非所有DL出版物都正确承认早期相关工作, 因此采用了额外的全局搜索策略, 并咨询了众多神经网络专家。结果, 这份预印本主要由参考文献组成。尽管如此, 由于我可能因专家选择偏差而遗漏了重要工作, 这肯定也引入了相关偏差。与我过去二十五年里对自身DL研究组的特殊熟悉程度有关。因此, 这项工作应该被视为一个正在进行中的信用分配过程的快照。为了帮助改进它, 请毫不犹豫地更正和建议发送至juergen@idsia.ch。

Deep Learning in Neural Networks: An Overview

Technical Report IDSIA-03-14 / arXiv:1404.7828 v4 [cs.NE] (88 pages, 888 references)

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Abstract

In recent years, deep artificial neural networks (including recurrent ones) have won numerous contests in pattern recognition and machine learning. This historical survey compactly summarises relevant work, much of it from the previous millennium. Shallow and deep learners are distinguished by the depth of their *credit assignment paths*, which are chains of possibly learnable, causal links between actions and effects. I review deep supervised learning (also recapitulating the history of backpropagation), unsupervised learning, reinforcement learning & evolutionary computation, and indirect search for short programs encoding deep and large networks.

LATEX source: <http://www.idsia.ch/~juergen/DeepLearning8Oct2014.tex>
Complete BIBTEX file (888 kB): <http://www.idsia.ch/~juergen/deep.bib>

Preface

This is the preprint of an invited *Deep Learning* (DL) overview. One of its goals is to assign credit to those who contributed to the present state of the art. I acknowledge the limitations of attempting to achieve this goal. The DL research community itself may be viewed as a continually evolving, deep network of scientists who have influenced each other in complex ways. Starting from recent DL results, I tried to trace back the origins of relevant ideas through the past half century and beyond, sometimes using “local search” to follow citations of citations backwards in time. Since not all DL publications properly acknowledge earlier relevant work, additional global search strategies were employed, aided by consulting numerous neural network experts. As a result, the present preprint mostly consists of references. Nevertheless, through an expert selection bias I may have missed important work. A related bias was surely introduced by my special familiarity with the work of my own DL research group in the past quarter-century. For these reasons, this work should be viewed as merely a snapshot of an ongoing credit assignment process. To help improve it, please do not hesitate to send corrections and suggestions to juergen@idsia.ch.

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按字母顺序排列的缩写

AE: 自动编码器 AI: 人工智能 ANN: 人工神经网络 BFGS: Broyden-Fletcher-Goldfarb-Shanno BNN: 生物神经网络 BM: 玻尔兹曼机 BP: 反向传播 BRNN: 双向循环神经网络 CAP: 信用分配路径 CEC: 恒定误差转盘 CFL: 上下文无关语言 CMA-ES: 协方差矩阵估计 ES CNN: 卷积神经网络 CoSyNE: 共突触神经进化 CSL: 上下文敏感语言 CTC: 连接时序分类 DBN: 深度信念网络 DCT: 离散余弦变换 DL: 深度学习 DP: 动态规划 DS: 直接策略搜索 EA: 进化算法 EM: 期望最大化 ES: 进化策略 FMS: 平坦最小值搜索 FNN: 前馈神经网络 FSA: 有限状态自动机 GMDH: 群方法数据处理 GOFAI: 老式人工智能 GP: 遗传编程 GPU: 图形处理器 GPU-MPCNN: 基于GPU的MPCNN HMM: 隐马尔可夫模型 HRL: 分层强化学习

HTM: 层次化时间记忆; HMAX: 层次化模型 “与X” ; LSTM: 长短期记忆 (RNN) ; MDL: 最小描述长度; MDP: 马尔可夫决策过程; MNIST: 混合美国国家标准与技术研究院数据库; MP: 最大池化; MPCNN: 最大池化CNN; NE: 神经进化; NEAT: 增强拓扑结构的神经网络; NES: 自然进化策略; NFQ: 神经拟合Q学习; NN: 神经网络; OCR: 光学字符识别; PCC: 潜在因果连接; PDCC: 潜在直接因果连接; PM: 可预测性最小化; POMDP: 部分可观察马尔可夫决策过程; RAAM: 递归自联想记忆; RBM: 受限玻尔兹曼机; ReLU: 修正线性单元; RL: 强化学习; RNN: 循环神经网络; R-prop: 弹性反向传播; SL: 监督学习; SLIM NN: 自限定神经网络; SOTA: 自组织树算法; SVM: 支持向量机; TDNN: 时间延迟神经网络; TIMIT: TI/SRI/MIT声学-语音连续语料库; UL: 无监督学习; WTA: 赢家通吃

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Abbreviations in Alphabetical Order

AE: Autoencoder AI: Artificial Intelligence ANN: Artificial Neural Network BFGS: Broyden-Fletcher-Goldfarb-Shanno BNN: Biological Neural Network BM: Boltzmann Machine BP: Backpropagation BRNN: Bi-directional Recurrent Neural Network CAP: Credit Assignment Path CEC: Constant Error Carousel CFL: Context Free Language CMA-ES: Covariance Matrix Estimation ES CNN: Convolutional Neural Network CoSyNE: Co-Synaptic Neuro-Evolution CSL: Context Senistive Language CTC : Connectionist Temporal Classification DBN: Deep Belief Network DCT: Discrete Cosine Transform DL: Deep Learning DP: Dynamic Programming DS: Direct Policy Search EA: Evolutionary Algorithm EM: Expectation Maximization ES: Evolution Strategy FMS: Flat Minimum Search FNN: Feedforward Neural Network FSA: Finite State Automaton GMDH: Group Method of Data Handling GOFAI: Good Old-Fashioned AI GP: Genetic Programming GPU: Graphics Processing Unit GPU-MPCNN: GPU-Based MPCNN HMM: Hidden Markov Model HRL: Hierarchical Reinforcement Learning

HTM: Hierarchical Temporal Memory HMAX: Hierarchical Model “and X” LSTM: Long Short-Term Memory (RNN) MDL: Minimum Description Length MDP: Markov Decision Process MNIST: Mixed National Institute of Standards and Technology Database MP: Max-Pooling MPCNN: Max-Pooling CNN NE: NeuroEvolution NEAT: NE of Augmenting Topologies NES: Natural Evolution Strategies NFQ: Neural Fitted Q-Learning NN: Neural Network OCR: Optical Character Recognition PCC: Potential Causal Connection PDCC: Potential Direct Causal Connection PM: Predictability Minimization POMDP: Partially Observable MDP RAAM: Recursive Auto-Associative Memory RBM: Restricted Boltzmann Machine ReLU: Rectified Linear Unit RL: Reinforcement Learning RNN: Recurrent Neural Network R-prop: Resilient Backpropagation SL: Supervised Learning SLIM NN: Self-Delimiting Neural Network SOTA: Self-Organising Tree Algorithm SVM: Support Vector Machine TDNN: Time-Delay Neural Network TIMIT: TI/SRI/MIT Acoustic-Phonetic Continuous Speech Corpus UL: Unsupervised Learning WTA: Winner-Take-All

1 深度学习（DL）在神经网络（NNs）中的简介

一个学习系统的哪些可修改组件对其成功或失败负责？如何改变它们来提高性能？这被称为<样式 id='1'>基本信用分配问题</样式>（明斯基，1963年）。对于<样式 id='3'>通用问题解决器</样式>存在通用信用分配方法，在多种理论意义上是时间最优的（第6.8节）。然而，本综述将重点关注更狭窄但现已商业化的子领域——<样式 id='5'>深度学习</样式>（DL）在<样式 id='7'>人工神经网络</样式>（NNs）中的应用。

一个标准的神经网络（NN）由许多简单的、相互连接的处理单元（称为神经元）组成，每个神经元产生一系列实值激活。输入神经元通过感知环境的传感器被激活，其他神经元通过来自先前激活神经元的加权连接被激活（详情见第2节）。某些神经元可以通过触发动作来影响环境。学习 或 信用分配 是关于寻找权重，使神经网络表现出期望 行为，例如驾驶汽车。根据问题和神经元如何连接，这种行为可能需要很长的计算阶段的因果链（第3节），其中每个阶段都转换（通常以非线性方式）网络的总体激活。深度学习是关于在许多这样的阶段中准确分配信用。

浅层神经网络模型，如果不算世纪之交的话，已经存在很多年了（第5.1节）。具有多个连续非线性神经层的模型至少可以追溯到1960年代（第5.3节）和1970年代（第5.5节）。一种用于离散、可微分的任意深度网络中的基于教师指导的<样式 id='5'>监督学习</样式>（SL）的高效梯度下降方法，称为<样式 id='7'>反向传播</样式>（BP），在1960年代和1970年代被开发出来，并在1981年被应用于神经网络（第5.5节）。然而，到1980年代末（第5.6节），基于BP的具有<样式 id='9'>多层</样式>的神经网络训练在实践中被发现非常困难，并在1990年代初（第5.9节）成为明确的研究课题。深度学习在一定程度上通过<样式 id='13'>无监督学习</样式>（UL）的帮助变得实用，例如第5.10节（1991年）、第5.15节（2006年）。1990年代和2000年代也见证了纯粹监督式深度学习的许多改进（第5节）。在新千年里，深度神经网络终于引起了广泛关注，主要是因为它们在众多重要应用中优于替代的机器学习方法，例如核机（瓦普尼克，1995年；施洛维克等人，1998年）。事实上，自2009年以来，监督式深度神经网络赢得了许多官方的国际模式识别竞赛（例如，第5.17、5.19、5.21、5.22节），在有限领域内实现了首个超越人类的视觉模式识别结果（第5.19节，2011年）。深度神经网络也对更广泛的<样式 id='16'>强化学习</样式>（RL）领域变得相关，在这个领域没有监督教师（第6节）。

前馈神经网络（FNNs）和循环神经网络（RNNs）都曾赢得竞赛（第5.12、5.14、5.17、5.19、5.21、5.22节）。从某种意义上说，RNNs是所有神经网络中最深的（第3节）——它们比FNNs更强大的通用计算机，原则上可以创建和处理任意输入模式序列的记忆（例如，Siegelmann和Sontag，1991年；Schmidhuber，1990a年）。与传统的自动顺序程序合成方法（例如，Waldinger和Lee，1969年；Balzer，1985年；Soloway，1986年；Deville和Lau，1994年）不同，RNNs能够以自然且高效的方式学习混合顺序和并行信息处理的程序，利用被视为过去75年间计算成本快速下降的关键因素的大量并行性。

本文其余部分的结构如下。第2节介绍了一种紧凑的、面向事件的符号表示法，它既简单又通用，足以涵盖前馈神经网络（FNNs）和循环神经网络（RNNs）。第3节引入了<样式 id='1'>信用分配路径</样式>（CAPs）的概念，用于衡量特定神经网络应用中的学习是深度学习还是浅层学习。第4节列出了深度学习在监督学习、无监督学习和强化学习中的常见主题。第5节重点讨论了监督学习和无监督学习，以及无监督学习如何促进监督学习，尽管纯监督学习在最近的比赛中已成为主流（第5.17–5.23节）。第5节采用历史时间线格式，包含关于重要启发和技术贡献的子节。第6节讨论深度强化学习，涉及传统的<样式 id='7'>动态规划</样式>（DP）强化学习，以及用于监督学习或无监督学习的基于梯度的搜索技术，还包括深度前馈神经网络和循环神经网络权重空间中直接和间接搜索的通用方法，包括成功的策略梯度和进化方法。

1 Introduction to Deep Learning (DL) in Neural Networks (NNs)

Which modifiable components of a learning system are responsible for its success or failure? What changes to them improve performance? This has been called the *fundamental credit assignment problem* (Minsky, 1963). There are general credit assignment methods for *universal problem solvers* that are time-optimal in various theoretical senses (Sec. 6.8). The present survey, however, will focus on the narrower, but now commercially important, subfield of *Deep Learning* (DL) in *Artificial Neural Networks* (NNs).

A standard neural network (NN) consists of many simple, connected processors called neurons, each producing a sequence of real-valued activations. Input neurons get activated through sensors perceiving the environment, other neurons get activated through weighted connections from previously active neurons (details in Sec. 2). Some neurons may influence the environment by triggering actions. *Learning* or *credit assignment* is about finding weights that make the NN exhibit *desired* behavior, such as driving a car. Depending on the problem and how the neurons are connected, such behavior may require long causal chains of computational stages (Sec. 3), where each stage transforms (often in a non-linear way) the aggregate activation of the network. Deep Learning is about accurately assigning credit across *many* such stages.

Shallow NN-like models with *few* such stages have been around for many decades if not centuries (Sec. 5.1). Models with several successive nonlinear layers of neurons date back at least to the 1960s (Sec. 5.3) and 1970s (Sec. 5.5). An efficient gradient descent method for teacher-based *Supervised Learning* (SL) in discrete, differentiable networks of arbitrary depth called *backpropagation* (BP) was developed in the 1960s and 1970s, and applied to NNs in 1981 (Sec. 5.5). BP-based training of *deep* NNs with *many* layers, however, had been found to be difficult in practice by the late 1980s (Sec. 5.6), and had become an explicit research subject by the early 1990s (Sec. 5.9). DL became practically feasible to some extent through the help of *Unsupervised Learning* (UL), e.g., Sec. 5.10 (1991), Sec. 5.15 (2006). The 1990s and 2000s also saw many improvements of purely supervised DL (Sec. 5). In the new millennium, deep NNs have finally attracted wide-spread attention, mainly by outperforming alternative machine learning methods such as kernel machines (Vapnik, 1995; Schölkopf et al., 1998) in numerous important applications. In fact, since 2009, supervised deep NNs have won many official international pattern recognition competitions (e.g., Sec. 5.17, 5.19, 5.21, 5.22), achieving the first superhuman visual pattern recognition results in limited domains (Sec. 5.19, 2011). Deep NNs also have become relevant for the more general field of *Reinforcement Learning* (RL) where there is no supervising teacher (Sec. 6).

Both *feedforward* (acyclic) NNs (FNNs) and *recurrent* (cyclic) NNs (RNNs) have won contests (Sec. 5.12, 5.14, 5.17, 5.19, 5.21, 5.22). In a sense, RNNs are the deepest of all NNs (Sec. 3)—they are general computers more powerful than FNNs, and can in principle create and process memories of arbitrary sequences of input patterns (e.g., Siegelmann and Sontag, 1991; Schmidhuber, 1990a). Unlike traditional methods for automatic sequential program synthesis (e.g., Waldinger and Lee, 1969; Balzer, 1985; Soloway, 1986; Deville and Lau, 1994), RNNs can learn programs that mix sequential and parallel information processing in a natural and efficient way, exploiting the massive parallelism viewed as crucial for sustaining the rapid decline of computation cost observed over the past 75 years.

The rest of this paper is structured as follows. Sec. 2 introduces a compact, event-oriented notation that is simple yet general enough to accommodate both FNNs and RNNs. Sec. 3 introduces the concept of *Credit Assignment Paths* (CAPs) to measure whether learning in a given NN application is of the *deep* or *shallow* type. Sec. 4 lists recurring themes of DL in SL, UL, and RL. Sec. 5 focuses on SL and UL, and on how UL can facilitate SL, although pure SL has become dominant in recent competitions (Sec. 5.17–5.23). Sec. 5 is arranged in a historical timeline format with subsections on important inspirations and technical contributions. Sec. 6 on deep RL discusses traditional *Dynamic Programming* (DP)-based RL combined with gradient-based search techniques for SL or UL in deep

神经网络，以及深度前馈神经网络和循环神经网络权重空间中直接和间接搜索的通用方法，包括成功的策略梯度和进化方法。

2 事件驱动符号表示神经网络中的激活传播

全文中，令 i, j, k, t, p, q, r 表示在给定上下文中隐含范围的正整数变量。令 n, m, T 表示正整数常量。

神经网络的结构可能随时间变化（例如，第 5.3 节、5.6.3 节）。在任何给定时刻，它可以描述为单元（或节点或神经元）的有限子集 $N = \{u_1, u_2, \dots\}$ 和节点之间有向边或连接的有限集 $H \subseteq N \times N$ 。前馈神经网络是无环图，循环神经网络是循环的。第一层（输入层）是输入单元的集合，是 N 的一个子集。在前馈神经网络中，第 k 层（ $k > 1$ ）是所有满足以下条件的节点 $u \in N$ 的集合：存在长度为 $k - 1$ （但不更长）的边路径，连接某个输入单元和 u 。可能存在连接不同层的捷径连接。在序列处理中，全连接循环神经网络中，所有单元都与所有非输入单元相连。

神经网络的行为或程序由一组实值、可能可修改的参数或权重 w_i ($i = 1, \dots, n$) 决定。我们现在专注于信息处理和激活传播的一个单个有限场景或周期，而不通过权重变化进行学习。以下这种略微非常规的符号旨在紧凑地描述系统运行时发生的情况。

在一个场景中，存在一个部分因果序列 x_t ($t = 1, \dots, T$) 的实数值，我称之为事件。每个 x_t 要么是环境设置的输入，要么是一个单元的激活，该单元可能直接依赖于其他 x_k ($k < t$)，通过当前神经网络拓扑相关的索引集 in_t 表示传入的因果连接或链接。令函数 v 编码拓扑信息，并将此类事件索引对 (k, t) 映射到权重索引。例如，在非输入情况下，我们可能有 $x_t = f_t(net_t)$ ，其中 $net_t = \sum_{k \in in_t} x_k w_{v(k,t)}$ 是实值的（加性情况）或 $net_t = \prod_{k \in in_t} x_k w_{v(k,t)}$ （乘性情况），其中 f_t 是一个通常非线性实值的激活函数，例如 $\tanh$ 。在许多最近的竞赛获胜神经网络（第 5.19、5.21、5.22 节）中，也存在类型为 $x_t = \max_{k \in in_t}(x_k)$ 的事件；某些网络类型也可能使用复多项式激活函数（第 5.3 节）。 x_t 可能通过通过当前索引集 out_t 表示的传出连接或链接直接影响某些 x_k ($k > t$)，其中 $t \in in_k$ 。其中一些非输入事件被称为输出事件。

请注意，许多的 x_t 可能指序列处理RNNs（例如，Williams, 1989年，同一单元的不同、随时间变化的激活（“时间展开”），或也指在FNNs中顺序暴露于大型训练集编码为输入事件的时间变化输入模式中。在一个回合期间，相同的权重可能会以拓扑依赖的方式反复使用，例如，在RNNs中，或在卷积神经网络（第5.4节、5.8节）中。我称这种权重共享为跨空间和/或时间。权重共享可以大大降低NN的描述复杂度，即描述NN所需的比特数信息（第4.4节）。

在监督学习（SL）中，某些NN输出事件 x_t 可能与教师给出的、实数值标签或目标 d_t 相关联，产生错误 e_t ，例如， $e_t = 1/2(x_t - d_t)^2$ 。监督NN训练的一个典型目标是为产生具有较小总错误 E （所有此类 e_t 的总和）的回合找到权重。希望NN在后续回合中能够很好地泛化，对先前未见过的输入事件序列只产生小的错误。对于SL和UL，许多替代错误函数都是可能的。

监督学习假设输入事件独立于较早的输出事件（这些事件可能通过行动影响环境，进而影响后续感知）。这一假设在更广泛的序列决策和强化学习(RL)（Kaelbling 等人，1996；Sutton 和 Barto，1998；哈特，2005；Wiering 和 van Otterlo，2012）（第 6 节）领域不成立。在强化学习中，一些输入事件可能编码由环境给出的实值奖励信号，并且典型的

NNs, as well as general methods for direct and indirect search in the weight space of deep FNNs and RNNs, including successful policy gradient and evolutionary methods.

2 Event-Oriented Notation for Activation Spreading in NNs

Throughout this paper, let i, j, k, t, p, q, r denote positive integer variables assuming ranges implicit in the given contexts. Let n, m, T denote positive integer constants.

An NN’s topology may change over time (e.g., Sec. 5.3, 5.6.3). At any given moment, it can be described as a finite subset of units (or nodes or neurons) $N = \{u_1, u_2, \dots\}$ and a finite set $H \subseteq N \times N$ of directed edges or connections between nodes. FNNs are acyclic graphs, RNNs cyclic. The first (input) layer is the set of input units, a subset of N . In FNNs, the k -th layer ($k > 1$) is the set of all nodes $u \in N$ such that there is an edge path of length $k - 1$ (but no longer path) between some input unit and u . There may be shortcut connections between distant layers. In sequence-processing, fully connected RNNs, all units have connections to all non-input units.

The NN’s behavior or program is determined by a set of real-valued, possibly modifiable, parameters or weights w_i ($i = 1, \dots, n$). We now focus on a single finite *episode* or *epoch* of information processing and activation spreading, without learning through weight changes. The following slightly unconventional notation is designed to compactly describe what is happening *during the runtime* of the system.

During an episode, there is a *partially causal sequence* x_t ($t = 1, \dots, T$) of real values that I call events. Each x_t is either an input set by the environment, or the activation of a unit that may directly depend on other x_k ($k < t$) through a current NN topology-dependent set in_t of indices k representing incoming causal connections or links. Let the function v encode topology information and map such event index pairs (k, t) to weight indices. For example, in the non-input case we may have $x_t = f_t(net_t)$ with real-valued $net_t = \sum_{k \in in_t} x_k w_{v(k,t)}$ (additive case) or $net_t = \prod_{k \in in_t} x_k w_{v(k,t)}$ (multiplicative case), where f_t is a typically nonlinear real-valued *activation function* such as $\tanh$. In many recent competition-winning NNs (Sec. 5.19, 5.21, 5.22) there also are events of the type $x_t = \max_{k \in in_t}(x_k)$; some network types may also use complex polynomial activation functions (Sec. 5.3). x_t may directly affect certain x_k ($k > t$) through outgoing connections or links represented through a current set out_t of indices k with $t \in in_k$. Some of the non-input events are called *output events*.

Note that many of the x_t may refer to different, time-varying activations of the *same* unit in sequence-processing RNNs (e.g., Williams, 1989, “*unfolding in time*”), or also in FNNs sequentially exposed to time-varying input patterns of a large training set encoded as input events. During an episode, the same weight may get reused over and over again in topology-dependent ways, e.g., in RNNs, or in convolutional NNs (Sec. 5.4, 5.8). I call this weight sharing *across space and/or time*. Weight sharing may greatly reduce the NN’s descriptive complexity, which is the number of bits of information required to describe the NN (Sec. 4.4).

In *Supervised Learning* (SL), certain NN output events x_t may be associated with teacher-given, real-valued labels or targets d_t yielding errors e_t , e.g., $e_t = 1/2(x_t - d_t)^2$. A typical goal of supervised NN training is to find weights that yield episodes with small total error E , the sum of all such e_t . The hope is that the NN will generalize well in later episodes, causing only small errors on previously unseen sequences of input events. Many alternative error functions for SL and UL are possible.

SL assumes that input events are independent of earlier output events (which may affect the environment through actions causing subsequent perceptions). This assumption does not hold in the broader fields of *Sequential Decision Making* and *Reinforcement Learning* (RL) (Kaelbling et al., 1996; Sutton and Barto, 1998; Hutter, 2005; Wiering and van Otterlo, 2012) (Sec. 6). In RL, some of the input events may encode real-valued reward signals given by the environment, and a typical

目标是通过一系列适当的输出动作来找到能够产生具有高奖励信号总和的轨迹的权重。

第5.5节将使用上述符号来紧凑地描述DL的一个中心算法，即用于监督权重共享的FNNs和RNNs的反向传播（BP）。（FNNs可以看作是某些固定零权重的RNNs。）第6节将处理更一般的RL情况。

3 信用分配路径（CAPs）及其问题

为了衡量给定神经网络应用中的信用分配是 深度 还是 浅层 类型，我引入了 信用分配路径 或 CAPs 的概念，它们是第 2 节事件之间可能的因果链，例如，在 FNNs 中从输入层通过隐藏层到输出层，或在 RNNs 中随时间变化的转换。

让我们首先关注 SL。考虑两个事件 x_p 和 x_q ($1 \leq p < q \leq T$)。根据应用场景，它们可能具有 潜在直接因果连接(PDCC)，由布尔谓词 $pdcc(p, q)$ 表示，当且仅当 $p \in in_q$ 时为真。然后定义 2 元素列表 (p, q) 为从 p 到 q 的 CAP（一个最小的）。学习算法可能被允许改变 $w_{v(p, q)}$ 以在未来场景中提高性能。

更一般地，可能间接的<样式 id='1'>潜在因果连接</样式>(PCC)由递归定义的布尔谓词 $pcc(p<样式 id='5'>, </样式> q)$ 表示，在监督学习(SL)的情况下，它仅当 $pdcc(p<样式 id='10'>, </样式> q)$ 时为真，或者对于某些 k 和 $pdcc(k<样式 id='21'>, </样式> q)$ 时为真。在后一种情况下，将 q 附加到从 p 到 k 的任何CAP，将产生从 p 到 q 的CAP（这也是一个递归定义）。这样的CAP集合可能很大但有限。请注意，相同的权重可能会影响给定CAP列出的连续事件之间的许多不同的PDCC，例如在循环神经网络(RNN)或权重共享前馈神经网络(FNN)的情况下。

假设一个CAP的形式为(<样式 id='1'>..., </样式> $k<样式 id='4'>$, </样式> $t<样式 id='7'>$, ..., </样式> q)，其中 k 和 t （可能是 $t = q$ ）是第一个具有<样式 id='13'>可修改</样式> $w_{v(k, t)}$ 的连续元素。则后缀列表($t<样式 id='17'>$, ..., </样式> q)的长度称为CAP的<样式 id='20'>深度</样式>（如果没有可修改的连接则为0）。这个深度限制了信用分配可以向后移动多远，以在因果链中找到一个可修改的权重。¹

假设一个场景及其事件序列 $x_1, \dots, x_T$ 满足一个可计算的准则，该准则用于判断给定问题是否已被解决（例如，总误差 E 低于某个阈值）。那么，所使用的权重集合被称为该问题的解，而序列中最深的 CAP 的深度被称为 解深度。可能存在其他解（产生不同的事件序列），并且具有不同的深度。对于给定的固定 NN 架构，任何解的最小深度被称为 问题深度。

有时我们也会提到 架构的深度：具有固定拓扑的 SL 前馈神经网络意味着一个与问题无关的最大问题深度，该深度受非输入层数量的限制。某些具有固定权重（除输出单元连接外）的 SL 循环神经网络（Jaeger, 2001; Maass 等人, 2002; Jaeger, 2004; Schrauwen 等人, 2007）具有最大问题深度为 1，因为只有相应 CAPs 中的最终连接是可修改的。然而，一般来说，循环神经网络可以学习解决潜在深度无限的问题。

请注意，上述定义完全基于因果链的深度，并且对事件之间的时间距离是无知的。例如，感知输入事件大型“时间窗口”的浅层 前馈神经网络 可以通过适当的输出事件正确分类 长输入序列，从而解决涉及 浅层 相关事件之间存在 长时间延迟的问题。

在什么问题深度上浅层学习结束，而深度学习开始？与深度学习专家的讨论尚未对这个问题的答案得出结论。与其做出明确的回答，让我只为本次概述定义：深度为的问题需要极深度学习。

¹An另一种选择是在测量深度时仅计算可修改的 链接。在许多典型的NN应用中这会造成差异，但在某些情况下会，例如，第 6.1 节。

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goal is to find weights that yield episodes with a high sum of reward signals, through sequences of appropriate output actions.

Sec. 5.5 will use the notation above to compactly describe a central algorithm of DL, namely, backpropagation (BP) for supervised weight-sharing FNNs and RNNs. (FNNs may be viewed as RNNs with certain fixed zero weights.) Sec. 6 will address the more general RL case.

3 Depth of Credit Assignment Paths (CAPs) and of Problems

To measure whether credit assignment in a given NN application is of the *deep* or *shallow* type, I introduce the concept of *Credit Assignment Paths* or CAPs, which are chains of possibly causal links between the events of Sec. 2, e.g., from input through hidden to output layers in FNNs, or through transformations over time in RNNs.

Let us first focus on SL. Consider two events x_p and x_q ($1 \leq p < q \leq T$). Depending on the application, they may have a *Potential Direct Causal Connection* (PDCC) expressed by the Boolean predicate $pdcc(p, q)$, which is true if and only if $p \in in_q$. Then the 2-element list (p, q) is defined to be a CAP (a minimal one) from p to q . A learning algorithm may be allowed to change $w_{v(p, q)}$ to improve performance in future episodes.

More general, possibly indirect, *Potential Causal Connections* (PCC) are expressed by the recursively defined Boolean predicate $pcc(p, q)$, which in the SL case is true only if $pdcc(p, q)$, or if $pcc(p, k)$ for some k and $pdcc(k, q)$. In the latter case, appending q to any CAP from p to k yields a CAP from p to q (this is a recursive definition, too). The set of such CAPs may be large but is finite. Note that the *same* weight may affect *many* different PDCCs between successive events listed by a given CAP, e.g., in the case of RNNs, or weight-sharing FNNs.

Suppose a CAP has the form $(\dots, k, t, \dots, q)$, where k and t (possibly $t = q$) are the first successive elements with *modifiable* $w_{v(k, t)}$. Then the length of the suffix list $(t, \dots, q)$ is called the CAP’s *depth* (which is 0 if there are no modifiable links at all). This depth limits how far backwards credit assignment can move down the causal chain to find a modifiable weight.¹

Suppose an episode and its event sequence $x_1, \dots, x_T$ satisfy a computable criterion used to decide whether a given problem has been solved (e.g., total error E below some threshold). Then the set of used weights is called a *solution* to the problem, and the depth of the deepest CAP within the sequence is called the *solution depth*. There may be other solutions (yielding different event sequences) with different depths. Given some fixed NN topology, the smallest depth of any solution is called the *problem depth*.

Sometimes we also speak of the *depth of an architecture*: SL FNNs with fixed topology imply a problem-independent maximal problem depth bounded by the number of non-input layers. Certain SL RNNs with fixed weights for all connections except those to output units (Jaeger, 2001; Maass et al., 2002; Jaeger, 2004; Schrauwen et al., 2007) have a maximal problem depth of 1, because only the final links in the corresponding CAPs are modifiable. In general, however, RNNs may learn to solve problems of potentially unlimited depth.

Note that the definitions above are solely based on the depths of causal chains, and agnostic to the temporal distance between events. For example, *shallow* FNNs perceiving large “time windows” of input events may correctly classify *long* input sequences through appropriate output events, and thus solve *shallow* problems involving *long* time lags between relevant events.

At which problem depth does *Shallow Learning* end, and *Deep Learning* begin? Discussions with DL experts have not yet yielded a conclusive response to this question. Instead of committing myself

¹An alternative would be to count only *modifiable* links when measuring depth. In many typical NN applications this would not make a difference, but in some it would, e.g., Sec. 6.1.

让我只为本次概述定义：深度为 > 10 的问题需要极深度学习。

一个问题的难度可能与它的深度关系不大。一些神经网络可以快速学会解决某些深度问题，例如，通过随机权重猜测（第5.9节）或其他类型的直接搜索（第6.6节）或间接搜索（第6.7节）在权重空间中，或者通过先在浅层问题上训练神经网络，其解决方案随后可能泛化到深度问题上，或者通过将（非）线性操作的序列折叠成一个（非）线性操作（但请参见对深度线性网络非平凡方面的分析，Baldi和Hornik, 1994年，第B节）。然而，一般来说，找到一个精确地模拟给定训练集的神经网络是一个NP完全问题（Judd, 1990年；Blum和Rivest, 1992年），在深度神经网络的情况下也是如此（Sîma, 1994年；de Souto等人, 1999年；Windisch, 2005年）；参见Sîma, 2002年，第1节中的负面结果调查。

前面我们主要关注了监督学习。在未知环境中强化学习的一般情况下， $pcc(p, q)$ 也成立，如果 x_p 是一个输出事件，而 x_q 是任何后续的输入事件——任何动作都可能影响环境，从而影响任何后续的感知。（在现实世界中，环境甚至可能影响与整个宇宙纠缠的物理硬件上计算的<样式 id='8'>非输入</样式>事件，但在这里忽略这一点。）可以通过NN的一部分来建模和替换这种不可修改的<样式 id='10'>环境</样式>部分条件控制（PCCs），该部分已经学会通过其某些单元从先前的输入事件和动作中预测输入事件（包括奖励信号）（第6.1节）。它的权重被冻结，但可以帮助将信用分配给其他仍然可修改的权重，这些权重用于计算动作（第6.1节）。这种方法可能会导致非常深的CAPs，尽管如此。

一些深度学习研究致力于自动重述问题，以降低其深度（第4节）。特别是，有时无监督学习被用来使监督学习问题变得更浅，例如第5.10节。通常<样式 id='1'>动态规划</样式>（第4.1节）被用来促进某些传统的强化学习问题，例如第6.2节。第5节专注于监督学习的CAPs，第6节则关注更复杂的强化学习情况。

深度学习的四个反复出现的主题

4.1 用于监督/强化学习（SL/RL）的动态规划

深度学习的一个反复出现的主题是 动态规划 (DP) (Bellman, 1957)，它可以在某些假设下帮助实现信用分配。例如，在监督学习（SL）神经网络中，反向传播本身可以被视为一种基于动态规划的算法（第5.5节）。在基于强马尔可夫假设的传统强化学习（RL）中，基于动态规划的算法可以帮助大大减少问题深度（第6.2节）。动态规划算法对于结合神经网络和图模型概念的系统也至关重要，例如 隐马尔可夫模型 (HMMs) (Stratonovich, 1960; Baum and Petrie, 1966) 和 期望最大化(EM) (Dempster et al., 1977; Friedman et al., 2001)，例如 (Bottou, 1991; Bengio, 1991; Bourlard and Morgan, 1994; Baldi and Chauvin, 1996; Jordan and Sejnowski, 2001; Bishop, 2006; Hastie et al., 2009; Poon and Domingos, 2011; Dahl et al., 2012; Hinton et al., 2012a; Wu and Shao, 2014)。

4.2 无监督学习 (UL) 促进监督学习与强化学习

另一个反复出现的主题是，无监督学习（UL，第 5 节）如何促进监督学习（SL，第 5 节）和强化学习（RL，第 6 节）。无监督学习（UL，第 5.6.4 节）通常用于将原始输入数据（如视频或语音流）编码成更便于后续目标导向学习的形式。特别是，那些以更少冗余或更紧凑方式描述原始数据的码，可以被输入到监督学习（SL，第 5.10 节、5.15 节）或强化学习机器（RL，第 6.4 节）中，从而使它们的搜索空间可能比处理原始数据所需的更小（并且它们的 CAPs 更浅）。无监督学习与 正则化（第 4.4 节、5.6.3 节）和压缩（第 4.4 节、5.6.3 节）等主题密切相关。

to a precise answer, let me just define for the purposes of this overview: problems of depth > 10 require *Very Deep Learning*.

The *difficulty* of a problem may have little to do with its depth. Some NNs can quickly learn to solve certain deep problems, e.g., through random weight guessing (Sec. 5.9) or other types of direct search (Sec. 6.6) or indirect search (Sec. 6.7) in weight space, or through training an NN first on shallow problems whose solutions may then generalize to deep problems, or through collapsing sequences of (non)linear operations into a single (non)linear operation (but see an analysis of non-trivial aspects of deep linear networks, Baldi and Hornik, 1994, Section B). In general, however, finding an NN that precisely models a given training set is an NP-complete problem (Judd, 1990; Blum and Rivest, 1992), also in the case of deep NNs (Sîma, 1994; de Souto et al., 1999; Windisch, 2005); compare a survey of negative results (Sîma, 2002, Section 1).

Above we have focused on SL. In the more general case of RL in unknown environments, $pcc(p, q)$ is also true if x_p is an output event and x_q any later input event—any action may affect the environment and thus any later perception. (In the real world, the environment may even influence *non-input* events computed on a physical hardware entangled with the entire universe, but this is ignored here.) It is possible to model and replace such unmodifiable *environmental* PCCs through a part of the NN that has already learned to predict (through some of its units) input events (including reward signals) from former input events and actions (Sec. 6.1). Its weights are frozen, but can help to assign credit to other, still modifiable weights used to compute actions (Sec. 6.1). This approach may lead to very deep CAPs though.

Some DL research is about automatically rephrasing problems such that their depth is reduced (Sec. 4). In particular, sometimes UL is used to make SL problems less deep, e.g., Sec. 5.10. Often *Dynamic Programming* (Sec. 4.1) is used to facilitate certain traditional RL problems, e.g., Sec. 6.2. Sec. 5 focuses on CAPs for SL, Sec. 6 on the more complex case of RL.

4 Recurring Themes of Deep Learning

4.1 Dynamic Programming for Supervised/Reinforcement Learning (SL/RL)

One recurring theme of DL is *Dynamic Programming* (DP) (Bellman, 1957), which can help to facilitate credit assignment under certain assumptions. For example, in SL NNs, backpropagation itself can be viewed as a DP-derived method (Sec. 5.5). In traditional RL based on strong Markovian assumptions, DP-derived methods can help to greatly reduce problem depth (Sec. 6.2). DP algorithms are also essential for systems that combine concepts of NNs and graphical models, such as *Hidden Markov Models* (HMMs) (Stratonovich, 1960; Baum and Petrie, 1966) and *Expectation Maximization* (EM) (Dempster et al., 1977; Friedman et al., 2001), e.g., (Bottou, 1991; Bengio, 1991; Bourlard and Morgan, 1994; Baldi and Chauvin, 1996; Jordan and Sejnowski, 2001; Bishop, 2006; Hastie et al., 2009; Poon and Domingos, 2011; Dahl et al., 2012; Hinton et al., 2012a; Wu and Shao, 2014).

4.2 Unsupervised Learning (UL) Facilitating SL and RL

Another recurring theme is how UL can facilitate both SL (Sec. 5) and RL (Sec. 6). UL (Sec. 5.6.4) is normally used to encode raw incoming data such as video or speech streams in a form that is more convenient for subsequent goal-directed learning. In particular, codes that describe the original data in a less redundant or more compact way can be fed into SL (Sec. 5.10, 5.15) or RL machines (Sec. 6.4), whose search spaces may thus become smaller (and whose CAPs shallower) than those necessary for dealing with the raw data. UL is closely connected to the topics of *regularization* and compression (Sec. 4.4, 5.6.3).

4.3 通过深度监督学习、无监督学习和强化学习学习层次化表示

许多 传统老式人工智能 (GOF AI) (Nilsson, 1980) 的方法, 以及更近期的 AI (Russell 等人, 1995) 和 机器学习 (Mitchell, 1997) 的方法, 都学习越来越抽象的数据表示层次结构。例如, 某些句法模式识别 (Fu, 1977) 的方法, 如 语法归纳, 会发现用于建模观察的正式规则层次结构。部分 (非) 监督的 自动数学家 / *EURISKO* (Lenat, 1983; Lenat 和 Brown, 1984) 通过组合先前学习到的概念来不断学习概念。这种层次表示学习 (Ring, 1994; Bengio 等人, 2013; Deng 和 Yu, 2014) 也是深度学习神经网络 (DL NNs) 用于监督学习 (SL, 第 5 节)、无监督学习辅助的监督学习 (UL 辅助的 SL, 第 5.7 节、5.10 节、5.15 节) 和层次强化学习 (RL, 第 6.5 节) 的反复出现的主题。通常, 抽象的层次表示是数据压缩 (第 4.4 节) 的自然副产品, 例如第 5.10 节。

4.4 奥卡姆剃刀: 压缩和最小描述长度 (MDL)

奥卡姆剃刀倾向于选择简单的解决方案而非复杂的方案。给定某种编程语言, 最小描述长度(MDL) 的原则可以用来通过计算它的最短程序长度来衡量一个解决方案候选的复杂性 (例如, Solomonoff, 1964; Kolmogorov, 1965b; Chaitin, 1966; Wallace and Boulton, 1968; Levin, 1973a; Solomonoff, 1978; Rissanen, 1986; Blumer et al., 1987; Li and Vitányi, 1997; Grünwald et al., 2005)。一些方法明确地考虑了程序运行时间 (Allender, 1992; Watanabe, 1992; Schmidhuber, 1997, 2002) ; 许多方法只考虑具有恒定运行时间的程序, 这些程序是用非通用编程语言编写的 (例如, Rissanen, 1986; Hinton and van Camp, 1993)。在神经网络的情况下, MDL 原则表明在贝叶斯观点中, 低神经网络权重复杂性对应于高神经网络概率 (例如, MacKay, 1992; Buntine and Weigend, 1991; Neal, 1995; De Freitas, 2003), 并且对应于高泛化性能 (例如, Baum and Haussler, 1989), 而不会过拟合训练数据。许多方法已被提出用于 正则化 神经网络, 即寻找计算解决方案但简单、低复杂性的监督学习神经网络 (第 5.6.3 节) 和强化学习神经网络 (第 6.7 节)。这与某些无监督方法 (第 4.2 节、5.6.4 节) 密切相关。

4.5 快速图形处理单元 (GPU) 用于神经网络中的深度学习

虽然在上个千年中, 人们曾尝试创建针对神经网络的快速专用硬件 (例如, Jackel 等人, 1990 年; Faggin, 1992 年; Ramacher 等人, 1993 年; Widrow 等人, 1994 年; Heemskerk, 1995 年; Korkin 等人, 1997 年; Urlbe, 1999 年), 并尝试利用标准硬件 (例如, Anguita 等人, 1994 年; Muller 等人, 1995 年; Anguita 和 Gomes, 1996 年), 但新千年带来了以廉价的多处理器图形卡或 GPU 形式的深度学习突破。GPU 广泛用于电子游戏, 这是一个庞大且竞争激烈的市场, 它推动了硬件价格的下降。GPU 在所需的快速矩阵和向量乘法方面表现出色, 这不仅适用于令人信服的虚拟现实, 也适用于神经网络训练, 其中它们可以将学习速度提高 50 倍以上。一些基于 GPU 的 FNN 实现对最近在模式识别 (第 5.19 节至第 5.22 节) 竞赛、图像分割 (第 5.21 节) 和目标检测 (第 5.21 节至第 5.22 节) 方面的成功做出了巨大贡献。

5 个监督神经网络, 一些由无监督神经网络帮助

当前实际应用的主要焦点在于 监督学习 (SL), 它在最近的模式识别竞赛中占据主导地位 (第5.17–5.23节)。然而, 一些方法使用额外的 无监督学习 (UL) 来辅助 SL (第5.7、5.10、5.15节)。将 SL 和 UL 放在同一部分讨论是有道理的: 通常基于梯度的方法, 如 BP (第5.5.1节), 被用于优化 UL 和 SL 的目标函数, 并且 SL 和 UL 之间的界限可能会变得模糊, 例如在时间序列预测和序列分类的情况下, 比如第5.10、5.12节。

4.3 Learning Hierarchical Representations Through Deep SL, UL, RL

Many methods of *Good Old-Fashioned Artificial Intelligence* (GOF AI) (Nilsson, 1980) as well as more recent approaches to AI (Russell et al., 1995) and *Machine Learning* (Mitchell, 1997) learn hierarchies of more and more abstract data representations. For example, certain methods of syntactic pattern recognition (Fu, 1977) such as *grammar induction* discover hierarchies of formal rules to model observations. The partially (un)supervised *Automated Mathematician* / *EURISKO* (Lenat, 1983; Lenat and Brown, 1984) continually learns concepts by combining previously learnt concepts. Such hierarchical representation learning (Ring, 1994; Bengio et al., 2013; Deng and Yu, 2014) is also a recurring theme of DL NNs for SL (Sec. 5), UL-aided SL (Sec. 5.7, 5.10, 5.15), and hierarchical RL (Sec. 6.5). Often, abstract hierarchical representations are natural by-products of data compression (Sec. 4.4), e.g., Sec. 5.10.

4.4 Occam’s Razor: Compression and Minimum Description Length (MDL)

Occam’s razor favors simple solutions over complex ones. Given some programming language, the principle of *Minimum Description Length* (MDL) can be used to measure the complexity of a solution candidate by the length of the shortest program that computes it (e.g., Solomonoff, 1964; Kolmogorov, 1965b; Chaitin, 1966; Wallace and Boulton, 1968; Levin, 1973a; Solomonoff, 1978; Rissanen, 1986; Blumer et al., 1987; Li and Vitányi, 1997; Grünwald et al., 2005). Some methods explicitly take into account program runtime (Allender, 1992; Watanabe, 1992; Schmidhuber, 1997, 2002); many consider only programs with constant runtime, written in non-universal programming languages (e.g., Rissanen, 1986; Hinton and van Camp, 1993). In the NN case, the MDL principle suggests that low NN weight complexity corresponds to high NN probability in the Bayesian view (e.g., MacKay, 1992; Buntine and Weigend, 1991; Neal, 1995; De Freitas, 2003), and to high generalization performance (e.g., Baum and Haussler, 1989), without overfitting the training data. Many methods have been proposed for *regularizing* NNs, that is, searching for solution-computing but simple, low-complexity SL NNs (Sec. 5.6.3) and RL NNs (Sec. 6.7). This is closely related to certain UL methods (Sec. 4.2, 5.6.4).

4.5 Fast Graphics Processing Units (GPUs) for DL in NNs

While the previous millennium saw several attempts at creating fast NN-specific hardware (e.g., Jackel et al., 1990; Faggin, 1992; Ramacher et al., 1993; Widrow et al., 1994; Heemskerk, 1995; Korkin et al., 1997; Urlbe, 1999), and at exploiting standard hardware (e.g., Anguita et al., 1994; Muller et al., 1995; Anguita and Gomes, 1996), the new millennium brought a DL breakthrough in form of cheap, multi-processor graphics cards or GPUs. GPUs are widely used for video games, a huge and competitive market that has driven down hardware prices. GPUs excel at the fast matrix and vector multiplications required not only for convincing virtual realities but also for NN training, where they can speed up learning by a factor of 50 and more. Some of the GPU-based FNN implementations (Sec. 5.16–5.19) have greatly contributed to recent successes in contests for pattern recognition (Sec. 5.19–5.22), image segmentation (Sec. 5.21), and object detection (Sec. 5.21–5.22).

5 Supervised NNs, Some Helped by Unsupervised NNs

The main focus of current practical applications is on *Supervised Learning* (SL), which has dominated recent pattern recognition contests (Sec. 5.17–5.23). Several methods, however, use additional *Unsupervised Learning* (UL) to facilitate SL (Sec. 5.7, 5.10, 5.15). It does make sense to treat SL and

UL 在同一部分中讨论: 通常基于梯度的方法, 如 BP (第5.5.1节), 被用于优化 UL 和 SL 的目标函数, 并且 SL 和 UL 之间的界限可能会变得模糊, 例如在时间序列预测和序列分类的情况下, 比如第5.10、5.12节。

一个历史时间轴格式将有助于安排关于重要启发和技术贡献的子章节 (尽管这样的子章节可能跨越多年的时间区间)。第5.1节简要提到了自1940年代 (和1800年代) 以来出现的早期、浅层神经网络模型, 第5.2节提到了与当今深度学习 (DL) 相关的早期神经生物学启发。第5.3节是关于GMDH网络 (自1965年以来), 据我所知, 这是第一个 (前馈) 深度学习系统。第5.4节是关于相对较深的新认知器 神经网络 (1979年), 它与某些现代深度前馈神经网络 (FNN) 架构非常相似, 因为它结合了卷积神经网络 (CNN)、权重模式复制和子采样机制。第5.5节使用第2节的符号, 简洁地描述了学习的核心算法, 即监督权重共享FNN和RNN的反向传播(BP)。它还总结了反向传播 (BP) 在1960-1981年及之后的历史。第5.6节描述了1980年代末深度神经网络 (NN) 中反向传播 (BP) 遇到的问题, 并提到了几个上一千年提出的想法来克服这些问题。第5.7节讨论了第一个层次化堆栈 (1987年) 的耦合无监督学习 (UL) 自动编码器 (AEs) ——这个概念在新千年中再次出现 (第5.15节)。第5.8节是关于将反向传播 (BP) 应用于卷积神经网络 (CNN) (1989年), 这对当今的深度学习 (DL) 应用非常重要。第5.9节解释了反向传播 (BP) 的深度学习基本问题 (梯度消失/梯度爆炸), 该问题是在1991年发现的。第5.10节解释了1991年的深度循环神经网络 (RNN) 堆栈 (即历史压缩机) 如何通过无监督学习 (UL) 进行预训练, 从而解决了之前无法学习的深度学习 (DL) 基准, 这些基准需要深度为1000及以上的信用分配路径(CAPs, 第3节)。第5.11节讨论了一种特定的胜者通吃(WTA)方法, 称为最大池化(MP, 1992年), 该方法在如今的深度前馈神经网络 (FNN) 中广泛使用。第5.12节提到了监督学习 (SL) 神经网络 (NN) 在1994年赢得的第一个重要比赛。第5.13节描述了一种纯粹监督的深度学习 (DL) 循环神经网络 (RNN) (即长短期记忆, LSTM, 1995年), 用于深度为1000及以上的问题。第5.14节提到了2003年早期由浅层前馈神经网络 (FNN) 集合赢得的比赛, 以及卷积神经网络 (CNN) 和深度前馈神经网络 (FNN) 以及长短期记忆循环神经网络 (LSTM RNN) 在2003年取得的良好模式识别结果。第5.15节主要关于深度信念网络(DBNs, 2006年)以及相关的自动编码器 (AEs, 第5.7节) 堆栈, 它们都通过无监督学习 (UL) 进行预训练, 以促进后续的基于反向传播 (BP) 的监督学习 (SL) (比较第5.6.1, 5.10节)。第5.16节提到了第一个基于监督学习的GPU-CNN (2006年), 反向传播 (BP) 训练的MPCNN (2007年), 以及LSTM堆栈 (2007年)。第5.17-5.22节重点介绍了自2009年以来, 在序列识别、图像分类、图像分割和目标检测方面, 由 (主要是纯粹监督的) 深度神经网络 (NN) 赢得的官方比赛, 这些比赛使用了秘密测试集。许多循环神经网络 (RNN) 结果依赖于长短期记忆 (LSTM) (第5.13节); 许多前馈神经网络 (FNN) 结果依赖于自2004年以来开发的基于GPU的前馈神经网络 (FNN) 代码 (第5.16, 5.17, 5.18, 5.19节), 特别是GPU-MPCNN (第5.19节)。第5.24节提到了最近用于改进神经网络 (NN) 中深度学习 (DL) 的技巧, 其中许多技巧与上一千年提出的技巧密切相关 (例如, 第5.6.2, 5.6.3节)。第5.25节讨论了人工神经网络 (NN) 如何有助于理解生物神经网络 (NN); 第5.26节探讨了脉冲神经元 (NN) 的神经网络 (NN) 中深度学习 (DL) 的可能性。

5.1 早期神经网络 (自1940年代 (及1800年代) 起)

早期的神经网络架构 (麦卡洛克和皮茨, 1943年) 并未学习。关于无监督学习的第一批想法是在几年后发表的 (赫布, 1949年)。接下来的几十年带来了由监督学习训练的简单神经网络 (例如, 罗森布拉特, 1958年, 1962年; Widrow 和 Hoff, 1962年; Narendra 和 Thathatchar, 1974年) 以及无监督学习 (例如, Grossberg, 1969年; Kohonen, 1972年; von der Malsburg, 1973年; Willshaw 和 von der Malsburg, 1976年), 以及与之密切相关的联想记忆 (例如, Palm, 1980年; Hopfield, 1982年)。

从某种意义上说, 神经网络的历史甚至更悠久, 因为早期的监督式神经网络本质上是线性回归方法的变体, 其历史至少可以追溯到19世纪初 (例如, 勒让德, 1805年; 高斯, 1809年, 1821年); 高斯也提到了他1795年的研究成果。早期的神经网络具有最大CAP深度为1 (第3节)。

UL in the same section: often gradient-based methods, such as BP (Sec. 5.5.1), are used to optimize objective functions of both UL and SL, and the boundary between SL and UL may blur, for example, when it comes to time series prediction and sequence classification, e.g., Sec. 5.10, 5.12.

A historical timeline format will help to arrange subsections on important inspirations and technical contributions (although such a subsection may span a time interval of many years). Sec. 5.1 briefly mentions early, shallow NN models since the 1940s (and 1800s), Sec. 5.2 additional early neurobiological inspiration relevant for modern Deep Learning (DL). Sec. 5.3 is about GMDH networks (since 1965), to my knowledge the first (feedforward) DL systems. Sec. 5.4 is about the relatively deep *Neocognitron* NN (1979) which is very similar to certain modern deep FNN architectures, as it combines convolutional NNs (CNNs), weight pattern replication, and subsampling mechanisms. Sec. 5.5 uses the notation of Sec. 2 to compactly describe a central algorithm of DL, namely, *backpropagation* (BP) for supervised weight-sharing FNNs and RNNs. It also summarizes the history of BP 1960-1981 and beyond. Sec. 5.6 describes problems encountered in the late 1980s with BP for deep NNs, and mentions several ideas from the previous millennium to overcome them. Sec. 5.7 discusses a first hierarchical stack (1987) of coupled UL-based Autoencoders (AEs)—this concept resurfaced in the new millennium (Sec. 5.15). Sec. 5.8 is about applying BP to CNNs (1989), which is important for today’s DL applications. Sec. 5.9 explains BP’s *Fundamental DL Problem* (of vanishing/exploding gradients) discovered in 1991. Sec. 5.10 explains how a deep RNN stack of 1991 (the *History Compressor*) pre-trained by UL helped to solve previously unlearnable DL benchmarks requiring *Credit Assignment Paths* (CAPs, Sec. 3) of depth 1000 and more. Sec. 5.11 discusses a particular *winner-take-all* (WTA) method called *Max-Pooling* (MP, 1992) widely used in today’s deep FNNs. Sec. 5.12 mentions a first important contest won by SL NNs in 1994. Sec. 5.13 describes a purely supervised DL RNN (*Long Short-Term Memory*, LSTM, 1995) for problems of depth 1000 and more. Sec. 5.14 mentions an early contest of 2003 won by an ensemble of shallow FNNs, as well as good pattern recognition results with CNNs and deep FNNs and LSTM RNNs (2003). Sec. 5.15 is mostly about *Deep Belief Networks* (DBNs, 2006) and related stacks of *Autoencoders* (AEs, Sec. 5.7), both pre-trained by UL to facilitate subsequent BP-based SL (compare Sec. 5.6.1, 5.10). Sec. 5.16 mentions the first SL-based GPU-CNNs (2006), BP-trained MPCNNs (2007), and LSTM stacks (2007). Sec. 5.17–5.22 focus on official competitions with secret test sets won by (mostly purely supervised) deep NNs since 2009, in sequence recognition, image classification, image segmentation, and object detection. Many RNN results depended on LSTM (Sec. 5.13); many FNN results depended on GPU-based FNN code developed since 2004 (Sec. 5.16, 5.17, 5.18, 5.19), in particular, GPU-MPCNNs (Sec. 5.19). Sec. 5.24 mentions recent tricks for improving DL in NNs, many of them closely related to earlier tricks from the previous millennium (e.g., Sec. 5.6.2, 5.6.3). Sec. 5.25 discusses how artificial NNs can help to understand biological NNs; Sec. 5.26 addresses the possibility of DL in NNs with spiking neurons.

5.1 Early NNs Since the 1940s (and the 1800s)

Early NN architectures (McCulloch and Pitts, 1943) did not learn. The first ideas about UL were published a few years later (Hebb, 1949). The following decades brought simple NNs trained by SL (e.g., Rosenblatt, 1958, 1962; Widrow and Hoff, 1962; Narendra and Thathatchar, 1974) and UL (e.g., Grossberg, 1969; Kohonen, 1972; von der Malsburg, 1973; Willshaw and von der Malsburg, 1976), as well as closely related associative memories (e.g., Palm, 1980; Hopfield, 1982).

In a sense NNs have been around even longer, since early supervised NNs were essentially variants of linear regression methods going back at least to the early 1800s (e.g., Legendre, 1805; Gauss, 1809, 1821); Gauss also refers to his work of 1795. Early NNs had a maximal CAP depth of 1 (Sec. 3).

5.2 约1960年：视觉皮层为深度学习提供灵感（第5.4节、第5.11节）

在猫的视觉皮层中发现了简单细胞和复杂细胞（例如，Hubel 和 Wiesel, 1962年；Wiesel 和 Hubel, 1959年）。这些细胞会对视觉感觉输入的某些特性做出反应，例如边缘的方向。复杂细胞比简单细胞表现出更强的空间不变性。这启发了后来用于某些现代获奖深度学习器（第5.19–5.22节）的深度神经网络架构（第5.4节，5.11节）。

5.3 1965年：基于数据处理群方法的深度网络

由<样式 id='1'>数据处理群方法</样式>(GMDH) (Ivakhnenko和Lapa, 1965年；Ivakhnenko等人, 1967年；Ivakhnenko, 1968年, 1971年) 训练的网络可能是第一种<样式 id='3'>前馈多层感知器</样式>类型的深度学习系统，尽管之前已有关于单隐藏层神经网络的研究（例如，Joseph, 1961年；Viglione, 1970年）。GMDH网络的单元可能有多项式激活函数，实现<样式 id='5'>柯尔莫哥洛夫-高斯多项式</样式> >（比其他广泛使用的神经网络激活函数更通用，第2节）。给定一个训练集，层会通过回归分析（例如，Legendre, 1805年；Gauss, 1809年, 1821年）（第5.1节）逐层增长和训练，然后借助一个单独的<样式 id='7'>验证集</样式>（用今天的术语来说），使用<样式 id='9'>决策正则化</样式>来剔除多余的单元（比较第5.6.3节）。每层的层数和单元数量可以按问题相关的方式进行学习。据我所知，这是神经网络中开放式、分层表示学习的第一个例子（第4.3节）。1971年的论文已经描述了一个有8层的深度GMDH网络（Ivakhnenko, 1971年）。GMDH风格网络有许多应用，例如（Ikeda等人, 1976年；Farlow, 1984年；Madala和Ivakhnenko, 1994年；Ivakhnenko, 1995年；Kondo, 1998年；Kordík等人, 2003年；Witczak等人, 2006年；Kondo和Ueno, 2008年）。

5.4 1979：卷积 + 权重复制 + 下采样（新认知器）

除了第5.3节的深度GMDH网络之外，新认知器 (Fukushima, 1979, 1980, 2013a)或许是第一个值得拥有深度属性的人工神经网络，也是第一个结合了第5.2节神经生理学见解的网络。它引入了卷积神经网络 (如今通常称为CNN或卷积网络)，其中具有给定权重向量（一个过滤器）的卷积单元的（通常为矩形）感受野会逐步在二维输入值数组（如图像的像素）上移动（通常有多个这样的过滤器）。该单元后续的激活事件形成的二维数组可以提供输入给更高级别的单元，如此类推。由于巨大的权重复制 (第2节)，相对较少的参数（第4.4节）可能就足以描述这种卷积层的行为。

下采样或降采样层由其固定权重的连接源自下方卷积层中物理相邻单元的单元组成。如果它们的至少一个输入处于激活状态，下采样单元就会激活；它们的响应对某些小的图像偏移不敏感（比较第5.2节）。

新认知器与现代、竞赛获胜的纯监督学习、前馈、基于梯度的深度学习器非常相似，这些深度学习器交替使用卷积层和下采样层（例如，第5.19节至第5.22节）。然而，福井康雄并没有通过监督反向传播来设置权重（第5.5节、第5.8节），而是通过基于WTA的局部无监督学习规则（例如，福井康雄, 2013b）或通过预连接。从这个意义上说，他对深度学习问题并不在意（第5.9节），尽管他的架构确实相当深。为了下采样目的，他使用了空间平均（福井康雄, 1980, 2011），而不是最大池化（MP, 第5.11节），后者是目前一种特别方便且流行的WTA机制。今天的深度学习组合，如卷积神经网络和最大池化以及反向传播，也从后来的工作（例如，第5.8节、第5.16节、第5.16节、第5.19节）中获益匪浅。

5.2 Around 1960: Visual Cortex Provides Inspiration for DL (Sec. 5.4, 5.11)

Simple cells and complex cells were found in the cat’s visual cortex (e.g., Hubel and Wiesel, 1962; Wiesel and Hubel, 1959). These cells fire in response to certain properties of visual sensory inputs, such as the orientation of edges. Complex cells exhibit more spatial invariance than simple cells. This inspired later deep NN architectures (Sec. 5.4, 5.11) used in certain modern award-winning Deep Learners (Sec. 5.19–5.22).

5.3 1965: Deep Networks Based on the Group Method of Data Handling

Networks trained by the *Group Method of Data Handling* (GMDH) (Ivakhnenko and Lapa, 1965; Ivakhnenko et al., 1967; Ivakhnenko, 1968, 1971) were perhaps the first DL systems of the *Feed-forward Multilayer Perceptron* type, although there was earlier work on NNs with a single hidden layer (e.g., Joseph, 1961; Viglione, 1970). The units of GMDH nets may have polynomial activation functions implementing *Kolmogorov-Gabor polynomials* (more general than other widely used NN activation functions, Sec. 2). Given a training set, layers are incrementally grown and trained by regression analysis (e.g., Legendre, 1805; Gauss, 1809, 1821) (Sec. 5.1), then pruned with the help of a separate *validation set* (using today’s terminology), where *Decision Regularisation* is used to weed out superfluous units (compare Sec. 5.6.3). The numbers of layers and units per layer can be learned in problem-dependent fashion. To my knowledge, this was the first example of open-ended, hierarchical representation learning in NNs (Sec. 4.3). A paper of 1971 already described a deep GMDH network with 8 layers (Ivakhnenko, 1971). There have been numerous applications of GMDH-style nets, e.g. (Ikeda et al., 1976; Farlow, 1984; Madala and Ivakhnenko, 1994; Ivakhnenko, 1995; Kondo, 1998; Kordík et al., 2003; Witczak et al., 2006; Kondo and Ueno, 2008).

5.4 1979: Convolution + Weight Replication + Subsampling (Neocognitron)

Apart from deep GMDH networks (Sec. 5.3), the *Neocognitron* (Fukushima, 1979, 1980, 2013a) was perhaps the first artificial NN that deserved the attribute *deep*, and the first to incorporate the neurophysiological insights of Sec. 5.2. It introduced *convolutional NNs* (today often called CNNs or convnets), where the (typically rectangular) receptive field of a *convolutional unit* with given weight vector (a *filter*) is shifted step by step across a 2-dimensional array of input values, such as the pixels of an image (usually there are several such filters). The resulting 2D array of subsequent activation events of this unit can then provide inputs to higher-level units, and so on. Due to massive *weight replication* (Sec. 2), relatively few parameters (Sec. 4.4) may be necessary to describe the behavior of such a *convolutional layer*.

Subsampling or *downsampling* layers consist of units whose fixed-weight connections originate from physical neighbours in the convolutional layers below. Subsampling units become active if at least one of their inputs is active; their responses are insensitive to certain small image shifts (compare Sec. 5.2).

The Neocognitron is very similar to the architecture of modern, contest-winning, purely *supervised*, feedforward, gradient-based Deep Learners with alternating convolutional and downsampling layers (e.g., Sec. 5.19–5.22). Fukushima, however, did not set the weights by supervised backpropagation (Sec. 5.5, 5.8), but by local, WTA-based *unsupervised* learning rules (e.g., Fukushima, 2013b), or by pre-wiring. In that sense he did not care for the DL problem (Sec. 5.9), although his architecture was comparatively deep indeed. For downsampling purposes he used *Spatial Averaging* (Fukushima, 1980, 2011) instead of *Max-Pooling* (MP, Sec. 5.11), currently a particularly convenient and popular WTA mechanism. Today’s DL combinations of CNNs and MP and BP also profit a lot from later work (e.g., Sec. 5.8, 5.16, 5.16, 5.19).

5.5 1960-1981及以后：反向传播（BP）在神经网络（NNs）中的发展

在复杂、非线性、可微分的多阶段、与神经网络相关的系统参数空间中通过梯度下降（哈达玛，1908）最小化误差至少自1960年代初（例如，凯利，1960；布莱森，1961；布莱森和登汉姆，1961；庞特里亚金等人，1961；德雷福斯，1962；威尔金森，1965；天野，1967；布莱森和霍，1969；迪克森和罗赫尔，1969）就有讨论，最初是在变分法（例如，欧拉，1744）的欧拉-拉格朗日方程框架内。

最速下降在这样系统的权重空间中可以通过迭代链式法则（Leibniz, 1676; L’Hôpital, 1696）à la 动态规划(DP)(Bellman, 1957)来执行（Bryson, 1961; Kelley, 1960; Bryson and Ho, 1969）。这种反向传播方法的简化推导仅使用链式法则（Dreyfus, 1962）。

1960年代的系统在动态规划（DP）意义上已经非常高效。然而，它们通过标准的雅可比矩阵计算将导数信息从一层反向传播到前一层，而没有明确处理跨多层的直接链接或由于网络稀疏性可能带来的额外效率提升（但也许这些增强对作者来说似乎是显而易见的）。考虑到在多层神经网络类似系统中的学习方面的所有先前工作（另见第5.3节关于1965年以来的深度非线性网络），从后见之明来看，一本关于简单单层线性感知器局限性的书（Minsky and Papert, 1969）（第5.1节）让一些研究人员不再进一步研究神经网络，这似乎令人惊讶。

显式、高效的误差反向传播（反向传播）在任意、离散、可能稀疏连接的类似神经网络中显然最早在1970年的硕士学位论文（Linnainmaa, 1970, 1976）中被描述，尽管当时并未提及神经网络。反向传播也被称为自动微分反向模式（Griewank, 2012），其中正向激活传播的成本基本上等于反向导数计算的成本。参见早期的FORTRAN代码（Linnainmaa, 1970）以及密切相关的工作（Ostrovskii等人，1971）。

高效的反向传播很快被明确用于通过调整控制参数（权重）来最小化成本函数（Dreyfus, 1973）。比较一些初步的、针对神经网络的讨论（Werbos, 1974, 第5.5.1节）、一种用于多层阈值神经网络的方法（Bobrowski, 1978），以及一个用于自动推导和实现给定可微系统的反向传播的计算机程序（Speelpenning, 1980）。

据我所知，上述高效反向传播（BP）作为特定于神经网络（NN）的应用，最早于1981年（Werbos, 1981, 2006）被描述。相关研究随后几年发表（Parker, 1985; LeCun, 1985, 1988）。1986年的一篇论文显著推动了反向传播在神经网络中的应用普及（Rumelhart 等人，1986），通过实验验证了隐藏层中出现有用内部表示的现象。参见序列处理循环神经网络（NN）的泛化（例如，Williams, 1989; Robinson 和 Fallside, 1987; Werbos, 1988; Williams 和 Zipser, 1988, 1989b,a; Rohwer, 1989; Pearlmutter, 1989; Gherrity, 1989; Williams 和 Peng, 1990; Schmidhuber, 1992a; Pearlmutter, 1995; Baldi, 1995; Kremer 和 Kolen, 2001; Atiya 和 Parlos, 2000），也适用于具有平稳输入的平衡循环神经网络（RNNs）（Almeida, 1987; Pineda, 1987）。

5.5.1 用于共享权重的前馈神经网络（FNNs）和循环神经网络（RNNs）的反向传播

使用第2节中关于共享权重的FNNs或RNNs的符号，在通过可微分的 f_i 激活传播的一个周期后，反向传播的单次梯度下降迭代计算所有 w_i 的变化，其与 $\frac{\partial E}{\partial w_i} = \sum_t \frac{\partial E}{\partial net_t} \frac{\partial net_t}{\partial w_i}$ 成比例，如算法 5.5.1（对于加性情况），其中每个权重 w_i 都与一个实值变量 $\triangle_i$ 相关联，该变量初始值为0。

反向传播（BP）的运算成本本质上与正向传播（第2节）相同。正向和反向传播会重复迭代，直到达到足够的性能。

5.5 1960-1981 and Beyond: Development of Backpropagation (BP) for NNs

The minimisation of errors through *gradient descent* (Hadamard, 1908) in the parameter space of complex, nonlinear, differentiable (Leibniz, 1684), multi-stage, NN-related systems has been discussed at least since the early 1960s (e.g., Kelley, 1960; Bryson, 1961; Bryson and Denham, 1961; Pontryagin et al., 1961; Dreyfus, 1962; Wilkinson, 1965; Amari, 1967; Bryson and Ho, 1969; Director and Rohrer, 1969), initially within the framework of Euler-LaGrange equations in the *Calculus of Variations* (e.g., Euler, 1744).

Steepest descent in the weight space of such systems can be performed (Bryson, 1961; Kelley, 1960; Bryson and Ho, 1969) by iterating the chain rule (Leibniz, 1676; L’Hôpital, 1696) à la *Dynamic Programming* (DP) (Bellman, 1957). A simplified derivation of this backpropagation method uses the chain rule only (Dreyfus, 1962).

The systems of the 1960s were already efficient in the DP sense. However, they backpropagated derivative information through standard Jacobian matrix calculations from one “layer” to the previous one, without explicitly addressing either direct links across several layers or potential additional efficiency gains due to network sparsity (but perhaps such enhancements seemed obvious to the authors). Given all the prior work on learning in multilayer NN-like systems (see also Sec. 5.3 on deep nonlinear nets since 1965), it seems surprising in hindsight that a book (Minsky and Papert, 1969) on the limitations of simple linear perceptrons with a single layer (Sec. 5.1) discouraged some researchers from further studying NNs.

Explicit, efficient error backpropagation (BP) in arbitrary, discrete, possibly sparsely connected, NN-like networks apparently was first described in a 1970 master’s thesis (Linnainmaa, 1970, 1976), albeit without reference to NNs. BP is also known as the reverse mode of automatic differentiation (Griewank, 2012), where the costs of forward activation spreading essentially equal the costs of backward derivative calculation. See early FORTRAN code (Linnainmaa, 1970) and closely related work (Ostrovskii et al., 1971).

Efficient BP was soon explicitly used to minimize cost functions by adapting control parameters (weights) (Dreyfus, 1973). Compare some preliminary, NN-specific discussion (Werbos, 1974, section 5.5.1), a method for multilayer threshold NNs (Bobrowski, 1978), and a computer program for automatically deriving and implementing BP for given differentiable systems (Speelpenning, 1980).

To my knowledge, the first NN-specific application of efficient BP as above was described in 1981 (Werbos, 1981, 2006). Related work was published several years later (Parker, 1985; LeCun, 1985, 1988). A paper of 1986 significantly contributed to the popularisation of BP for NNs (Rumelhart et al., 1986), experimentally demonstrating the emergence of useful internal representations in hidden layers. See generalisations for sequence-processing recurrent NNs (e.g., Williams, 1989; Robinson and Fallside, 1987; Werbos, 1988; Williams and Zipser, 1988, 1989b,a; Rohwer, 1989; Pearlmutter, 1989; Gherrity, 1989; Williams and Peng, 1990; Schmidhuber, 1992a; Pearlmutter, 1995; Baldi, 1995; Kremer and Kolen, 2001; Atiya and Parlos, 2000), also for equilibrium RNNs (Almeida, 1987; Pineda, 1987) with stationary inputs.

5.5.1 BP for Weight-Sharing Feedforward NNs (FNNs) and Recurrent NNs (RNNs)

Using the notation of Sec. 2 for weight-sharing FNNs or RNNs, after an episode of activation spreading through differentiable f_t , a single iteration of gradient descent through BP computes changes of all w_i in proportion to $\frac{\partial E}{\partial w_i} = \sum_t \frac{\partial E}{\partial net_t} \frac{\partial net_t}{\partial w_i}$ as in Algorithm 5.5.1 (for the additive case), where each weight w_i is associated with a real-valued variable $\triangle_i$ initialized by 0.

The computational costs of the backward (BP) pass are essentially those of the forward pass (Sec. 2). Forward and backward passes are re-iterated until sufficient performance is reached.

算法 5.5.1: 用于权重共享前馈神经网络或循环神经网络的反向传播的一次迭代

为 $t = T, \dots, 1$ 执行以计算 $\frac{\partial E}{\partial net_t}$, 初始化实值误差信号变量 δ_t 为 0; 如果 x_t 是一个输入事件, 则继续下一个迭代; 如果有错误 e_t , 则 $\delta_t := x_t - d_t$; 将值 $\sum_{k \in out_t} w_{v(t,k)} \delta_k$ 添加到 δ_t ; (这是优雅且高效的递归链式法则应用, 收集对 net_t 未来事件的影响) 将 δ_t 乘以 $f'_t(net_t)$; 对于所有 $k \in in_t$, 将值 $x_k \delta_t$ 添加到 $\Delta_{w_{v(k,t)}}$ **结束循环**按比例改变每个 w_i , 相对于 Δ_i 和一个小的实值学习率

截至 2014 年, 这种简单的反向传播方法仍然是前馈神经网络和循环神经网络的核心学习算法。值得注意的是, 截至 2014 年 (第 5.12、5.14、5.17、5.19、5.21、5.22 节), 大多数获奖神经网络并没有像第 5.7、5.10、5.15 节讨论的那样, 通过某种无监督学习来增强监督学习。不会通过某种无监督学习来增强监督学习

5.6 1980 年代末至 2000 年及以后: 神经网络的众多改进

到 1980 年代末, 人们似乎已经明确, 仅靠反向传播 (第 5.5 节) 并不能解决所有问题。大多数前馈神经网络应用都集中在具有很少隐藏层的前馈神经网络上。额外的隐藏层似乎并没有提供实证上的好处。许多从业者从以下定理中找到了慰藉 (科尔莫戈罗夫, 1965a; 赫赫特-尼尔森, 1989; 霍尼克等人, 1989), 该定理指出, 具有足够隐藏单元的单层神经网络可以任意精度地逼近任何多元连续函数。

同样地, 大多数循环神经网络 (RNN) 应用并不需要将误差反向传播得很远。许多研究人员通过首先在浅层问题上训练他们的 RNN (第 3 节), 然后让解决方案泛化到更深层的问题, 来帮助他们的 RNN。事实上, 一些流行的 RNN 算法将信用分配限制为单步反向 (Elman, 1990; Jordan, 1986, 1997), 这在更近期的研究中 (Jaeger, 2001; Maass 等人, 2002; Jaeger, 2004) 也是如此。

总的来说, 虽然反向传播 (BP) 原则上允许处理深层问题, 但它似乎只适用于<样式 id='1'>浅层</样式>问题。在 1980 年代末和 1990 年代初, 出现了一些有潜力克服这个问题的新思路, 这个问题直到 1991 年 (第 5.9 节) 才被完全理解。

5.6.1 处理长时滞和深层 CAPs 的思路

为了处理相关事件之间的长时滞, 提出了几种序列处理方法, 包括基于 RNN 中单元激活衰减因子的<样式 id='1'>聚焦反向传播</样式> (Mozer, 1989, 1992)、<样式 id='3'>时间延迟神经网络</样式> (TDNNs) (Lang 等人, 1990) 及其自适应扩展 (Bodenhausen 和 Waibel, 1991)、<样式 id='5'>非线性自回归外生输入</样式> (NARX) RNN (Lin 等人, 1996)、某些层次化 RNN (Hihi 和 Bengio, 1996) (比较第 5.10 节, 1991)、具有 WTA 单元和局部学习规则的 RNN 中的强化学习经济 (Schmidhuber, 1989b), 以及其他方法 (例如, Ring, 1993, 1994; Plate, 1993; de Vries 和 Principe, 1991; Sun 等人, 1993a; Bengio 等人, 1994)。然而, 这些算法要么只适用于浅层 CAPs, 要么无法泛化到未见过的 CAP 深度, 要么在相关事件之间时间滞差异很大时存在问题, 要么需要外部微调延迟常数, 要么受到其他问题的困扰。事实上, 结果表明, 某些简单但深层的基准问题, 用于评估这些方法, 实际上是通过随机猜测 RNN 权重直到找到解决方案来更快解决的 (Hochreiter 和 Schmidhuber, 1996)。

虽然上述 RNN 方法是为时间序列的深度学习设计的, 但 Schmidhuber (1990c) 提出的 神经热交换器 由两个流向相反的并行 深度前馈神经网络 组成。

Alg. 5.5.1: One iteration of BP for weight-sharing FNNs or RNNs

for $t = T, \dots, 1$ **do**
 to compute $\frac{\partial E}{\partial net_t}$, initialize real-valued error signal variable δ_t by 0;
 if x_t is an input event then continue with next iteration;
 if there is an error e_t then $\delta_t := x_t - d_t$;
 add to δ_t the value $\sum_{k \in out_t} w_{v(t,k)} \delta_k$; (*this is the elegant and efficient recursive chain rule application collecting impacts of net_t on future events*)
 multiply δ_t by $f'_t(net_t)$;
 for all $k \in in_t$ add to $\Delta_{w_{v(k,t)}}$ the value $x_k \delta_t$
end for
change each w_i in proportion to Δ_i and a small real-valued learning rate

As of 2014, this simple BP method is still the central learning algorithm for FNNs and RNNs. Notably, most contest-winning NNs up to 2014 (Sec. 5.12, 5.14, 5.17, 5.19, 5.21, 5.22) did *not* augment supervised BP by some sort of *unsupervised* learning as discussed in Sec. 5.7, 5.10, 5.15.

5.6 Late 1980s-2000 and Beyond: Numerous Improvements of NNs

By the late 1980s it seemed clear that BP by itself (Sec. 5.5) was no panacea. Most FNN applications focused on FNNs with few hidden layers. Additional hidden layers often did not seem to offer empirical benefits. Many practitioners found solace in a theorem (Kolmogorov, 1965a; Hecht-Nielsen, 1989; Hornik et al., 1989) stating that an NN with a single layer of enough hidden units can approximate any multivariate continous function with arbitrary accuracy.

Likewise, most RNN applications did not require backpropagating errors far. Many researchers helped their RNNs by first training them on shallow problems (Sec. 3) whose solutions then generalized to deeper problems. In fact, some popular RNN algorithms restricted credit assignment to a single step backwards (Elman, 1990; Jordan, 1986, 1997), also in more recent studies (Jaeger, 2001; Maass et al., 2002; Jaeger, 2004).

Generally speaking, although BP allows for deep problems in principle, it seemed to work only for *shallow* problems. The late 1980s and early 1990s saw a few ideas with a potential to overcome this problem, which was fully understood only in 1991 (Sec. 5.9).

5.6.1 Ideas for Dealing with Long Time Lags and Deep CAPs

To deal with long time lags between relevant events, several sequence processing methods were proposed, including *Focused BP* based on decay factors for activations of units in RNNs (Mozer, 1989, 1992), *Time-Delay Neural Networks* (TDNNs) (Lang et al., 1990) and their adaptive extension (Bodenhausen and Waibel, 1991), *Nonlinear AutoRegressive with eXogenous inputs* (NARX) RNNs (Lin et al., 1996), certain hierarchical RNNs (Hihi and Bengio, 1996) (compare Sec. 5.10, 1991), RL economies in RNNs with WTA units and local learning rules (Schmidhuber, 1989b), and other methods (e.g., Ring, 1993, 1994; Plate, 1993; de Vries and Principe, 1991; Sun et al., 1993a; Bengio et al., 1994). However, these algorithms either worked for shallow CAPs only, could not generalize to unseen CAP depths, had problems with greatly varying time lags between relevant events, needed external fine tuning of delay constants, or suffered from other problems. In fact, it turned out that certain simple but deep benchmark problems used to evaluate such methods are more quickly solved by *randomly guessing* RNN weights until a solution is found (Hochreiter and Schmidhuber, 1996).

While the RNN methods above were designed for DL of temporal sequences, the *Neural Heat Exchanger* (Schmidhuber, 1990c) consists of two parallel *deep FNNs* with opposite flow directions.

输入模式进入第一个前馈神经网络并“向上”传播。期望输出（目标）进入“相反”的前馈神经网络并“向下”传播。使用局部学习规则，每个网络中的每一层都试图与前一层以及另一网络的相邻层在信息内容上相似。进入第一个网络的输入会缓慢“升温”以成为目标。进入相反网络的目标会缓慢“降温”以成为输入。亥姆霍兹机 (Dayan 等人, 1995; Dayan 和 Hinton, 1996) 可以看作是监督学习（第 5.6.4 节）的一种无监督变体（Peter Dayan, 1994 年个人通信）。

一种混合方法（Shavlik 和 Towell, 1989; Towell 和 Shavlik, 1994）通过命题逻辑中的领域理论初始化一个潜在的深度 FNN，该领域理论可以通过基于解释的学习（Mitchell 等人, 1986; DeJong 和 Mooney, 1986; Minton 等人, 1989）获得。然后通过 BP（第 5.5 节）对 NN 进行微调。NN 的深度反映了原始逻辑规则集中推理的最长链。这种方法的扩展（Maclin 和 Shavlik, 1993; Shavlik, 1994）通过表示为有限状态自动机（FSA）的领域知识初始化 RNN。基于 BP 的微调对于后来通过 UL 预训练的 DL 系统变得很重要，例如第 5.10 节、5.15 节。

5.6.2 通过高级梯度下降实现更好的 BP（比较第 5.24 节）

通过反向传播（Sec. 5.5）对最速下降法提出了许多改进。最小二乘法（高斯-牛顿法、Levenberg-Marquardt 法）（高斯, 1809 年; 牛顿, 1687 年; Levenberg, 1944 年; Marquardt, 1963 年; Schaback 和 Werner, 1992 年）和拟牛顿法（Broyden-Fletcher-Goldfarb-Shanno 法, BFGS）（Broyden 等人, 1965 年; Fletcher 和 Powell, 1963 年; Goldfarb, 1970 年; Shanno, 1970 年）对于大型神经网络来说计算成本过高。部分 BFGS（Battiti, 1992 年; Saito 和 Nakano, 1997 年）和共轭梯度法（Hestenes 和 Stiefel, 1952 年; Møller, 1993 年）以及其他方法（Solla, 1988 年; Schmidhuber, 1989a; Cauwenberghs, 1993 年）有时能提供有用的快速替代方案。反向传播可以被视为一个线性最小二乘问题（Biegler-König 和 Bärman, 1993 年），其中二阶梯度信息被传递回前一层。

为加速反向传播，动量 被引入（Rumelhart 等人, 1986 年），在线性化激活函数的斜率上添加了自定义常数（Fahlman, 1988 年），或夸张了斜率的非线性（West 和 Saad, 1995 年）。

成功且广泛使用的反向传播变体 $R\text{-prop}$ （Riedmiller 和 Braun, 1993 年）以及稳健变体 $iRprop+$ （Igel 和 Hüsken, 2003 年）仅考虑误差导数的符号，这些变体也成功应用于循环神经网络。

基于神经网络架构，可以对局部梯度进行归一化（Schraudolph 和 Sejnowski, 1996 年），通过对角化 Hessian 方法（Becker 和 Le Cun, 1989 年），或相关高效方法（Schraudolph, 2002 年）。

一些控制反向传播步长的算法采用全局学习率（Lapedes 和 Farber, 1986 年; Vogl 等人, 1988 年; Battiti, 1989 年; LeCun 等人, 1993 年; Yu 等人, 1995 年），而另一些算法为每个权重计算个体学习率（Jacobs, 1988 年; Silva 和 Almeida, 1990 年）。在在线学习中，反向传播在每次模式呈现后应用，vario- η 算法（Neuneier 和 Zimmermann, 1996 年）将每个权重的学习率与其局部梯度的经验标准差成反比，从而归一化随机权重波动。比较非线性神经网络局部在线步长自适应方法（Almeida 等人, 1997 年）。

许多用于改进神经网络（NNs）的额外技巧已被描述（例如，Orr 和 Müller, 1998; Montavon 等人, 2012）。比较第 5.6.3 节和第 5.24 节中提到的最新进展。

Input patterns enter the first FNN and are propagated “up”. Desired outputs (targets) enter the “opposite” FNN and are propagated “down”. Using a local learning rule, each layer in each net tries to be similar (in information content) to the preceding layer and to the adjacent layer of the other net. The input entering the first net slowly “heats up” to become the target. The target entering the opposite net slowly “cools down” to become the input. The *Helmholtz Machine* (Dayan et al., 1995; Dayan and Hinton, 1996) may be viewed as an unsupervised (Sec. 5.6.4) variant thereof (Peter Dayan, personal communication, 1994).

A hybrid approach (Shavlik and Towell, 1989; Towell and Shavlik, 1994) initializes a potentially deep FNN through a domain theory in propositional logic, which may be acquired through explanation-based learning (Mitchell et al., 1986; DeJong and Mooney, 1986; Minton et al., 1989). The NN is then fine-tuned through BP (Sec. 5.5). The NN’s depth reflects the longest chain of reasoning in the original set of logical rules. An extension of this approach (Maclin and Shavlik, 1993; Shavlik, 1994) initializes an RNN by domain knowledge expressed as a Finite State Automaton (FSA). BP-based fine-tuning has become important for later DL systems pre-trained by UL, e.g., Sec. 5.10, 5.15.

5.6.2 Better BP Through Advanced Gradient Descent (Compare Sec. 5.24)

Numerous improvements of steepest descent through BP (Sec. 5.5) have been proposed. Least-squares methods (Gauss-Newton, Levenberg-Marquardt) (Gauss, 1809; Newton, 1687; Levenberg, 1944; Marquardt, 1963; Schaback and Werner, 1992) and quasi-Newton methods (Broyden-Fletcher-Goldfarb-Shanno, BFGS) (Broyden et al., 1965; Fletcher and Powell, 1963; Goldfarb, 1970; Shanno, 1970) are computationally too expensive for large NNs. Partial BFGS (Battiti, 1992; Saito and Nakano, 1997) and conjugate gradient (Hestenes and Stiefel, 1952; Møller, 1993) as well as other methods (Solla, 1988; Schmidhuber, 1989a; Cauwenberghs, 1993) provide sometimes useful fast alternatives. BP can be treated as a linear least-squares problem (Biegler-König and Bärman, 1993), where second-order gradient information is passed back to preceding layers.

To speed up BP, *momentum* was introduced (Rumelhart et al., 1986), ad-hoc constants were added to the slope of the linearized activation function (Fahlman, 1988), or the nonlinearity of the slope was exaggerated (West and Saad, 1995).

Only the signs of the error derivatives are taken into account by the successful and widely used BP variant $R\text{-prop}$ (Riedmiller and Braun, 1993) and the robust variation $iRprop+$ (Igel and Hüsken, 2003), which was also successfully applied to RNNs.

The local gradient can be normalized based on the NN architecture (Schraudolph and Sejnowski, 1996), through a diagonalized Hessian approach (Becker and Le Cun, 1989), or related efficient methods (Schraudolph, 2002).

Some algorithms for controlling BP step size adapt a global learning rate (Lapedes and Farber, 1986; Vogl et al., 1988; Battiti, 1989; LeCun et al., 1993; Yu et al., 1995), while others compute individual learning rates for each weight (Jacobs, 1988; Silva and Almeida, 1990). In online learning, where BP is applied after each pattern presentation, the vario- η algorithm (Neuneier and Zimmermann, 1996) sets each weight’s learning rate inversely proportional to the empirical standard deviation of its local gradient, thus normalizing the stochastic weight fluctuations. Compare a local online step size adaptation method for nonlinear NNs (Almeida et al., 1997).

Many additional tricks for improving NNs have been described (e.g., Orr and Müller, 1998; Montavon et al., 2012). Compare Sec. 5.6.3 and recent developments mentioned in Sec. 5.24.

5.6.3 寻找简单、低复杂度、问题解决型神经网络（第5.24节）

许多研究人员使用类似反向传播（BP）的方法来寻找“简单”、低复杂度的神经网络（NNs）（第4.4节），并具有高泛化能力。大多数方法通过强先验假设来解决偏差/方差困境（Geman等人，1992）。例如，权重衰减（weight decay）（Hanson和Pratt，1989；Weigend等人，1991；Krogh和Hertz，1992）通过惩罚大的权重来鼓励接近零的权重。在贝叶斯框架（Bayes，1763）中，权重衰减可以从高斯或拉普拉斯权重先验（Gauss，1809；Laplace，1774）中推导出来（Hinton和van Camp，1993）；另见（Mur-ray和Edwards，1993）。这种方法的扩展假设具有许多相似权重的网络分布由高斯混合模型生成是“更好”的先验（a priori）（Nowlan和Hinton，1992）。

通常权重先验体现在附加惩罚项中（MacKay，1992年）或体现在基于<样式 id='1'>验证集</样式>的方法中（Mosteller和Tukey，1968年；Stone，1974年；Eubank，1988年；Hastie和Tibshirani，1990年；Craven和Wahba，1979年；Golub等，1979年），赤池信息准则和<样式 id='3'>最终预测误差</样式>（赤池，1970年、1973年、1974年），或<样式 id='5'>广义预测误差</样式>（Moody和Utans，1994年；Moody，1992年）。另见（Holden，1994年；Wang等，1994年；Amari和Murata，1993年；Wang等，1994年；Guyon等，1992年；Vapnik，1992年；Wolpert，1994年）。类似的先验（或对简单性的偏好）隐含在构造和剪枝算法中，例如逐层<样式 id='7'>顺序网络构造</样式>（例如，Ivakhnenko，1968年、1971年；Ash，1989年；Moody，1989年；Gallant，1988年；Honavar和Uhr，1988年；Ring，1991年；Fahlman，1991年；Weng等，1992年；Honavarand Uhr，1993年；Burgess，1994年；Fritzke，1994年；Parekh等，2000年；Utgoff和Stracuzzi，2002年）（另见第5.3节、第5.11节），<样式 id='9'>输入剪枝</样式>（Moody，1992年；Refenes等，1994年），<样式 id='11'>单元剪枝</样式>（例如，Ivakhnenko，1968年、1971年；White，1989年；Mozer和Smolenisky，1989年；Levin等，1994年），<样式 id='13'>权重剪枝</样式>，例如<样式 id='15'>最优脑损伤</样式>（LeCun等，1990b），和<样式 id='17'>最优脑外科医生</样式>（Hassibi和Stork，1993年）。

一种非常通用但不总是实用的方法，用于发现低复杂度的监督学习神经网络或强化学习神经网络，这种方法是在用通用编程语言编写的权重矩阵计算程序中搜索，并倾向于快速和简短的程序（Schmidhuber，1997）（第6.7节）。

平坦最小值搜索(FMS)（Hochreiter和Schmidhuber，1997a，1999）搜索误差函数的“平坦”最小值：权重空间中一个大型连通区域，其中误差较低且保持近似恒定，也就是说，用高方差描述低精度权重只需要很少的信息。比较扰动容忍条件(Minai和Williams，1994；Murray和Edwards，1993；Hanson，1990；Neti等人，1992；Matsuoka，1992；Bishop，1993；Ker-lirzin和Vallet，1993；Carter等人，1990)。基于MDL的贝叶斯论证表明，平坦最小值对应于“简单”的神经网络和低预期过拟合。比较第5.6.4节以及第5.24节中提到的最新发展。

5.6.4 无监督学习对监督学习的潜在益处（参见第5.7、5.10、5.15节）

第2节引入了教师提供的标签 d_t 。然而，上一个世纪的许多论文却是关于无监督学习（UL）且无需教师(例如，Hebb，1949；von der Mals-burg，1973；Kohonen，1972，1982，1988；Willshaw and von der Malsburg，1976；Grossberg，1976a,b；Watanabe，1985；Pearlmutter and Hinton，1986；Barrow，1987；Field，1987；Oja，1989；Barlow et al.，1989；Baldi and Hornik，1989；Sanger，1989；Ritter and Kohonen，1989；Rubner and Schulten，1990；Földiák，1990；Martinetz et al.，1990；Kosko，1990；Mozer，1991；Palm，1992；Atick et al.，1992；Miller，1994；Saund，1994；Földiák and Young，1995；Deco and Parra，1997)；另见2000年后的工作（例如，Carreira-Perpinan，2001；Wiskott and Sejnowski，2002；Franzius et al.，2007；Waydo and Koch，2008）。

许多无监督学习方法旨在最大化与熵相关的、信息论的目标（例如玻尔兹曼，1909年；香农，1948年；库尔贝克和莱布勒，1951年）（例如林斯克，1988年；巴洛等人，1989年；麦凯和米勒，1990年；普伦比，1991年；Schmidhuber，1992b,c；施劳多夫和塞乔夫斯基，

5.6.3 Searching For Simple, Low-Complexity, Problem-Solving NNs (Sec. 5.24)

Many researchers used BP-like methods to search for “simple,” low-complexity NNs (Sec. 4.4) with high generalization capability. Most approaches address the *bias/variance dilemma* (Geman et al., 1992) through strong prior assumptions. For example, *weight decay* (Hanson and Pratt, 1989; Weigend et al., 1991; Krogh and Hertz, 1992) encourages near-zero weights, by penalizing large weights. In a Bayesian framework (Bayes, 1763), weight decay can be derived (Hinton and van Camp, 1993) from Gaussian or Laplacian weight priors (Gauss, 1809; Laplace, 1774); see also (Murray and Edwards, 1993). An extension of this approach postulates that a distribution of networks with many similar weights generated by Gaussian mixtures is “better” *a priori* (Nowlan and Hinton, 1992).

Often weight priors are implicit in additional penalty terms (MacKay, 1992) or in methods based on *validation sets* (Mosteller and Tukey, 1968; Stone, 1974; Eubank, 1988; Hastie and Tibshirani, 1990; Craven and Wahba, 1979; Golub et al., 1979), Akaike’s information criterion and *final prediction error* (Akaike, 1970, 1973, 1974), or *generalized prediction error* (Moody and Utans, 1994; Moody, 1992). See also (Holden, 1994; Wang et al., 1994; Amari and Murata, 1993; Wang et al., 1994; Guyon et al., 1992; Vapnik, 1992; Wolpert, 1994). Similar priors (or biases towards simplicity) are implicit in constructive and pruning algorithms, e.g., layer-by-layer *sequential network construction* (e.g., Ivakhnenko, 1968, 1971; Ash, 1989; Moody, 1989; Gallant, 1988; Honavar and Uhr, 1988; Ring, 1991; Fahlman, 1991; Weng et al., 1992; Honavar and Uhr, 1993; Burgess, 1994; Fritzke, 1994; Parekh et al., 2000; Utgoff and Stracuzzi, 2002) (see also Sec. 5.3, 5.11), *input pruning* (Moody, 1992; Refenes et al., 1994), *unit pruning* (e.g., Ivakhnenko, 1968, 1971; White, 1989; Mozer and Smolen-sky, 1989; Levin et al., 1994), *weight pruning*, e.g., *optimal brain damage* (LeCun et al., 1990b), and *optimal brain surgeon* (Hassibi and Stork, 1993).

A very general but not always practical approach for discovering low-complexity SL NNs or RL NNs searches among weight matrix-computing programs written in a universal programming language, with a bias towards fast and short programs (Schmidhuber, 1997) (Sec. 6.7).

Flat Minimum Search (FMS) (Hochreiter and Schmidhuber, 1997a, 1999) searches for a “flat” minimum of the error function: a large connected region in weight space where error is low and remains approximately constant, that is, few bits of information are required to describe low-precision weights with high variance. Compare *perturbation tolerance conditions* (Minai and Williams, 1994; Murray and Edwards, 1993; Hanson, 1990; Neti et al., 1992; Matsuoka, 1992; Bishop, 1993; Ker-lirzin and Vallet, 1993; Carter et al., 1990). An MDL-based, Bayesian argument suggests that flat minima correspond to “simple” NNs and low expected overfitting. Compare Sec. 5.6.4 and more recent developments mentioned in Sec. 5.24.

5.6.4 Potential Benefits of UL for SL (Compare Sec. 5.7, 5.10, 5.15)

The notation of Sec. 2 introduced teacher-given labels d_t . Many papers of the previous millennium, however, were about *unsupervised learning (UL) without a teacher* (e.g., Hebb, 1949; von der Mals-burg, 1973; Kohonen, 1972, 1982, 1988; Willshaw and von der Malsburg, 1976; Grossberg, 1976a,b; Watanabe, 1985; Pearlmutter and Hinton, 1986; Barrow, 1987; Field, 1987; Oja, 1989; Barlow et al., 1989; Baldi and Hornik, 1989; Sanger, 1989; Ritter and Kohonen, 1989; Rubner and Schulten, 1990; Földiák, 1990; Martinetz et al., 1990; Kosko, 1990; Mozer, 1991; Palm, 1992; Atick et al., 1992; Miller, 1994; Saund, 1994; Földiák and Young, 1995; Deco and Parra, 1997); see also post-2000 work (e.g., Carreira-Perpinan, 2001; Wiskott and Sejnowski, 2002; Franzius et al., 2007; Waydo and Koch, 2008).

Many UL methods are designed to maximize entropy-related, information-theoretic (Boltzmann, 1909; Shannon, 1948; Kullback and Leibler, 1951) objectives (e.g., Linsker, 1988; Barlow et al., 1989; MacKay and Miller, 1990; Plumbley, 1991; Schmidhuber, 1992b,c; Schraudolph and Sejnowski,

1993年; Redlich, 1993年; Zemel, 1993年; Zemel和Hinton, 1994年; Field, 1994年; Hinton等人, 1995年; Dayan和Zemel, 1995年; Amari等人, 1996年; Deco和Parra, 1997年)。许多人这样做是为了揭示和分离信号隐藏的潜在来源(例如, Jutten和Herault, 1991年; Schuster, 1992年; Andrade等人, 1993年; Molgedey和Schuster, 1994年; Comon, 1994年; Cardoso, 1994年; Bell和Sejnowski, 1995年; Karhunen和Joutsensalo, 1995年; Belouchrani等人, 1997年; Hyvärinen等人, 2001年; Szabó等人, 2006年; Shan等人, 2007年; Shan和Cottrell, 2014年)。

许多UL方法通过众所周知的特征检测器(例如, Olshausen和Field, 1996年; Schmidhuber等人, 1996年), 如中心偏移-环绕样结构、以及方向敏感的边缘检测器和Gabor滤波器(Gabor, 1946年), 自动且稳健地生成输入模式的分布式、稀疏表示(Földiák, 1990年; Hinton和Ghahramani, 1997年; Lewicki和Olshausen, 1998年; Hyvärinen等人, 1999年; Hochreiter和Schmidhuber, 1999年; Falconbridge等人, 2006年)。它们提取与生物系统早期视觉预处理阶段观察到的简单特征相关的特征(例如, De Valois等人, 1982年; Jones和Palmer, 1987年)。

无监督学习也可以通过观察两个不同的输入(Schmidhuber 和 Prelinger, 1992), 利用耦合神经网络从不同的数据项中提取不变特征(例如, Becker, 1991), 这种网络也被称为孪生神经网络(例如, Bromley 等人, 1993; Hadsell 等人, 2006; Taylor 等人, 2011; Chen 和 Salman, 2011)。

无监督学习可以帮助以有利于进一步处理的形式对输入数据进行编码。在深度学习的背景下, 无监督学习的一个重要目标是冗余减少。理想情况下, 通过深度神经网络进行冗余减少, 将为输入模式集合创建一个因子码(一个具有统计独立分量的码)(Barlow 等人, 1989; Barlow, 1989), 以分离未知的变化因素(比较 Bengio 等人, 2013)。这样的码可能是稀疏的, 并且有利于: (1) 数据压缩, (2) 加速随后的反向传播(Becker, 1991), (3) 简化随后的朴素但最优的贝叶斯分类器的任务(Schmidhuber 等人, 1996)。

早期的许多无监督学习前馈神经网络只有一个层级。用于构建更深层数的无监督学习前馈神经网络的方法包括层级<第4.3节>自组织Kohonen映射(例如, Koikkalainen和Oja, 1990年; Lampinen和Oja, 1992年; Versino和Gambardella, 1996年; Dittenbach等人, 2000年; Rauber等人, 2002年), 层级<样式 id='1'>高斯函数</样式>网络(Lee和Kil, 1991年), 将特征层级逐层输入监督学习分类器的无监督学习(Behnke, 1999年, 2003a年), <样式 id='3'>自组织树算法</样式>(SOTA)(Herrero等人, 2001年), 以及非线性<样式 id='5'>自动编码器</样式>(AEs)具有3层以上(例如, 5层)(Kramer, 1991年; Oja, 1991年; DeMers和Cottrell, 1993年)。此类AE神经网络(Rumelhart等人, 1986年)可以被训练以将输入模式映射到自身, 例如, 通过窄瓶颈隐藏层单元的激活来紧凑地编码它们。某些非线性AE存在某些局限性(Baldi, 2012年)。

LOCOCODE(Hochreiter和Schmidhuber, 1999)使用FMS(Sec. 5.6.3)来寻找具有低复杂度的AE, 其权重精度低, 可以用少量信息描述, 通常会产生稀疏或因子码。可预测性最小化(PM)(Schmidhuber, 1992c)通过非线性特征检测器搜索因子码, 这些检测器与非线性预测器对抗, 试图同时变得尽可能信息和不可预测。基于PM的无监督学习不仅应用于前馈神经网络, 也应用于循环神经网络(例如, Schmidhuber, 1993b; Lindstädt, 1993)。比较关于基于UL的RNN堆栈的第5.10节(1991年), 以及后来的UL RNN(例如, Klapper-Rybicka等人, 2001年; Steil, 2007年)。

5.7 1987: 通过自动编码器(AE)层次的无监督学习(比较Sec. 5.15)

或许, 第一篇研究UL(无监督学习)预训练潜在益处的工作是在1987年发表的。它提出了无监督AE(自动编码器)层次结构(Ballard, 1987), 与某些2000年后基于UL的前馈深度学习器密切相关(第5.15节)。最低级别的具有单个隐藏层的AE(自动编码器)神经网络被训练以将输入模式映射到自身。然后将其隐藏层代码输入到更高一级的同类型AE(自动编码器)中, 以此类推。希望隐藏的AE(自动编码器)层中的代码具有某些属性

1993; Redlich, 1993; Zemel, 1993; Zemel and Hinton, 1994; Field, 1994; Hinton et al., 1995; Dayan and Zemel, 1995; Amari et al., 1996; Deco and Parra, 1997).

Many do this to uncover and disentangle hidden underlying sources of signals (e.g., Jutten and Herault, 1991; Schuster, 1992; Andrade et al., 1993; Molgedey and Schuster, 1994; Comon, 1994; Cardoso, 1994; Bell and Sejnowski, 1995; Karhunen and Joutsensalo, 1995; Belouchrani et al., 1997; Hyvärinen et al., 2001; Szabó et al., 2006; Shan et al., 2007; Shan and Cottrell, 2014).

Many UL methods automatically and robustly generate distributed, sparse representations of input patterns (Földiák, 1990; Hinton and Ghahramani, 1997; Lewicki and Olshausen, 1998; Hyvärinen et al., 1999; Hochreiter and Schmidhuber, 1999; Falconbridge et al., 2006) through well-known feature detectors (e.g., Olshausen and Field, 1996; Schmidhuber et al., 1996), such as *off-center-on-surround*-like structures, as well as orientation sensitive edge detectors and Gabor filters (Gabor, 1946). They extract simple features related to those observed in early visual pre-processing stages of biological systems (e.g., De Valois et al., 1982; Jones and Palmer, 1987).

UL can also serve to extract invariant features from different data items (e.g., Becker, 1991) through *coupled NNs* observing two different inputs (Schmidhuber and Prelinger, 1992), also called *Siamese NNs* (e.g., Bromley et al., 1993; Hadsell et al., 2006; Taylor et al., 2011; Chen and Salman, 2011).

UL can help to encode input data in a form advantageous for further processing. In the context of DL, one important goal of UL is redundancy reduction. Ideally, given an ensemble of input patterns, redundancy reduction through a deep NN will create a *factorial code* (a code with statistically independent components) of the ensemble (Barlow et al., 1989; Barlow, 1989), to disentangle the unknown factors of variation (compare Bengio et al., 2013). Such codes may be sparse and can be advantageous for (1) data compression, (2) speeding up subsequent BP (Becker, 1991), (3) trivialising the task of subsequent naive yet optimal Bayes classifiers (Schmidhuber et al., 1996).

Most early UL FNNs had a single layer. Methods for deeper UL FNNs include hierarchical (Sec. 4.3) self-organizing Kohonen maps (e.g., Koikkalainen and Oja, 1990; Lampinen and Oja, 1992; Versino and Gambardella, 1996; Dittenbach et al., 2000; Rauber et al., 2002), hierarchical *Gaussian potential function* networks (Lee and Kil, 1991), layer-wise UL of feature hierarchies fed into SL classifiers (Behnke, 1999, 2003a), the *Self-Organising Tree Algorithm* (SOTA) (Herrero et al., 2001), and nonlinear *Autoencoders* (AEs) with more than 3 (e.g., 5) layers (Kramer, 1991; Oja, 1991; DeMers and Cottrell, 1993). Such AE NNs (Rumelhart et al., 1986) can be trained to map input patterns to themselves, for example, by compactly encoding them through activations of units of a narrow bottleneck hidden layer. Certain nonlinear AEs suffer from certain limitations (Baldi, 2012).

LOCOCODE (Hochreiter and Schmidhuber, 1999) uses FMS (Sec. 5.6.3) to find low-complexity AEs with low-precision weights describable by few bits of information, often producing sparse or factorial codes. *Predictability Minimization* (PM) (Schmidhuber, 1992c) searches for factorial codes through nonlinear feature detectors that fight nonlinear predictors, trying to become both as informative and as unpredictable as possible. PM-based UL was applied not only to FNNs but also to RNNs (e.g., Schmidhuber, 1993b; Lindstädt, 1993). Compare Sec. 5.10 on UL-based RNN stacks (1991), as well as later UL RNNs (e.g., Klapper-Rybicka et al., 2001; Steil, 2007).

5.7 1987: UL Through Autoencoder (AE) Hierarchies (Compare Sec. 5.15)

Perhaps the first work to study potential benefits of UL-based pre-training was published in 1987. It proposed unsupervised AE hierarchies (Ballard, 1987), closely related to certain post-2000 feedforward Deep Learners based on UL (Sec. 5.15). The lowest-level AE NN with a single hidden layer is trained to map input patterns to themselves. Its hidden layer codes are then fed into a higher-level AE of the same type, and so on. The hope is that the codes in the hidden AE layers have properties that

比较一篇早期的调查（Hinton, 1989）和与之 somewhat 相关的 递归自联想记忆 （RAAM）（Pollack, 1988, 1990; Melnik等人, 2000），最初用于通过固定数量的隐藏单元编码任意大小的序列语言结构。最近，RAAM也被用作无监督预处理器，以促进强化学习的深度信用分配（Gisslen等人, 2011）（第6.4节）。

原则上，许多无监督学习方法（第5.6.4节）可以像上述自编码器一样堆叠起来，也可以是第5.10节中用于历史压缩的循环神经网络、第5.15节的<样式 id='1'>限制玻尔兹曼机</样式>(RBMs)，或是第5.6.4节中的分层Kohonen网络，以促进后续的监督学习。比较<样式 id='3'>堆叠泛化</样式>(Wolpert, 1992; Ting和Witten, 1997)，以及那些通过竞争性无监督学习（例如，Rumelhart和Zipser, 1986）进行预训练，然后通过基于反向传播的微调进行优化的前馈神经网络（Maclin和Shavlik, 1995）。另见更多使用无监督学习来改进后续监督学习的新方法（例如，Behnke, 1999, 2003a; Escalante-B.和Wiskott, 2013）。

5.8 1989: 反向传播用于卷积神经网络(CNNs, 第5.4节)

1989年，反向传播（第5.5节）被应用于类似Neocognitron、权值共享的卷积神经网络层（第5.4节），并具有自适应连接。这种组合，通过增强 最大池化 （MP，第5.11节、第5.16节），并在图形卡上加速（第5.19节），已成为许多现代、竞赛获胜的前馈视觉深度学习器（第5.19–5.23节）的基本要素。这项工作还引入了手写数字的MNIST数据集（Le-Cun等人, 1989年），随着时间的推移，它可能成为机器学习最著名的基准之一。卷积神经网络帮助在MNIST（LeCun等人, 1990a）（CAP深度5）和指纹识别（Baldi和Chauvin, 1993年）上取得了良好的性能；类似的卷积神经网络在20世纪90年代被商业使用。

5.9 1991: 梯度下降的深度学习基本问题

一篇学位论文（Hochreiter, 1991）代表了显式深度学习研究的里程碑。如第5.6节所述，到20世纪80年代末，实验表明传统的深度前馈或循环网络很难通过反向传播（BP）（第5.5节）进行训练。Hochreiter的工作正式确定了一个主要原因：典型的深度神经网络受到如今著名的梯度消失或梯度爆炸问题的困扰。使用标准激活函数（第1节），累积反向传播的错误信号（第5.5.1节）要么迅速缩小，要么超出范围。事实上，它们在层数或CAP深度（第3节）上呈指数级衰减，或者会爆炸。这也被称为<样式 id='1'>长时间滞后问题</样式>。20世纪90年代和21世纪初的许多后续深度学习研究都受到了这一见解的启发。后来的工作（Bengio等人, 1994）也从动力系统角度研究了吸引域及其在噪声下的稳定性：要么动力学对噪声不鲁棒，要么梯度消失。另见（Hochreiter等人, 2001a; Ti^ˆno和Hammer, 2004）。多年来，人们探索了部分克服<样式 id='3'>深度学习基本问题</样式>的方法：

我是1991年的一位非常深度学习者（历史压缩器，第5.10节），通过无监督预训练来缓解问题，用于RNN的层次结构。这极大地简化了后续通过BP（第5.5节）进行的监督信用分配。在FNN的情况下，可以通过概念上相关的AE堆栈（第5.7节、5.15节）和深度信念网络(DBNs, 第5.15节)实现类似的效果。

facilitate subsequent learning. In one experiment, a particular AE-specific learning algorithm (different from traditional BP of Sec. 5.5.1) was used to learn a mapping in an AE stack pre-trained by this type of UL (Ballard, 1987). This was faster than learning an equivalent mapping by BP through a single deeper AE without pre-training. On the other hand, the task did not really require a deep AE, that is, the benefits of UL were not that obvious from this experiment. Compare an early survey (Hinton, 1989) and the somewhat related *Recursive Auto-Associative Memory* (RAAM) (Pollack, 1988, 1990; Melnik et al., 2000), originally used to encode sequential linguistic structures of arbitrary size through a fixed number of hidden units. More recently, RAAMs were also used as unsupervised pre-processors to facilitate deep credit assignment for RL (Gisslen et al., 2011) (Sec. 6.4).

In principle, many UL methods (Sec. 5.6.4) could be stacked like the AEs above, the history-compressing RNNs of Sec. 5.10, the *Restricted Boltzmann Machines* (RBMs) of Sec. 5.15, or hierarchical Kohonen nets (Sec. 5.6.4), to facilitate subsequent SL. Compare *Stacked Generalization* (Wolpert, 1992; Ting and Witten, 1997), and FNNs that profit from pre-training by *competitive* UL (e.g., Rumelhart and Zipser, 1986) prior to BP-based fine-tuning (Maclin and Shavlik, 1995). See also more recent methods using UL to improve subsequent SL (e.g., Behnke, 1999, 2003a; Escalante-B. and Wiskott, 2013).

5.8 1989: BP for Convolutional NNs (CNNs, Sec. 5.4)

In 1989, backpropagation (Sec. 5.5) was applied (LeCun et al., 1989, 1990a, 1998) to Neocognitron-like, weight-sharing, convolutional neural layers (Sec. 5.4) with adaptive connections. This combination, augmented by *Max-Pooling* (MP, Sec. 5.11, 5.16), and sped up on graphics cards (Sec. 5.19), has become an essential ingredient of many modern, competition-winning, feedforward, visual Deep Learners (Sec. 5.19–5.23). This work also introduced the MNIST data set of handwritten digits (LeCun et al., 1989), which over time has become perhaps the most famous benchmark of Machine Learning. CNNs helped to achieve good performance on MNIST (LeCun et al., 1990a) (CAP depth 5) and on fingerprint recognition (Baldi and Chauvin, 1993); similar CNNs were used commercially in the 1990s.

5.9 1991: Fundamental Deep Learning Problem of Gradient Descent

A diploma thesis (Hochreiter, 1991) represented a milestone of explicit DL research. As mentioned in Sec. 5.6, by the late 1980s, experiments had indicated that traditional deep feedforward or recurrent networks are hard to train by backpropagation (BP) (Sec. 5.5). Hochreiter’s work formally identified a major reason: Typical deep NNs suffer from the now famous problem of vanishing or exploding gradients. With standard activation functions (Sec. 1), cumulative backpropagated error signals (Sec. 5.5.1) either shrink rapidly, or grow out of bounds. In fact, they decay exponentially in the number of layers or CAP depth (Sec. 3), or they explode. This is also known as the *long time lag problem*. Much subsequent DL research of the 1990s and 2000s was motivated by this insight. Later work (Bengio et al., 1994) also studied basins of attraction and their stability under noise from a dynamical systems point of view: either the dynamics are not robust to noise, or the gradients vanish. See also (Hochreiter et al., 2001a; Tiño and Hammer, 2004). Over the years, several ways of partially overcoming the *Fundamental Deep Learning Problem* were explored:

I A Very Deep Learner of 1991 (the *History Compressor*, Sec. 5.10) alleviates the problem through unsupervised pre-training for a hierarchy of RNNs. This greatly facilitates subsequent supervised credit assignment through BP (Sec. 5.5). In the FNN case, similar effects can be achieved through conceptually related AE stacks (Sec. 5.7, 5.15) and *Deep Belief Networks* (DBNs, Sec. 5.15).

II 类似LSTM的网络（第5.13节、5.16节、5.17节、5.21–5.23节），通过一种不受其影响的特殊架构来缓解问题。

III 如今基于GPU的计算机比1990年代初的台式机计算能力高出百万倍。这使得即使在传统的神经网络中（第5.18节），也能在合理的时间内将误差传播几层。基本上这就是现在赢得许多图像识别竞赛的原因（第5.19、5.21、5.22节）。（尽管这并没有从根本上解决该问题。）

IV 无Hessian优化（第5.6.2节）可以缓解前馈神经网络（Møller, 1993; Pearlmutter, 1994; Schraudolph, 2002; Martens, 2010）（第5.6.2节）和循环神经网络（Martens和Sutskever, 2011）（第5.20节）的问题。

V 神经网络权重矩阵的空间也可以在不依赖误差梯度的前提下进行搜索，从而完全避免深度学习基本问题。随机权重猜测有时比更复杂的方法效果更好（Hochreiter和Schmidhuber, 1996）。某些更复杂的问题更适合使用通用搜索（Levin, 1973b）来为用通用编程语言编写的权重矩阵计算程序进行求解（Schmidhuber, 1997）。有些问题更适合使用线性方法来为连接到输出事件的权重获得最优解（第2节），以及进化连接到其他事件的权重——这被称为*Evolino*（Schmidhuber等人, 2007）。也请比较由某些无监督学习规则预训练的相关循环神经网络（Steil, 2007），在脉冲神经元（Yin等人, 2012; Klampfl和Maass, 2013）（第5.26节）的情况下也是如此。直接搜索方法不仅与监督学习相关，也与更一般的强化学习相关，将在第6.6节中更详细地讨论。

5.10 1991：基于无监督学习的压缩历史记录通过深度堆叠的循环神经网络

1991年的一项工作 非常深度学习器（第3节）（Schmidhuber, 1992b, 2013a）可以通过使用 无监督预训练 对RNN的层次结构进行信用分配，跨越数百个非线性算子或神经网络层。

基本思想至今仍然相关。每个RNN都以无监督的方式训练一段时间，以预测其下一个输入（例如，Connor等人, 1994年；Dorffner, 1996年）。从那时起，只有意外的输入（错误）会传递新信息并传递给下一个更高层的RNN，从而使它在更慢的自组织时间尺度上运行。可以很容易地证明信息不会丢失。它只是被压缩了（机器学习的很大一部分本质上就是关于压缩，例如，第4.4节，5.6.3节，6.7节）。对于每个单独的输入序列，我们在这个 历史压缩器 或 神经序列块化器 的更深层次上得到一系列越来越冗余的编码，它可以在空间（类似于前馈神经网络）和时间上进行压缩。这是层次化表示学习的另一个好例子（第4.3节）。还有一个历史压缩器的连续变体（Schmidhuber等人, 1993年）。

RNN堆栈本质上是一种数据的深度生成模型，可以从其压缩形式中重建。向堆栈中添加另一个RNN可以提高数据描述长度的界限——相当于其概率的负对数（Huffman, 1952; Shannon, 1948）——只要在数据表示的相应层次上仍有剩余的局部可学习预测性。比较一下关于前馈深度信念网络(DBNs, 2006, 第5.15节)的类似观察。

该系统能够学习许多先前无法学习的深度学习任务。一个古老的说明性深度学习实验（Schmidhuber, 1993b）需要深度为1200的CAPs（第3节）。然而，最初的无监督RNN堆栈顶层代码变得如此紧凑，以至于通过额外的基于BP的监督学习（先前不可行）的序列分类成为可能。本质上，该系统使用了无监督学习来

II LSTM-like networks (Sec. 5.13, 5.16, 5.17, 5.21–5.23) alleviate the problem through a special architecture unaffected by it.

III Today’s GPU-based computers have a million times the computational power of desktop machines of the early 1990s. This allows for propagating errors a few layers further down within reasonable time, even in traditional NNs (Sec. 5.18). That is basically what is winning many of the image recognition competitions now (Sec. 5.19, 5.21, 5.22). (Although this does not really overcome the problem in a fundamental way.)

IV Hessian-free optimization (Sec. 5.6.2) can alleviate the problem for FNNs (Møller, 1993; Pearlmutter, 1994; Schraudolph, 2002; Martens, 2010) (Sec. 5.6.2) and RNNs (Martens and Sutskever, 2011) (Sec. 5.20).

V The space of NN weight matrices can also be searched without relying on error gradients, thus avoiding the *Fundamental Deep Learning Problem* altogether. Random weight guessing sometimes works better than more sophisticated methods (Hochreiter and Schmidhuber, 1996). Certain more complex problems are better solved by using *Universal Search* (Levin, 1973b) for weight matrix-computing programs written in a universal programming language (Schmidhuber, 1997). Some are better solved by using linear methods to obtain optimal weights for connections to output events (Sec. 2), and *evolving* weights of connections to other events—this is called *Evolino* (Schmidhuber et al., 2007). Compare also related RNNs pre-trained by certain UL rules (Steil, 2007), also in the case of *spiking neurons* (Yin et al., 2012; Klampfl and Maass, 2013) (Sec. 5.26). Direct search methods are relevant not only for SL but also for more general RL, and are discussed in more detail in Sec. 6.6.

5.10 1991: UL-Based History Compression Through a Deep Stack of RNNs

A working *Very Deep Learner* (Sec. 3) of 1991 (Schmidhuber, 1992b, 2013a) could perform credit assignment across hundreds of nonlinear operators or neural layers, by using *unsupervised pre-training* for a hierarchy of RNNs.

The basic idea is still relevant today. Each RNN is trained for a while in unsupervised fashion to predict its next input (e.g., Connor et al., 1994; Dorffner, 1996). From then on, only unexpected inputs (errors) convey new information and get fed to the next higher RNN which thus ticks on a slower, self-organising time scale. It can easily be shown that no information gets lost. It just gets compressed (much of machine learning is essentially about compression, e.g., Sec. 4.4, 5.6.3, 6.7). For each individual input sequence, we get a series of less and less redundant encodings in deeper and deeper levels of this *History Compressor* or *Neural Sequence Chunker*, which can compress data in both space (like feedforward NNs) and time. This is another good example of hierarchical representation learning (Sec. 4.3). There also is a continuous variant of the history compressor (Schmidhuber et al., 1993).

The RNN stack is essentially a deep generative model of the data, which can be reconstructed from its compressed form. Adding another RNN to the stack improves a bound on the data’s description length—equivalent to the negative logarithm of its probability (Huffman, 1952; Shannon, 1948)—as long as there is remaining local learnable predictability in the data representation on the corresponding level of the hierarchy. Compare a similar observation for feedforward *Deep Belief Networks* (DBNs, 2006, Sec. 5.15).

The system was able to learn many previously unlearnable DL tasks. One ancient illustrative DL experiment (Schmidhuber, 1993b) required CAPs (Sec. 3) of depth 1200. The top level code of the initially unsupervised RNN stack, however, got so compact that (previously infeasible) sequence classification through additional BP-based SL became possible. Essentially the system used UL to

极大降低问题深度。比较早期基于反向传播的、由命题逻辑规则初始化的神经网络的微调（Shavlik和Towell, 1989）（第5.6.1节）。

存在一种将高层信息压缩到低层的方法，从而完全或部分地折叠RNN堆栈。诀窍是重新训练一个低层RNN，使其能够持续地模仿（预测）一个已经训练好、较慢、较高层的RNN（即“有意识的分块器”）的隐藏单元，通过额外的预测输出神经元（Schmidhuber, 1992b）。这有助于低层RNN（即自动化器）发展适当且很少变化的记忆，从而能够跨越非常长的时间间隔。同样，此过程可以大大减少反向传播过程所需的深度。

1991年的系统是一个符合现代2000年后定义的深度学习器，也是一个首批神经分层时间记忆(HTM)。它在概念上与早期的AE层次结构（1987年，第5.7节）以及后来的深度信念网络(2006年，第5.15节)相似，但在使用序列处理循环神经网络（RNNs）而不是具有不变输入的前馈神经网络（FNNs）方面更为通用。最近，一些知名的企业家（Hawkins和George, 2006年；Kurzweil, 2012年）也对HTM产生了兴趣；比较一下层次化隐马尔可夫模型（HMMs）（例如，Fine等人，1998年），以及后来的基于无监督学习（UL）的循环系统（Klapper-Rybicka等人，2001年；Steil, 2007年；Klampfl和Maass, 2013年；Young等人，2014年）。钟表循环神经网络（Koutník等人，2014年）也由具有不同时钟速率的交互式循环神经网络模块组成，但它们不使用UL来设置这些速率。循环神经网络的堆栈在后来的监督学习（SL）工作中取得了巨大成功，例如，第5.13节、5.16节、5.17节、5.22节。

5.11 1992：最大池化（MP）：迈向MPCNNs（比较第5.16节、5.19节）

风格标签为'1'的Neocognitron（第5.4节）启发了风格标签为'3'的Cresceptron（Weng等人，1992年），它在训练过程中会调整其拓扑结构（第5.6.3节）；比较一下逐渐增长和收缩的GMDH网络（1965年，第5.3节）。

Cresceptron没有使用替代性的局部子采样或WTA方法（例如Fukushima, 1980年；Schmidhuber, 1989b；Maass, 2000年；Fukushima, 2013a年），而是使用了风格标签为'1'的最大池化（MP）层。在这里，一个二维单元激活层或数组被划分为更小的矩形数组。每个矩形数组在降采样层中被替换为其最大激活单元的激活值。Cresceptron的一个更复杂的后续版本（Weng等人，1997年）还包括了风格标签为'3'的'模糊'层，以提高对象位置容忍度。

前馈HMAX模型（Riesenhuber和Poggio, 1999）的神经生理学上合理的拓扑结构非常类似于1992年的Cresceptron（因此也类似于1979年的Neocognitron）。HMAX并不会学习。它的单元具有手工设计的权重；后来为类似模型提出了生物上合理的学习规则（例如，Serre等人，2002年；Teichmann等人，2012年）。

当卷积神经网络（CNNs或convnets，第5.4、5.8节）与最大池化（MP）结合时，它们会变成类似Cresceptron或HMAX的*MPCNNs*，具有交替的卷积和最大池化层。然而，与Cresceptron和HMAX不同，MPCNNs通过反向传播（BP，第5.5、5.16节）（Ranzato等人，2007年）进行训练。随后指出了这样做的好处（Scherer等人，2010年）。反向传播训练的MPCNN已成为许多现代、竞赛获胜、前馈、视觉深度学习者的核心（第5.17、5.19–5.23节）。

5.12 1994：早期获奖神经网络

20世纪90年代，某些神经网络已经凭借秘密测试集赢得了某些受控的模式识别竞赛。值得注意的是，一个带有内部延迟线的神经网络赢得了1994年圣达菲时间序列竞赛，该竞赛研究的是氨激光混沌强度脉冲（Wan, 1994；Weigend和Gershenfeld, 1993）。不过，虽然如此，但并不需要非常深的CAPs（第3节）。

greatly reduce problem depth. Compare earlier BP-based fine-tuning of NNs initialized by rules of propositional logic (Shavlik and Towell, 1989) (Sec. 5.6.1).

There is a way of compressing higher levels down into lower levels, thus fully or partially collapsing the RNN stack. The trick is to retrain a lower-level RNN to continually imitate (predict) the hidden units of an already trained, slower, higher-level RNN (the “*conscious*” *chunker*), through additional predictive output neurons (Schmidhuber, 1992b). This helps the lower RNN (the *automatizer*) to develop appropriate, rarely changing memories that may bridge very long time lags. Again, this procedure can greatly reduce the required depth of the BP process.

The 1991 system was a working Deep Learner in the modern post-2000 sense, and also a first *Neural Hierarchical Temporal Memory* (HTM). It is conceptually similar to earlier AE hierarchies (1987, Sec. 5.7) and later *Deep Belief Networks* (2006, Sec. 5.15), but more general in the sense that it uses sequence-processing RNNs instead of FNNs with unchanging inputs. More recently, well-known entrepreneurs (Hawkins and George, 2006; Kurzweil, 2012) also got interested in HTMs; compare also hierarchical HMMs (e.g., Fine et al., 1998), as well as later UL-based recurrent systems (Klapper-Rybicka et al., 2001; Steil, 2007; Klampfl and Maass, 2013; Young et al., 2014). Clockwork RNNs (Koutník et al., 2014) also consist of interacting RNN modules with different clock rates, but do not use UL to set those rates. Stacks of RNNs were used in later work on SL with great success, e.g., Sec. 5.13, 5.16, 5.17, 5.22.

5.11 1992: Max-Pooling (MP): Towards MPCNNs (Compare Sec. 5.16, 5.19)

The *Neocognitron* (Sec. 5.4) inspired the *Cresceptron* (Weng et al., 1992), which adapts its topology during training (Sec. 5.6.3); compare the incrementally growing and shrinking GMDH networks (1965, Sec. 5.3).

Instead of using alternative local subsampling or WTA methods (e.g., Fukushima, 1980; Schmidhuber, 1989b; Maass, 2000; Fukushima, 2013a), the Cresceptron uses *Max-Pooling* (MP) layers. Here a 2-dimensional layer or array of unit activations is partitioned into smaller rectangular arrays. Each is replaced in a downsampling layer by the activation of its maximally active unit. A later, more complex version of the Cresceptron (Weng et al., 1997) also included “*blurring*” layers to improve object location tolerance.

The neurophysiologically plausible topology of the feedforward HMAX model (Riesenhuber and Poggio, 1999) is very similar to the one of the 1992 Cresceptron (and thus to the 1979 Neocognitron). HMAX does not learn though. Its units have hand-crafted weights; biologically plausible learning rules were later proposed for similar models (e.g., Serre et al., 2002; Teichmann et al., 2012).

When CNNs or convnets (Sec. 5.4, 5.8) are combined with MP, they become Cresceptron-like or HMAX-like *MPCNNs* with alternating convolutional and max-pooling layers. Unlike Cresceptron and HMAX, however, MPCNNs are trained by BP (Sec. 5.5, 5.16) (Ranzato et al., 2007). Advantages of doing this were pointed out subsequently (Scherer et al., 2010). BP-trained MPCNNs have become central to many modern, competition-winning, feedforward, visual Deep Learners (Sec. 5.17, 5.19–5.23).

5.12 1994: Early Contest-Winning NNs

Back in the 1990s, certain NNs already won certain controlled pattern recognition contests with secret test sets. Notably, an NN with internal delay lines won the Santa Fe time-series competition on chaotic intensity pulsations of an NH3 laser (Wan, 1994; Weigend and Gershenfeld, 1993). No very deep CAPs (Sec. 3) were needed though.

5.13 1995：监督循环超深度学习器（LSTM RNN）

监督型<样式 id='1'>长短期记忆</样式>(LSTM)循环神经网络（Hochreiter和Schmidhuber, 1997b; Gers等人, 2000; P´erez-Ortiz等人, 2003）最终能够像1991年的深度循环神经网络层次结构（第5.10节）那样完成类似的壮举，而无需任何无监督预训练即可克服<样式 id='4'>深度学习基本问题</样式>（第5.9节）。LSTM还可以学习深度学习任务<样式 id='6'>无需</样式>局部序列可预测性（因此<样式 id='8'>无法</样式>被部分无监督的1991年<样式 id='10'>历史压缩器</样式>学习，第5.10节），处理非常深的问题（第3节）（例如，Gers等人, 2002年）。

LSTM 的基本思想非常简单。其中一些单元被称为 恒定误差回环(CECs)。每个 CEC 使用 f 作为激活函数，即恒等函数，并且自身有一个固定权重为 1.0 的连接。由于 f 的恒定导数为 1.0，通过 CEC 反向传播的误差不会消失或爆炸（第 5.9 节），而是保持不变（除非它们“流”出 CEC 到其他、通常是自适应的神经网络部分）。CECs 连接到多个非线性自适应单元（其中一些具有乘法激活函数），用于学习非线性行为。这些单元的权重变化通常受益于通过 CECs 长时间反向传播的误差信号。CECs 是 LSTM 网络能够学习发现并记忆数千个离散时间步长前发生的事件重要性的主要原因，而之前的 RNNs 在最小时间延迟 10 步的情况下就已经失败了。

允许使用许多不同的LSTM变体和拓扑结构。可以进化出针对特定问题的良好拓扑结构（Bayer等人, 2009年）。一些LSTM变体还使用可修改的CEC自连接（Gers和Schmidhuber, 2001年）。

在某种程度上，LSTM具有生物学上的合理性（O’ Reilly, 2003年）。LSTM学会了解决许多以前无法学习的深度学习任务，包括：在嘈杂输入流中识别广泛分离事件的时间顺序；在长时间间隔内稳健存储高精度实数；对连续输入流进行算术运算；提取事件之间时间距离所传递的信息；在嘈杂输入序列中识别时间上扩展的模式（Hochreiter和Schmidhuber, 1997b; Gers等人, 2000年）；稳定生成精确计时节律，以及平滑和非平滑的周期性轨迹（Gers和Schmidhuber, 2000年）。在需要学习可由确定性<样式id='1'>有限状态自动机</样式>(FSAs)描述的规则语言的任务上，LSTM在可靠性和速度方面都明显优于之前的循环神经网络（Watrous和Kuhn, 1992年; Casey, 1996年; Siegelmann, 1992年; Blair和Pollack, 1997年; Kalinke和Lehmann, 1998年; Zeng等人, 1994年; Manolios和Fanelli, 1994年; Omlin和Giles, 1996年; Vahed和Omlin, 2004年）。

LSTM 也适用于涉及上下文无关语言（CFL）的任务，这些语言无法用 HMM 或 RNN 文献中讨论的类似 FSAs 表示（Sun 等人, 1993b; Wiles 和 Elman, 1995; Andrews 等人, 1995; Steijvers 和 Grunwald, 1996; Tonkes 和 Wiles, 1997; Rodriguez 等人, 1999; Rodriguez 和 Wiles, 1998）。CFL 识别（Lee, 1996）需要运行时堆栈的功能等效。一些先前的 RNN 无法学习小的 CFL 训练集（Rodriguez 和 Wiles, 1998）。那些没有成功的（Rodriguez 等人, 1999; Bod´en 和 Wiles, 2000）无法提取通用规则，并且在显著更大的测试集上泛化效果不佳。对于上下文敏感语言（CSL）（例如，Chalup 和 Blair, 2003）也是如此。然而 LSTM 泛化效果很好，只需要 CSL $a^n b^n c^n$ 的 30 个最短示例 ($n \leq 10$) 就能正确预测 n 长度达 1000 或更多的序列前缀的可能延续。一个解耦的扩展卡尔曼滤波器（Kalman, 1960; Williams, 1992b; Puskorius 和 Feldkamp, 1994; Feldkamp 等人, 1998; Haykin, 2001; Feld- kamp 等人, 2003）与 LSTM RNN（P´erez-Ortiz 等人, 2003）的组合学会了正确处理高达 1000 万甚至更多的 n 值。也就是说，经过训练后，网络能够一次读取 3000 万个符号以上的序列，并最终检测出 合法字符串（例如 $a^{10,000,000}b^{10,000,000}c^{10,000,000}$ ）和非常相似但 非法字符串（例如 $a^{10,000,000}b^{9,999,999}c^{10,000,000}$ ）之间的细微差别。也可以比较更近期的能够处理长时间滞后的 RNN 算法（Sch´afer 等人, 2006; Martens 和 Sutskever, 2011; Zimmermann 等人, 2012; Koutnm ≤ 10 k 等人, 2014）。

5.13 1995: Supervised Recurrent Very Deep Learner (LSTM RNN)

Supervised *Long Short-Term Memory* (LSTM) RNN (Hochreiter and Schmidhuber, 1997b; Gers et al., 2000; Pérez-Ortiz et al., 2003) could eventually perform similar feats as the deep RNN hierarchy of 1991 (Sec. 5.10), overcoming the *Fundamental Deep Learning Problem* (Sec. 5.9) without any unsupervised pre-training. LSTM could also learn DL tasks *without* local sequence predictability (and thus *unlearnable* by the partially unsupervised 1991 *History Compressor*, Sec. 5.10), dealing with very deep problems (Sec. 3) (e.g., Gers et al., 2002).

The basic LSTM idea is very simple. Some of the units are called *Constant Error Carousels* (CECs). Each CEC uses as an activation function f , the identity function, and has a connection to itself with fixed weight of 1.0. Due to f ’s constant derivative of 1.0, errors backpropagated through a CEC cannot vanish or explode (Sec. 5.9) but stay as they are (unless they “flow out” of the CEC to other, typically *adaptive* parts of the NN). CECs are connected to several *nonlinear adaptive* units (some with multiplicative activation functions) needed for learning nonlinear behavior. Weight changes of these units often profit from error signals propagated far back in time through CECs. CECs are the main reason why LSTM nets can learn to discover the importance of (and memorize) events that happened thousands of discrete time steps ago, while previous RNNs already failed in case of minimal time lags of 10 steps.

Many different LSTM variants and topologies are allowed. It is possible to evolve good problem-specific topologies (Bayer et al., 2009). Some LSTM variants also use *modifiable* self-connections of CECs (Gers and Schmidhuber, 2001).

To a certain extent, LSTM is biologically plausible (O’Reilly, 2003). LSTM learned to solve many previously unlearnable DL tasks involving: Recognition of the temporal order of widely separated events in noisy input streams; Robust storage of high-precision real numbers across extended time intervals; Arithmetic operations on continuous input streams; Extraction of information conveyed by the temporal distance between events; Recognition of temporally extended patterns in noisy input sequences (Hochreiter and Schmidhuber, 1997b; Gers et al., 2000); Stable generation of precisely timed rhythms, as well as smooth and non-smooth periodic trajectories (Gers and Schmidhuber, 2000). LSTM clearly outperformed previous RNNs on tasks that require learning the rules of regular languages describable by deterministic *Finite State Automata* (FSAs) (Watrous and Kuhn, 1992; Casey, 1996; Siegelmann, 1992; Blair and Pollack, 1997; Kalinke and Lehmann, 1998; Zeng et al., 1994; Manolios and Fanelli, 1994; Omlin and Giles, 1996; Vahed and Omlin, 2004), both in terms of reliability and speed.

LSTM also worked on tasks involving context free languages (CFLs) that cannot be represented by HMMs or similar FSAs discussed in the RNN literature (Sun et al., 1993b; Wiles and Elman, 1995; Andrews et al., 1995; Steijvers and Grunwald, 1996; Tonkes and Wiles, 1997; Rodriguez et al., 1999; Rodriguez and Wiles, 1998). CFL recognition (Lee, 1996) requires the functional equivalent of a run-time stack. Some previous RNNs failed to learn small CFL training sets (Rodriguez and Wiles, 1998). Those that did not (Rodriguez et al., 1999; Bodén and Wiles, 2000) failed to extract the general rules, and did not generalize well on substantially larger test sets. Similar for context-sensitive languages (CSLs) (e.g., Chalup and Blair, 2003). LSTM generalized well though, requiring only the 30 shortest exemplars ($n \leq 10$) of the CSL $a^n b^n c^n$ to correctly predict the possible continuations of sequence prefixes for n up to 1000 and more. A combination of a decoupled extended Kalman filter (Kalman, 1960; Williams, 1992b; Puskorius and Feldkamp, 1994; Feldkamp et al., 1998; Haykin, 2001; Feldkamp et al., 2003) and an LSTM RNN (Pérez-Ortiz et al., 2003) learned to deal correctly with values of n up to 10 million and more. That is, after training the network was able to read sequences of 30,000,000 symbols and more, one symbol at a time, and finally detect the subtle differences between *legal* strings such as $a^{10,000,000}b^{10,000,000}c^{10,000,000}$ and very similar but *illegal* strings such as $a^{10,000,000}b^{9,999,999}c^{10,000,000}$. Compare also more recent RNN algorithms able to deal with long

能够处理时间滞后 (Schäfer 等人, 2006; Martens 和 Sutskever, 2011; Zimmermann 等人, 2012; Koutník 等人, 2014)。

双向RNNs (BRNNs) (Schuster和Paliwal, 1997; Schuster, 1999) 是为输入序列设计, 其起始和结束是预先已知的, 例如需要根据其音素进行标注的语音句子; 比较 (Fukada等人, 1999)。为了考虑每个序列元素的过去和未来上下文, 一个RNN从开始到结束处理序列, 另一个从结束到开始反向处理。在每个时间步, 它们的组合输出预测相应的标签 (如果存在的话)。BRNNs已成功应用于二级蛋白质结构预测 (Baldi等人, 1999)。DAG-RNNs (Baldi和Pollastri, 2003; 吴和Baldi, 2008) 将BRNNs推广到多个维度。它们学会了预测小有机分子的性质 (Lusci等人, 2013), 以及蛋白质接触图 (Tegge等人, 2009), 也结合了不断增长的深度FNN (Di Lena等人, 2012) (第5.21节)。当与LSTM概念结合时 (Graves和Schmidhuber, 2005, 2009; Graves等人, 2009), BRNNs和DAG-RNNs才能充分发挥其全部潜力。

在最近的比赛中, 由 连接主义时序分类(CTC) (Graves等人, 2006) 训练的 LSTM 循环神经网络堆栈 (第 5.10 节) (Fernandez等人, 2007; Graves和Schmidhuber, 2009) 尤其成功。CTC 是一种基于梯度的方法, 用于寻找 RNN 权重, 以最大化给定的 (通常更长且更高维的) 实值输入向量序列的教师标签概率。CTC-LSTM 执行同时分割 (对齐) 和识别 (第 5.22 节)。

在 2000 年代初, 语音识别主要由 HMM 与 FNN 组合主导 (例如, Bourlard 和 Morgan, 1994)。然而, 当在 TIDIG-ITS 语音数据库的语音上从头开始训练时, 在 2003 年 LSTM 已经获得了与基于 HMM 的系统相当的结果 (Graves等人, 2003; Beringer等人, 2005; Graves等人, 2006)。在 2007 年, LSTM 在关键词识别任务中超越了 HMM (Fernández等人, 2007); 比较最近的改进 (Indermuhle等人, 2011; Wöllmer等人, 2013)。到 2013 年, LSTM 在著名的 TIMIT 音素识别基准测试上也取得了当时最佳结果 (Graves等人, 2013) (第 5.22 节)。最近, LSTM 循环神经网络 / HMM 混合体在中等词汇量 (Geiger等人, 2014) 和大规模词汇量语音识别 (Sak 等人, 2014a) 上获得了当时最佳性能。

LSTM也适用于机器人定位 (Förster等人, 2007年)、机器人控制 (Mayer等人, 2008年)、在线驾驶员分心检测 (Wöllmer等人, 2011年) 以及其他许多任务。例如, 它帮助改进了蛋白质分析 (Hochreiter和Obermayer, 2005年)、手写识别 (Graves等人, 2008年、2009年; Graves和Schmidhuber, 2009年; Bluche等人, 2014年)、语音活动检测 (Eyben等人, 2013年)、光学字符识别 (Breuel等人, 2013年)、语言识别 (Gonzalez-Dominguez等人, 2014年)、韵律轮廓预测 (Fernandez等人, 2014年)、音频起始检测 (Marchi等人, 2014年)、文本到语音合成 (Fan等人, 2014年)、社交信号分类 (Brueckner和Schulter, 2014年)、机器翻译 (Sutskever等人, 2014年) 以及其他许多应用领域的最先进技术。

循环神经网络 (RNNs) 也可用于元学习 (Schmidhuber, 1987; Schaul和Schmidhuber, 2010; Prokhorov等人, 2002), 因为它们原则上可以学习运行自己的权重变化算法 (Schmidhuber, 1993a)。一个成功的元学习器 (Hochreiter等人, 2001b) 使用LSTM循环神经网络快速学习一种学习算法 来处理二次函数 (比较第6.8节)。

近年来, 长短期记忆循环神经网络 (LSTM RNNs) 在多项国际模式识别竞赛中获胜, 并在大型复杂数据集上创下了多个基准记录, 例如第5.17节、5.21节、5.22节。尽管基于梯度的LSTM并非万能药——其他方法有时在特定任务上表现更优 (Jaeger, 2004; Schmidhuber等人, 2007; Martens和Sutskever, 2011; Pascanu等人, 2013b; Koutník等人, 2014); 对比第5.20节。

time lags (Schäfer et al., 2006; Martens and Sutskever, 2011; Zimmermann et al., 2012; Koutník et al., 2014).

Bi-directional RNNs (BRNNs) (Schuster and Paliwal, 1997; Schuster, 1999) are designed for input sequences whose starts and ends are known in advance, such as spoken sentences to be labeled by their phonemes; compare (Fukada et al., 1999). To take both past and future context of each sequence element into account, one RNN processes the sequence from start to end, the other backwards from end to start. At each time step their combined outputs predict the corresponding label (if there is any). BRNNs were successfully applied to secondary protein structure prediction (Baldi et al., 1999). DAG-RNNs (Baldi and Pollastri, 2003; Wu and Baldi, 2008) generalize BRNNs to multiple dimensions. They learned to predict properties of small organic molecules (Lusci et al., 2013) as well as protein contact maps (Tegge et al., 2009), also in conjunction with a growing deep FNN (Di Lena et al., 2012) (Sec. 5.21). BRNNs and DAG-RNNs unfold their full potential when combined with the LSTM concept (Graves and Schmidhuber, 2005, 2009; Graves et al., 2009).

Particularly successful in recent competitions are stacks (Sec. 5.10) of LSTM RNNs (Fernandez et al., 2007; Graves and Schmidhuber, 2009) trained by *Connectionist Temporal Classification* (CTC) (Graves et al., 2006), a gradient-based method for finding RNN weights that maximize the probability of teacher-given label sequences, given (typically much longer and more high-dimensional) streams of real-valued input vectors. CTC-LSTM performs simultaneous segmentation (alignment) and recognition (Sec. 5.22).

In the early 2000s, speech recognition was dominated by HMMs combined with FNNs (e.g., Bourlard and Morgan, 1994). Nevertheless, when trained from scratch on utterances from the TIDIG-ITS speech database, in 2003 LSTM already obtained results comparable to those of HMM-based systems (Graves et al., 2003; Beringer et al., 2005; Graves et al., 2006). In 2007, LSTM outperformed HMMs in keyword spotting tasks (Fernández et al., 2007); compare recent improvements (Indermuhle et al., 2011; Wöllmer et al., 2013). By 2013, LSTM also achieved best known results on the famous TIMIT phoneme recognition benchmark (Graves et al., 2013) (Sec. 5.22). Recently, LSTM RNN / HMM hybrids obtained best known performance on medium-vocabulary (Geiger et al., 2014) and large-vocabulary speech recognition (Sak et al., 2014a).

LSTM is also applicable to robot localization (Förster et al., 2007), robot control (Mayer et al., 2008), online driver distraction detection (Wöllmer et al., 2011), and many other tasks. For example, it helped to improve the state of the art in diverse applications such as protein analysis (Hochreiter and Obermayer, 2005), handwriting recognition (Graves et al., 2008, 2009; Graves and Schmidhuber, 2009; Bluche et al., 2014), voice activity detection (Eyben et al., 2013), optical character recognition (Breuel et al., 2013), language identification (Gonzalez-Dominguez et al., 2014), prosody contour prediction (Fernandez et al., 2014), audio onset detection (Marchi et al., 2014), text-to-speech synthesis (Fan et al., 2014), social signal classification (Brueckner and Schulter, 2014), machine translation (Sutskever et al., 2014), and others.

RNNs can also be used for metalearning (Schmidhuber, 1987; Schaul and Schmidhuber, 2010; Prokhorov et al., 2002), because they can in principle learn to run their own weight change algorithm (Schmidhuber, 1993a). A successful metalearner (Hochreiter et al., 2001b) used an LSTM RNN to quickly *learn a learning algorithm* for quadratic functions (compare Sec. 6.8).

Recently, LSTM RNNs won several international pattern recognition competitions and set numerous benchmark records on large and complex data sets, e.g., Sec. 5.17, 5.21, 5.22. Gradient-based LSTM is no panacea though—other methods sometimes outperformed it at least on certain tasks (Jaeger, 2004; Schmidhuber et al., 2007; Martens and Sutskever, 2011; Pascanu et al., 2013b; Koutník et al., 2014); compare Sec. 5.20.

5.14 2003：更多竞赛获胜/创纪录的神经网络；成功的深度神经网络

在2000年左右的那十年间，许多实际和商业模式识别应用主要由非神经网络机器学习方法主导，例如支持向量机 (SVMs) (Vapnik, 1995; Schölkopf等人, 1998)。尽管如此，至少在某些领域，神经网络 (NNs) 的表现优于其他技术。

一个基于神经网络集成 (Breiman, 1996; Schapire, 1990; Wolpert, 1992; Hashem and Schmeiser, 1992; Ueda, 2000; Dietterich, 2000a) 的<样式 id='1'>贝叶斯神经网络 </样式>(Neal, 2006)赢得了<样式 id='3'>NIPS 2003特征选择挑战 </样式>的秘测试集 (Nealand Zhang, 2006)。尽管这个神经网络不是很深——它有两个隐藏层，因此其CAPs (第3节) 的深度为3，相当浅。

对许多当前竞赛获胜的模式识别器 (第5.19、5.21、5.22节) 而言，CNN领域的发展非常重要。一个反向传播训练 (LeCun等人, 1989年) 的CNN (第5.4、5.8节) 创造了新的MNIST记录0.4% (Simard等人, 2003年)，使用了训练模式变形 (Baird, 1990年)，但没有无监督预训练 (第5.7、5.10、5.15节)。一个标准的反向传播网络达到了0.7% (Simard等人, 2003年)。同样，相应的CAP深度较低。比较第5.16、5.18、5.19节的进一步改进。

使用反向传播变体 *R-prop* (Riedmiller和Braun, 1993) 训练的相当深的神经网络 (Behnke, 2003b) 取得了良好的图像解释结果 (Sec. 5.6.2)；这里通过循环连接的反馈有助于提高图像解释。使用CAP深度高达6的前馈神经网络成功地对高维数据进行了分类 (Vieiraand Barradas, 2003)。

深度LSTM RNN开始获得与基于HMM系统的语音识别结果相当的第一批结果 (Graves et al.,2003)；比较Sec.5.13, 5.16, 5.21, 5.22。

5.15 2006/7: 无监督学习 对于深度信念网络/ AEStacks 通过BP微调

虽然学习具有多个非线性层的网络至少可以追溯到1965年 (Sec. 5.3)，并且显式深度学习研究结果至少自1991年以来就已发表 (Sec. 5.9, 5.10)，但“深度学习”这一术语实际上是在2006年左右被创造的，当时无监督预训练的深度前馈神经网络帮助通过BP加速了后续的监督学习 (Hinton and Salakhutdinov, 2006; Hinton et al., 2006)。比较早期的关于 加载深度网络 (Síma, 1994; Windisch, 2005) 和 学习深度记忆 (Gomez and Schmidhuber, 2005) 的术语。比较BP基础 (Sec. 5.5) 的微调 (Sec. 5.6.1) 对通过竞争性无监督学习预训练的 (不太深的) 前馈神经网络的微调 (Maclin and Shavlik, 1995)。

深度信念网络 (*DBN*) 是由 受限玻尔兹曼机 (*RBM*s) (Smolen-sky, 1986) 堆叠而成的，而RBMs本身是 玻尔兹曼机 (*BM*s) (Hinton 和 Sejnowski, 1986) 的一种，具有单层特征检测单元；也请比较 高阶*BM* (*Higher-Order BM*s) (Memisevic 和 Hinton, 2010)。每个RBM感知来自下一层的模式表示，并以无监督的方式学习对其进行编码。至少在满足某些假设的理论下，增加更多层可以提高数据负对数概率的界限 (Hinton 等人, 2006) (相当于数据的描述长度——比较RNN堆栈的相应观察结果，第5.10节)。有针对 时序*RBMs* (*Temporal RBMs*) (Sutskever 等人, 2008) 的扩展。

在没有任何训练模式变形 (第5.14节) 的情况下，通过BP微调的DBN在MNIST手写数字数据集 (第5.8节，第5.14节) 上达到了1.2%的错误率 (Hinton 和 Salakhutdinov, 2006)。这一结果帮助激发了人们对DBN的兴趣。DBN在语音素识别上也取得了良好结果，在TIMIT核心测试集上的错误率为26.7% (Mohamed 和 Hinton, 2010)；请进一步比较通过前馈神经网络 (Hinton 等人, 2012a; Deng 和 Yu, 2014) 和长短期记忆循环神经网络 (第5.22节) 实现的改进。

一种基于DBN的技术，称为<样式 id='1'>语义哈希</样式>(Salakhutdinov和Hinton, 2009)，将语义相似的文档 (大小可变) 映射到文档表示空间中的附近地址。它在相似文档的搜索方面优于之前的搜索器，例如<样式 id='3'>局部敏感哈希</样式>(Buhler, 2001; Datar等人, 2004)。请参阅RBM/DBN教程 (Fischer和Igel, 2014)。

5.14 2003: More Contest-Winning/Record-Setting NNs; Successful Deep NNs

In the decade around 2000, many practical and commercial pattern recognition applications were dominated by non-neural machine learning methods such as *Support Vector Machines* (SVMs) (Vapnik, 1995; Schölkopf et al., 1998). Nevertheless, at least in certain domains, NNs outperformed other techniques.

A *Bayes NN* (Neal, 2006) based on an ensemble (Breiman, 1996; Schapire, 1990; Wolpert, 1992; Hashem and Schmeiser, 1992; Ueda, 2000; Dietterich, 2000a) of NNs won the *NIPS 2003 Feature Selection Challenge* with secret test set (Neal and Zhang, 2006). The NN was not very deep though—it had two hidden layers and thus rather shallow CAPs (Sec. 3) of depth 3.

Important for many present competition-winning pattern recognisers (Sec. 5.19, 5.21, 5.22) were developments in the CNN department. A BP-trained (LeCun et al., 1989) CNN (Sec. 5.4, Sec. 5.8) set a new MNIST record of 0.4% (Simard et al., 2003), using training pattern deformations (Baird, 1990) but no unsupervised pre-training (Sec. 5.7, 5.10, 5.15). A standard BP net achieved 0.7% (Simard et al., 2003). Again, the corresponding CAP depth was low. Compare further improvements in Sec. 5.16, 5.18, 5.19.

Good image interpretation results (Behnke, 2003b) were achieved with rather deep NNs trained by the BP variant *R-prop* (Riedmiller and Braun, 1993) (Sec. 5.6.2); here feedback through recurrent connections helped to improve image interpretation. FNNs with CAP depth up to 6 were used to successfully classify high-dimensional data (Vieira and Barradas, 2003).

Deep LSTM RNNs started to obtain certain first speech recognition results comparable to those of HMM-based systems (Graves et al., 2003); compare Sec. 5.13, 5.16, 5.21, 5.22.

5.15 2006/7: UL For Deep Belief Networks / AE Stacks Fine-Tuned by BP

While learning networks with numerous non-linear layers date back at least to 1965 (Sec. 5.3), and explicit DL research results have been published at least since 1991 (Sec. 5.9, 5.10), the expression *Deep Learning* was actually coined around 2006, when unsupervised pre-training of deep FNNs helped to accelerate subsequent SL through BP (Hinton and Salakhutdinov, 2006; Hinton et al., 2006). Compare earlier terminology on *loading deep networks* (Síma, 1994; Windisch, 2005) and *learning deep memories* (Gomez and Schmidhuber, 2005). Compare also BP-based (Sec. 5.5) fine-tuning (Sec. 5.6.1) of (not so deep) FNNs pre-trained by competitive UL (Maclin and Shavlik, 1995).

The *Deep Belief Network* (DBN) is a stack of *Restricted Boltzmann Machines* (RBMs) (Smolen-sky, 1986), which in turn are *Boltzmann Machines* (BM)s (Hinton and Sejnowski, 1986) with a single layer of feature-detecting units; compare also *Higher-Order BM*s (Memisevic and Hinton, 2010). Each RBM perceives pattern representations from the level below and learns to encode them in unsupervised fashion. At least in theory under certain assumptions, adding more layers improves a bound on the data’s negative log probability (Hinton et al., 2006) (equivalent to the data’s description length—compare the corresponding observation for RNN stacks, Sec. 5.10). There are extensions for *Temporal RBMs* (Sutskever et al., 2008).

Without any training pattern deformations (Sec. 5.14), a DBN fine-tuned by BP achieved 1.2% error rate (Hinton and Salakhutdinov, 2006) on the MNIST handwritten digits (Sec. 5.8, 5.14). This result helped to arouse interest in DBNs. DBNs also achieved good results on phoneme recognition, with an error rate of 26.7% on the TIMIT core test set (Mohamed and Hinton, 2010); compare further improvements through FNNs (Hinton et al., 2012a; Deng and Yu, 2014) and LSTM RNNs (Sec. 5.22).

A DBN-based technique called *Semantic Hashing* (Salakhutdinov and Hinton, 2009) maps semantically similar documents (of variable size) to nearby addresses in a space of document representations. It outperformed previous searchers for similar documents, such as *Locality Sensitive Hashing* (Buhler, 2001; Datar et al., 2004). See the RBM/DBN tutorial (Fischer and Igel, 2014).

自动编码器（AE）堆栈（Ballard, 1987）（第5.7节）成为在无监督方式下预训练深度前馈神经网络（FNNs）的一种流行替代方法，然后再通过反向传播（BP）（第5.5节）（Bengio等人, 2007; Vincent等人, 2008; Erhan等人, 2010）对其进行微调（第5.6.1节）。稀疏编码（第5.6.4节）被表述为一组凸优化问题的组合（Lee等人, 2007a）。最近的堆叠RBM和AE方法的综述侧重于2006年后的进展（Bengio, 2009; Arel等人, 2010）。无监督DBNs和AE堆栈在概念上类似于1991年的无监督RNN堆栈式<样式 id='1'>历史压缩器</样式>（第5.10节），后者不仅可以处理和重新编码稳态输入模式，还可以处理整个模式序列。

5.16 2006/7: 改进的卷积神经网络 /GPU卷积神经网络 /反向传播用于MPCNNs / LSTM堆栈

2006年，一个反向传播训练的（李等, 1989年）卷积神经网络（第5.4节，第5.8节）创造了新的MNIST记录0.39%（Ranzato等, 2006年），使用了训练模式变形（第5.14节），但没有无监督预训练。比较第5.18节、第5.19节的进一步改进。类似的卷积神经网络被用于越野障碍物避让（李等, 2006年）。卷积神经网络和TDNNs的组合后来学会了将可变大小句子的固定大小表示映射到与语言处理相关的特征，使用了监督学习和无监督学习（Collobert和Weston, 2008年）。

2006年也出现了早期的基于GPU的卷积神经网络实现（Chellapilla等, 2006年），比CPU卷积神经网络快4倍；比较一下早期基于GPU的标准前馈神经网络的实现，据报道速度提升因子为20（Oh和Jung, 2004年）。GPU或显卡在随后的几年中对深度学习变得越来越重要（第5.18-5.22节）。

2007年，反向传播（Sec. 5.5）首次应用于受Neocognitron启发（Sec. 5.4）、类似Cresceptron（或HMAX）的MPCNNs（Sec. 5.11），这些网络交替使用卷积层和最大池化层。反向传播训练的MPCNNs已成为许多现代、竞赛获胜、前馈、视觉深度学习器（Sec. 5.17, 5.19–5.23）的重要组成部分。

同年（2007年），Fernandez等人（2007）引入了LSTM RNNs的层级堆栈。它们可以通过层级<样式 id='1'>连接主义时间分类</样式>(CTC)（Graves等人, 2006）进行训练。对于序列标注任务，每个LSTM RNN层级（Sec. 5.13）都会预测一个序列标签，并将其输入到下一层级。每一层的误差信号都会反向传播到所有较低层级。在语音数字识别任务中，尽管LSTM堆栈对领域的假设更少，但其性能优于HMMs。LSTM堆栈并不一定需要像Sec. 5.10中提到的早期基于无监督学习的RNN堆栈（Schmidhuber, 1992b）那样进行无监督预训练。

5.17 2009: 循环神经网络首次赢得官方国际模式识别比赛，并与MPCNNs一同参赛

由CTC（第5.13、5.16节）训练的LSTM循环神经网络堆栈成为首批赢得官方国际模式识别比赛的循环神经网络（测试集仅组织者知晓）。更具体地说，在ICDAR 2009上以三种不同语言（法语、阿拉伯语、波斯语）举办的三个连接手写比赛均由深度LSTM循环神经网络赢得，且无需任何先验语言知识，实现了同步分割和识别。比较（Graves和Schmidhuber, 2005; Graves等人, 2009; Schmidhuber等人, 2011; Graves等人, 2013; Graves和Jaitly, 2014）（第5.22节）。

为在监控视频中检测人类动作，一个三维CNN（例如Jain和Seung, 2009; Prokhorov, 2010）与SVMs结合，成为了一个更大系统（Yang等人, 2009）的一部分，该系统采用特征包方法（Nowak等人, 2006）提取感兴趣区域。该系统赢得了2009年三个TRECVID比赛。这些可能是首批借助（MP）CNNs（第5.16节）赢得的官方国际比赛。该方法的改进版本后来发表（Ji等人, 2013）。

2009年还出现了GPU-DBN实现（Raina等人, 2009），其速度比之前的CPU-DBN快了几个数量级（参见第5.15节）；另见（Coates等人, 2013）。卷积*DBN*（Lee等人, 2009a）（采用MP的概率变体，第5.11节）结合了CNNs和DBNs的思想，并成功应用于音频分类（Lee等人, 2009b）。

Autoencoder (AE) stacks (Ballard, 1987) (Sec. 5.7) became a popular alternative way of pre-training deep FNNs in unsupervised fashion, before fine-tuning (Sec. 5.6.1) them through BP (Sec. 5.5) (Bengio et al., 2007; Vincent et al., 2008; Erhan et al., 2010). Sparse coding (Sec. 5.6.4) was formulated as a combination of convex optimization problems (Lee et al., 2007a). Recent surveys of stacked RBM and AE methods focus on post-2006 developments (Bengio, 2009; Arel et al., 2010). Unsupervised DBNs and AE stacks are conceptually similar to, but in a certain sense less general than, the unsupervised RNN stack-based *History Compressor* of 1991 (Sec. 5.10), which can process and re-encode not only stationary input patterns, but entire pattern sequences.

5.16 2006/7: Improved CNNs / GPU-CNNs / BP for MPCNNs / LSTM Stacks

Also in 2006, a BP-trained (LeCun et al., 1989) CNN (Sec. 5.4, Sec. 5.8) set a new MNIST record of 0.39% (Ranzato et al., 2006), using training pattern deformations (Sec. 5.14) but no unsupervised pre-training. Compare further improvements in Sec. 5.18, 5.19. Similar CNNs were used for off-road obstacle avoidance (LeCun et al., 2006). A combination of CNNs and TDNNs later learned to map fixed-size representations of variable-size sentences to features relevant for language processing, using a combination of SL and UL (Collobert and Weston, 2008).

2006 also saw an early GPU-based CNN implementation (Chellapilla et al., 2006) up to 4 times faster than CPU-CNNs; compare also earlier GPU implementations of standard FNNs with a reported speed-up factor of 20 (Oh and Jung, 2004). GPUs or graphics cards have become more and more important for DL in subsequent years (Sec. 5.18–5.22).

In 2007, BP (Sec. 5.5) was applied for the first time (Ranzato et al., 2007) to Neocognitron-inspired (Sec. 5.4), Cresceptron-like (or HMAX-like) MPCNNs (Sec. 5.11) with alternating convolutional and max-pooling layers. BP-trained MPCNNs have become an essential ingredient of many modern, competition-winning, feedforward, visual Deep Learners (Sec. 5.17, 5.19–5.23).

Also in 2007, hierarchical stacks of LSTM RNNs were introduced (Fernandez et al., 2007). They can be trained by hierarchical *Connectionist Temporal Classification* (CTC) (Graves et al., 2006). For tasks of sequence labelling, every LSTM RNN level (Sec. 5.13) predicts a sequence of labels fed to the next level. Error signals at every level are back-propagated through all the lower levels. On spoken digit recognition, LSTM stacks outperformed HMMs, despite making fewer assumptions about the domain. LSTM stacks do not necessarily require unsupervised pre-training like the earlier UL-based RNN stacks (Schmidhuber, 1992b) of Sec. 5.10.

5.17 2009: First Official Competitions Won by RNNs, and with MPCNNs

Stacks of LSTM RNNs trained by CTC (Sec. 5.13, 5.16) became the first RNNs to win official international pattern recognition contests (with secret test sets known only to the organisers). More precisely, three connected handwriting competitions at ICDAR 2009 in three different languages (French, Arab, Farsi) were won by deep LSTM RNNs without any *a priori* linguistic knowledge, performing simultaneous segmentation and recognition. Compare (Graves and Schmidhuber, 2005; Graves et al., 2009; Schmidhuber et al., 2011; Graves et al., 2013; Graves and Jaitly, 2014) (Sec. 5.22).

To detect human actions in surveillance videos, a 3-dimensional CNN (e.g., Jain and Seung, 2009; Prokhorov, 2010), combined with SVMs, was part of a larger system (Yang et al., 2009) using a *bag of features* approach (Nowak et al., 2006) to extract regions of interest. The system won three 2009 TRECVID competitions. These were possibly the first official international contests won with the help of (MP)CNNs (Sec. 5.16). An improved version of the method was published later (Ji et al., 2013).

2009 also saw a GPU-DBN implementation (Raina et al., 2009) orders of magnitudes faster than previous CPU-DBNs (see Sec. 5.15); see also (Coates et al., 2013). The *Convolutional DBN* (Lee

et al., 2009a) (采用MP的概率变体，第5.11节) 结合了卷积神经网络和DBNs的思想，并成功应用于音频分类 (Lee等人，2009b)。

5.18 2010：普通反向传播（+扭曲）在GPU上打破MNIST记录

2010年，通过传统的反向传播（BP）（第5.5节）在深度但其他方面标准的神经网络（Ciresan等人，2010年）上，设定了新的MNIST（第5.8节）记录，错误率为0.35%。该方法既未使用无监督预训练（例如，第5.7、5.10、5.15节），也未使用卷积（例如，第5.4、5.8、5.14、5.16节）。然而，训练模式变形（例如，第5.14节）对于生成大型训练集和避免过拟合非常重要。这一成功主要得益于GPU实现的反向传播，其速度比标准CPU版本快50倍。在不使用扭曲的情况下（除了小的类似眼球快速移动的平移之外），获得了0.95%的良好值——比较第5.15节。

由于反向传播当时已有3-5十年的历史（第5.5节），而模式变形已有2十年（Baird, 1990）（第5.14节），这些结果似乎表明，利用现代计算硬件的进步比算法的进步更为重要。

5.19 2011：MPCNNs在GPU上实现超人类视觉性能

2011年，一种灵活的<样式 id='1'>GPU实现</样式>(Ciresan等人，2011a)的<样式 id='3'>最大池化(MP)卷积神经网络或卷积网络</样式>被描述（一个GPU-MPCNN），它基于早期的最大池化工作（Weng等人，1992）（第5.11节）卷积神经网络（Fukushima, 1979; LeCun等人，1989）（第5.4节，5.8节，5.16节），以及早期的基于GPU的卷积神经网络<样式 id='5'>不使用</样式>最大池化（Chellapilla等人，2006）（第5.16节）；比较早期的GPU神经网络（Oh和Jung，2004）和GPU-DBNs（Raina等人，2009）（第5.17节）。MPCNNs交替使用卷积层（第5.4节）和最大池化层（MP，第5.11节），顶部是标准的全连接层。所有权重都通过反向传播（第5.5节，5.8节，5.16节）（Ranzato等人，2007; Scherer等人，2010）进行训练。GPU-MPCNNs已成为许多竞赛获胜的前馈神经网络（第5.21节，第5.22节）的关键。

多列GPU-MPCNNs (Ciresan 等人，2011b) 是由 GPU-MPCNNs 组成的委员会（Breiman, 1996; Schapire, 1990; Wolpert, 1992; Hashem 和 Schmeiser, 1992; Ueda, 2000; Dietterich, 2000a），采用简单的民主输出平均法。多个 MPCNN 会看到相同的输入；它们的输出向量用于为各种可能的类别分配概率。平均概率最高的类别被选为系统对当前输入的分类。与更复杂的集成方法（Schapire, 1990）相比，包括第 5.14 节的获胜集成方法 Bayes-NN (Neal, 2006) 以及最近的相关研究（Shao 等人，2014）。

GPU-MPCNNs 集合是首个在受控竞赛中实现超人类视觉模式识别（Ciresan 等人，2011b, 2012b）的系统，即圣何塞（CA）的 IJCNN 2011 交通标志识别竞赛（Stallkamp 等人，2011, 2012）。这对于完全自主的自动驾驶汽车在交通中的应用（例如，Dickmanns 等人，1994）具有重要意义。GPU-MPCNNs 集合的错误率为 0.56%，表现优于人类测试对象两倍，优于最接近的人工神经网络竞争对手（Sermanet 和 LeCun，2011）三倍，优于最佳非神经网络方法六倍。

几个月前，预选赛是在第一阶段的在线比赛中获胜的，尽管优势要小得多：1.02%（Ciresan 等人，2011b）对比第二名的1.03%（Sermanet 和 LeCun，2011）。截止日期后，组织者透露人类在测试集上的表现是1.19%。也就是说，最好的方法已经看似具有人类竞争力。然而，在预选赛中，通过反复提交可以逐步获取关于测试集的信息。这一点通过不同团队随时间取得的越来越好（Stallkamp 等人，2012）（组织者最终设置了10次重新提交的限制）。在最终比赛中这是不可能的。

et al., 2009a) (with a probabilistic variant of MP, Sec. 5.11) combines ideas from CNNs and DBNs, and was successfully applied to audio classification (Lee et al., 2009b).

5.18 2010: Plain Backprop (+ Distortions) on GPU Breaks MNIST Record

In 2010, a new MNIST (Sec. 5.8) record of 0.35% error rate was set by good old BP (Sec. 5.5) in deep but otherwise standard NNs (Ciresan et al., 2010), using neither unsupervised pre-training (e.g., Sec. 5.7, 5.10, 5.15) nor convolution (e.g., Sec. 5.4, 5.8, 5.14, 5.16). However, training pattern deformations (e.g., Sec. 5.14) were important to generate a big training set and avoid overfitting. This success was made possible mainly through a GPU implementation of BP that was up to 50 times faster than standard CPU versions. A good value of 0.95% was obtained without distortions except for small saccadic eye movement-like translations—compare Sec. 5.15.

Since BP was 3-5 decades old by then (Sec. 5.5), and pattern deformations 2 decades (Baird, 1990) (Sec. 5.14), these results seemed to suggest that advances in exploiting modern computing hardware were more important than advances in algorithms.

5.19 2011: MPCNNs on GPU Achieve Superhuman Vision Performance

In 2011, a flexible *GPU-implementation* (Ciresan et al., 2011a) of *Max-Pooling (MP) CNNs or Convnets* was described (a GPU-MPCNN), building on earlier MP work (Weng et al., 1992) (Sec. 5.11) CNNs (Fukushima, 1979; LeCun et al., 1989) (Sec. 5.4, 5.8, 5.16), and on early GPU-based CNNs *without* MP (Chellapilla et al., 2006) (Sec. 5.16); compare early GPU-NNs (Oh and Jung, 2004) and GPU-DBNs (Raina et al., 2009) (Sec. 5.17). MPCNNs have alternating convolutional layers (Sec. 5.4) and max-pooling layers (MP, Sec. 5.11) topped by standard fully connected layers. All weights are trained by BP (Sec. 5.5, 5.8, 5.16) (Ranzato et al., 2007; Scherer et al., 2010). GPU-MPCNNs have become essential for many contest-winning FNNs (Sec. 5.21, Sec. 5.22).

Multi-Column GPU-MPCNNs (Ciresan et al., 2011b) are committees (Breiman, 1996; Schapire, 1990; Wolpert, 1992; Hashem and Schmeiser, 1992; Ueda, 2000; Dietterich, 2000a) of GPU-MPCNNs with simple democratic output averaging. Several MPCNNs see the same input; their output vectors are used to assign probabilities to the various possible classes. The class with the on average highest probability is chosen as the system’s classification of the present input. Compare earlier, more sophisticated ensemble methods (Schapire, 1990), the contest-winning ensemble Bayes-NN (Neal, 2006) of Sec. 5.14, and recent related work (Shao et al., 2014).

An ensemble of GPU-MPCNNs was the first system to achieve superhuman visual pattern recognition (Ciresan et al., 2011b, 2012b) in a controlled competition, namely, the IJCNN 2011 traffic sign recognition contest in San Jose (CA) (Stallkamp et al., 2011, 2012). This is of interest for fully autonomous, self-driving cars in traffic (e.g., Dickmanns et al., 1994). The GPU-MPCNN ensemble obtained 0.56% error rate and was twice better than human test subjects, three times better than the closest artificial NN competitor (Sermanet and LeCun, 2011), and six times better than the best non-neural method.

A few months earlier, the qualifying round was won in a 1st stage online competition, albeit by a much smaller margin: 1.02% (Ciresan et al., 2011b) vs 1.03% for second place (Sermanet and LeCun, 2011). After the deadline, the organisers revealed that human performance on the test set was 1.19%. That is, the best methods already seemed human-competitive. However, during the qualifying it was possible to incrementally gain information about the test set by probing it through repeated submissions. This is illustrated by better and better results obtained by various teams over time (Stallkamp et al., 2012) (the organisers eventually imposed a limit of 10 resubmissions). In the final competition this was not possible.

这说明了公开测试集的基准测试存在的一个普遍问题，或者至少可以在一定程度上探测到：竞争团队倾向于在测试集上过拟合，即使它不能直接用于训练，只能用于评估。

1997年，许多人认为IBM计算机击败国际象棋世界冠军卡斯帕罗夫是一件大事。但当时，计算机完全无法与小孩子竞争视觉模式识别能力，而从计算角度来看，视觉模式识别似乎比国际象棋要难得多。当然，交通标志领域非常受限，而孩子们仍然是更优秀的通用模式识别者。然而，到2011年，深度神经网络已经能够学会在重要的有限视觉领域与它们匹敌。

GPU-MPCNN的集成也是第一个在MNIST（Ciresan等人，2012c）上达到人类竞争水平性能（约0.2%）的方法。这代表了一次巨大的改进，因为当时MNIST记录已经徘徊在0.4%左右近十年（第5.14节、5.16节、5.18节）。鉴于之前在（MP）卷积神经网络（第5.4节、5.8节、5.11节、5.16节）和GPU卷积神经网络（第5.16节）方面的大量工作，GPU-MPCNN在科学意义上并非突破。但它们在高效编码方面具有商业相关性突破，自2011年以来已在多个竞赛中产生了影响。如今，大多数前馈竞赛获胜的深度神经网络都是（GPU-MPCNN的）集成（第5.21节至5.23节）。

5.20 2011：适用于循环神经网络的Hessian-Free优化

2011年还表明（Martens和Sutskever，2011年），Hessian-Free优化（例如Møller，1993年；Pearlmutter，1994年；Schraudolph，2002年）（第5.6.2节）可以缓解循环神经网络中的<样式 id='1'>基本深度学习问题</样式>（第5.9节），在多项任务上优于标准基于梯度的长短期记忆循环神经网络（第5.13节）。比较其他循环神经网络算法（Jaeger，2004年；Schmidhuber等人，2007年；Pascanu等人，2013b年；Koutník等人，2014年），这些算法至少有时在长短期记忆循环神经网络上的结果优于最速下降法。

5.21 2012：在ImageNet、目标检测和分割上首次赢得竞赛

在2012年，一个GPU-MPCNNs（第5.19节）的集成在流行的计算机视觉社区中的ImageNet 图像分类 基准测试（Krizhevsky等人，2012年）上取得了最佳结果。在这里，相对较大的图像尺寸256x256像素是必要的，而2011年的交通标志竞赛（第5.19节）只需要48x48像素。参见第5.22节的进一步改进。

也在2012年，目前最大的NN（ 10^9 自由参数）在无监督模式下（Sec. 5.7, 5.15）在无标签数据（Le et al., 2012）上训练，然后应用于ImageNet。它顶层的代码被用来训练一个简单的监督分类器，该分类器在20,000个类别上取得了迄今为止最佳结果。不是依赖高效的GPU编程，而是通过在1,000台标准机器的16,000个核心上进行暴力计算来完成。

因此，到2011/2012年，深度学习器在图像识别与分类(第5.19节、5.21节)方面已取得优异成绩。然而，计算机视觉领域特别关注大型图像中的目标检测，例如应用于基于图像的搜索引擎，或在生物医学诊断中自动检测人体组织图像中的肿瘤等。目标检测带来了额外的挑战。一种自然的方法是在大图像的图像块上训练深度神经网络分类器，然后将其作为特征检测器，在未知视觉场景中平移，并使用不同的旋转和缩放因子。产生高度活跃输出单元的图像部分很可能包含与神经网络训练时相似的物体。

2012年终于出现了首个深度学习系统（一个GPU-MPCNNs的集成，第5.19节），在数百万像素的大型图像中赢得了视觉<样式id='1'>目标检测</样式>（Ciresan等人，2013年）的比赛（ICPR 2012乳腺癌组织学图像中细胞分裂检测竞赛，2012年；Roux等人，2013年）。这类生物医学应用可能成为深度学习最重要的应用之一。世界

This illustrates a general problem with benchmarks whose test sets are public, or at least can be probed to some extent: competing teams tend to overfit on the test set even when it cannot be directly used for training, only for evaluation.

In 1997 many thought it a big deal that human chess world champion Kasparov was beaten by an IBM computer. But back then computers could not at all compete with little kids in visual pattern recognition, which seems much harder than chess from a computational perspective. Of course, the traffic sign domain is highly restricted, and kids are still much better general pattern recognisers. Nevertheless, by 2011, deep NNs could already learn to rival them in important limited visual domains.

An ensemble of GPU-MPCNNs was also the first method to achieve human-competitive performance (around 0.2%) on MNIST (Ciresan et al., 2012c). This represented a dramatic improvement, since by then the MNIST record had hovered around 0.4% for almost a decade (Sec. 5.14, 5.16, 5.18).

Given all the prior work on (MP)CNNs (Sec. 5.4, 5.8, 5.11, 5.16) and GPU-CNNs (Sec. 5.16), GPU-MPCNNs are not a breakthrough in the scientific sense. But they are a commercially relevant breakthrough in efficient coding that has made a difference in several contests since 2011. Today, most *feedforward* competition-winning deep NNs are (ensembles of) GPU-MPCNNs (Sec. 5.21–5.23).

5.20 2011: Hessian-Free Optimization for RNNs

Also in 2011 it was shown (Martens and Sutskever, 2011) that Hessian-free optimization (e.g., Møller, 1993; Pearlmutter, 1994; Schraudolph, 2002) (Sec. 5.6.2) can alleviate the *Fundamental Deep Learning Problem* (Sec. 5.9) in RNNs, outperforming standard gradient-based LSTM RNNs (Sec. 5.13) on several tasks. Compare other RNN algorithms (Jaeger, 2004; Schmidhuber et al., 2007; Pascanu et al., 2013b; Koutník et al., 2014) that also at least sometimes yield better results than steepest descent for LSTM RNNs.

5.21 2012: First Contests Won on ImageNet, Object Detection, Segmentation

In 2012, an ensemble of GPU-MPCNNs (Sec. 5.19) achieved best results on the *ImageNet* classification benchmark (Krizhevsky et al., 2012), which is popular in the computer vision community. Here relatively large image sizes of 256x256 pixels were necessary, as opposed to only 48x48 pixels for the 2011 traffic sign competition (Sec. 5.19). See further improvements in Sec. 5.22.

Also in 2012, the biggest NN so far (10^9 free parameters) was trained in unsupervised mode (Sec. 5.7, 5.15) on unlabeled data (Le et al., 2012), then applied to ImageNet. The codes across its top layer were used to train a simple supervised classifier, which achieved best results so far on 20,000 classes. Instead of relying on efficient GPU programming, this was done by brute force on 1,000 standard machines with 16,000 cores.

So by 2011/2012, excellent results had been achieved by Deep Learners in image *recognition and classification* (Sec. 5.19, 5.21). The computer vision community, however, is especially interested in *object detection* in large images, for applications such as image-based search engines, or for biomedical diagnosis where the goal may be to automatically detect tumors etc in images of human tissue. Object detection presents additional challenges. One natural approach is to train a deep NN classifier on patches of big images, then use it as a feature detector to be shifted across unknown visual scenes, using various rotations and zoom factors. Image parts that yield highly active output units are likely to contain objects similar to those the NN was trained on.

2012 finally saw the first DL system (an ensemble of GPU-MPCNNs, Sec. 5.19) to win a contest on visual *object detection* (Ciresan et al., 2013) in large images of several million pixels (ICPR 2012 Contest on Mitosis Detection in Breast Cancer Histological Images, 2012; Roux et al., 2013). Such biomedical applications may turn out to be among the most important applications of DL. The world

在医疗保健上花费了GDP的10%以上（每年> 6 万亿美元），其中大部分用于通过昂贵的专家进行医学诊断。这种部分自动化不仅能够节省大量资金，还能让许多目前无法负担得起专家诊断的人获得服务。令人欣慰的是，今天深度神经网络实际上可能有助于改善医疗保健，甚至挽救人类生命。

2012年还见证了深度学习（DL）在首个纯图像分割（image segmentation）竞赛中获胜（Ciresan 等人，2012a），再次通过 GPU-MPCNN 集成（神经元结构在电子显微镜堆栈中的分割挑战，2012）。² 电子显微镜堆栈与欧洲和美国最近批准的巨大脑项目相关（例如，Markram，2012）。给定动物脑薄片堆栈的电子显微镜图像，目标是构建大脑神经元和树突的详细 3D 模型。但人类专家需要许多小时、天和周来标注图像：哪些部分描绘了神经元膜？哪些部分是不相关的背景？这需要自动化（例如，Turaga 等人，2010）。深度多列 GPU-MPCNN 通过大量训练图像的经验学习解决了这项任务，并在所有三个评估指标上以巨大优势获胜，在像素误差方面表现出超人类性能。

目标检测（Ciresan 等人，2013）和图像分割（Ciresan 等人，2012a）都受益于快速基于 MPCNN 的图像扫描，避免了冗余计算。最近的 MPCNN 扫描器将朴素实现的速度提高了三个数量级（Masci 等人，2013；Giusti 等人，2013）；比较早期针对 CNNs 不使用 MP 的高效方法（Vaillant 等人，1994）。

同样在2012年，一个由深度前馈神经网络和2D-BRNNs（Di Lena 等人，2012）组成的系统赢得了 CASB 2012 蛋白质接触图预测竞赛。在 IAM-OnDoDB 基准测试中，长短期记忆循环神经网络（第 5.13 节）在在线模式检测（Otte 等人，2012；Indermuhle 等人，2012）和关键词识别（Indermuhle 等人，2011）方面优于所有其他方法（隐马尔可夫模型，SVMs）。在语言建模的长时间滞后问题中，长短期记忆循环神经网络在 IAM-DB 基准测试中优于所有统计方法（Frinken 等人，2012）；后来通过神经网络和隐马尔可夫模型的组合（Zamora-Martinez 等人，2014）获得了更好的结果。比较早期通过迭代图像解释进行目标识别的循环神经网络（Behnke 和 Rojas，1998；Behnke，2002，2003b）；另见更多近期发表的研究（Wyatte 等人，2012；O'Reilly 等人，2013），这些研究扩展了关于循环神经网络生物学上合理的学习规则的工作（O’ Reilly，1996）。

5.22 2013-: 更多竞赛和基准记录

一个由双向长短期记忆循环神经网络（Fernandez 等人，2007；Graves 和 Schmidhuber，2009）（第 5.10 节）堆叠而成的模型（Graves 和 Schmidhuber，2005），通过 CTC（第 5.13 节、5.17 节）训练，打破了著名的 TIMIT 语音（音素）识别记录，在测试集上实现了 17.7% 的错误率（Graves 等人，2013），尽管此前已有数千人年投入到基于<样式 id='1'>隐马尔可夫模型 </样式>（HMMs）的语音识别研究中。与早期的 DBN 结果（第 5.15 节）进行比较。

CTC-LSTM 还有助于在 NIST 的 OpenHaRT2013 评估中获得第一名（Bluche 等人，2014）。对于<样式 id='1'>光学字符识别</样式>（OCR），长短期记忆循环神经网络优于历史数据中的商业识别器（Breuel 等人，2013）。基于长短期记忆的系统还在<样式 id='3'>语言识别 </样式>（Gonzalez-Dominguez 等人，2014）、中等词汇量的<样式 id='5'>语音识别 </样式>（Geiger 等人，2014）、<样式 id='7'>韵律轮廓预测</样式>（Fernandez 等人，2014）、<样式 id='9'>音频起始检测 </样式>（Marchi 等人，2014）、<样式 id='11'>文本到语音合成 </样式>（Fan 等人，2014），以及<样式 id='13'>社交信号分类 </样式>（Brueckner 和 Schulter，2014）等方面创下了基准记录。

AnLSTM-RNN 被用于估计 HMM 的状态后验；该系统在大词汇量语音识别（Sak 等人，2014b,a）方面超越了之前的技术水平。另一个具有数亿连接的 LSTM RNN 被用于重新排序统计机器翻译的假设

²不过需要指出的是，长短期记忆循环神经网络在成为首个赢得官方国际模式识别比赛的循环深度学习模型时，就已经实现了同时分割和识别——参见第 5.17 节。

spends over 10% of GDP on healthcare (> 6 trillion USD per year), much of it on medical diagnosis through expensive experts. Partial automation of this could not only save lots of money, but also make expert diagnostics accessible to many who currently cannot afford it. It is gratifying to observe that today deep NNs may actually help to improve healthcare and perhaps save human lives.

2012 also saw the first pure *image segmentation* contest won by DL (Ciresan et al., 2012a), again through an GPU-MPCNN ensemble (Segmentation of Neuronal Structures in EM Stacks Challenge, 2012).² EM stacks are relevant for the recently approved huge brain projects in Europe and the US (e.g., Markram, 2012). Given electron microscopy images of stacks of thin slices of animal brains, the goal is to build a detailed 3D model of the brain’s neurons and dendrites. But human experts need many hours and days and weeks to annotate the images: Which parts depict neuronal membranes? Which parts are irrelevant background? This needs to be automated (e.g., Turaga et al., 2010). Deep Multi-Column GPU-MPCNNs learned to solve this task through experience with many training images, and won the contest on all three evaluation metrics by a large margin, with superhuman performance in terms of pixel error.

Both object detection (Ciresan et al., 2013) and image segmentation (Ciresan et al., 2012a) profit from fast MPCNN-based image scans that avoid redundant computations. Recent MPCNN scanners speed up naive implementations by up to three orders of magnitude (Masci et al., 2013; Giusti et al., 2013); compare earlier efficient methods for CNNs *without* MP (Vaillant et al., 1994).

Also in 2012, a system consisting of growing deep FNNs and 2D-BRNNs (Di Lena et al., 2012) won the CASP 2012 contest on protein contact map prediction. On the IAM-OnDoDB benchmark, LSTM RNNs (Sec. 5.13) outperformed all other methods (HMMs, SVMs) on online mode detection (Otte et al., 2012; Indermuhle et al., 2012) and keyword spotting (Indermuhle et al., 2011). On the long time lag problem of language modelling, LSTM RNNs outperformed all statistical approaches on the IAM-DB benchmark (Frinken et al., 2012); improved results were later obtained through a combination of NNs and HMMs (Zamora-Martinez et al., 2014). Compare earlier RNNs for object recognition through iterative image interpretation (Behnke and Rojas, 1998; Behnke, 2002, 2003b); see also more recent publications (Wyatte et al., 2012; O'Reilly et al., 2013) extending work on biologically plausible learning rules for RNNs (O’Reilly, 1996).

5.22 2013-: More Contests and Benchmark Records

A stack (Fernandez et al., 2007; Graves and Schmidhuber, 2009) (Sec. 5.10) of bi-directional LSTM RNNs (Graves and Schmidhuber, 2005) trained by CTC (Sec. 5.13, 5.17) broke a famous TIMIT speech (phoneme) recognition record, achieving 17.7% test set error rate (Graves et al., 2013), despite thousands of man years previously spent on *Hidden Markov Model* (HMMs)-based speech recognition research. Compare earlier DBN results (Sec. 5.15).

CTC-LSTM also helped to score first at NIST’s OpenHaRT2013 evaluation (Bluche et al., 2014). For *optical character recognition* (OCR), LSTM RNNs outperformed commercial recognizers of historical data (Breuel et al., 2013). LSTM-based systems also set benchmark records in *language identification* (Gonzalez-Dominguez et al., 2014), medium-vocabulary *speech recognition* (Geiger et al., 2014), *prosody contour prediction* (Fernandez et al., 2014), *audio onset detection* (Marchi et al., 2014), *text-to-speech synthesis* (Fan et al., 2014), and *social signal classification* (Brueckner and Schulter, 2014).

An LSTM RNN was used to estimate the state posteriors of an HMM; this system beat the previous state of the art in *large vocabulary speech recognition* (Sak et al., 2014b,a). Another LSTM RNN with hundreds of millions of connections was used to rerank hypotheses of a statistical machine translation

²It should be mentioned, however, that LSTM RNNs already performed simultaneous segmentation and recognition when they became the first recurrent Deep Learners to win official international pattern recognition contests—see Sec. 5.17.

系统；该系统在 英语到法语翻译 (Sutskever 等人, 2014) 方面超越了之前的技术水平。

在 *ICDAR* 中文手写识别基准测试 (超过3700个类别)上, 由GPU-MPCNN集成(第5.19节)在一台台式机上创下了新纪录, 其性能几乎与人类相当(Ciresan和Schmidhuber,2013); 参见(Yin等人, 2013)。

在 *MICCAI 2013* 胞质检测挑战赛 (Veta等人, 2013)中, 由目标检测GPU-MPCNN集成(Ciresan等人,2013)同样获胜。其数据集比ICPR 2012(第5.21节)的更大、更具挑战性: 一个包含许多模糊案例和常见问题的真实世界数据集, 例如不完美的切片染色问题。

三种2D-CNN(使用 均值池化 代替MP, 第5.11节)观察3D图像的三个正交投影, 在低场膝关节MRI扫描中分割胫骨软骨的任务上, 其性能优于传统的全3D方法(Prasoon等人, 2013)。

深度 GPU-MPCNNs (第 5.19 节) 也帮助在计算机视觉领域的重要基准测试上取得了新的最佳结果: ImageNet 分类 (Zeiler 和 Fergus, 2013; Szegedy 等人, 2014) 以及在传统方法结合下 PASCAL 目标检测 (Girshick 等人, 2013) 。它们还学会了预测 ImageNet 2013 数据库中物体的边界框坐标, 并在定位和检测任务上获得了最先进的结果 (Sermanet 等人, 2013) 。GPU-MPCNNs 还帮助在 Google Street View 图像中识别多位数 (Goodfellow 等人, 2014b) , 其中部分神经网络被训练用于计数可见数字; 对比早期通过 DBNs 检测 “数量” 的工作 (Stoianov 和 Zorzi, 2012) 。该系统在识别*reCAPTCHA* 谜题中扭曲的合成文本方面也表现出色。其他成功的 CNN 应用包括场景解析 (Farabet 等人, 2013) 、目标检测 (Szegedy 等人, 2013) 、阴影检测 (Khan 等人, 2014) 、视频分类 (Karpathy 等人, 2014) 以及阿尔茨海默病神经影像 (Li 等人, 2014) 。

瑞士AILab IDSIA、多伦多大学、纽约大学和蒙特利尔大学的网页中还提到了其他竞赛。

5.23 当前成功的技术：LSTMRNNs和GPU-MPCNNs

大多数赢得竞赛或创下基准记录的深度学习模型实际上使用其中两种 监督 技术: (a) 由CTC (2006) 训练的循环LSTM (1997) (第5.13、5.17、5.21、5.22节), 或 (b) 基于CNN (1979, 第5.4节) 的前馈GPU-MPCNNs (2011, 第5.19、5.21、5.22节), 其中MP (1992, 第5.11节) 通过BP (1989–2007, 第5.8、5.16节) 进行训练。

例外包括两个2011年的竞赛 (Goodfellow等人, 2011; Mesnil等人, 2011; Goodfellow等人, 2012), 这些竞赛专门针对从一个数据集到另一个数据集的 迁移学习 (例如, Caruana, 1997; Schmidhuber, 2004; Pan和Yang, 2010)。然而, 深度GPU-MPCNNs确实允许进行 纯*SL-based*迁移 (Ciresan等人, 2012d), 其中在某个训练集上的预训练可以显著提高在差异很大的其他数据集上的性能, 这在更近期的研究中也得到了验证 (Oquab等人, 2013; Donahue等人, 2013)。事实上, 通过SL预训练的深度MPCNNs可以从相当多样化的训练集外图像中提取有用特征, 在许多视觉任务上 (Razavian等人, 2014) 比传统的、广泛使用的特征 (如SIFT (Lowe, 1999, 2004)) 效果更好。为了应对数据集的变化, 缓慢学习的深度神经网络也被与快速适应的 “表面 ” 神经网络 (Kak 等人 , 2010) 相结合。

值得注意的是, 在20世纪90年代, 趋势从部分无监督的RNN堆栈 (第5.10节) 转向纯粹的监督长短期记忆循环神经网络 (第5.13节) , 就像在2000年代从部分无监督的前馈神经网络堆栈 (第5.15节) 转向纯粹的监督MPCNNs (第5.16–5.22节) 。然而, 在许多应用中, 结合两者的最佳之处——监督学习和无监督预训练 (第5.10节, 5.15节) 仍然可能有利。

system; this system beat the previous state of the art in *English to French translation* (Sutskever et al., 2014).

A new record on the *ICDAR Chinese handwriting recognition benchmark* (over 3700 classes) was set on a desktop machine by an ensemble of GPU-MPCNNs (Sec. 5.19) with almost human performance (Ciresan and Schmidhuber, 2013); compare (Yin et al., 2013).

The *MICCAI 2013 Grand Challenge on Mitosis Detection* (Veta et al., 2013) also was won by an object-detecting GPU-MPCNN ensemble (Ciresan et al., 2013). Its data set was even larger and more challenging than the one of ICPR 2012 (Sec. 5.21): a real-world dataset including many ambiguous cases and frequently encountered problems such as imperfect slide staining.

Three 2D-CNNs (with *mean-pooling* instead of MP, Sec. 5.11) observing three orthogonal projections of 3D images outperformed traditional full 3D methods on the task of segmenting tibial cartilage in low field knee MRI scans (Prasoon et al., 2013).

Deep GPU-MPCNNs (Sec. 5.19) also helped to achieve new best results on important benchmarks of the computer vision community: ImageNet classification (Zeiler and Fergus, 2013; Szegedy et al., 2014) and—in conjunction with traditional approaches—PASCAL object detection (Girshick et al., 2013). They also learned to predict bounding box coordinates of objects in the Imagenet 2013 database, and obtained state-of-the-art results on tasks of localization and detection (Sermanet et al., 2013). GPU-MPCNNs also helped to recognise multi-digit numbers in Google Street View images (Goodfellow et al., 2014b), where part of the NN was trained to *count* visible digits; compare earlier work on detecting “numerosity” through DBNs (Stoianov and Zorzi, 2012). This system also excelled at recognising distorted synthetic text in *reCAPTCHA* puzzles. Other successful CNN applications include scene parsing (Farabet et al., 2013), object detection (Szegedy et al., 2013), shadow detection (Khan et al., 2014), video classification (Karpathy et al., 2014), and Alzheimers disease neuroimaging (Li et al., 2014).

Additional contests are mentioned in the web pages of the Swiss AI Lab IDSIA, the University of Toronto, NY University, and the University of Montreal.

5.23 Currently Successful Techniques: LSTM RNNs and GPU-MPCNNs

Most competition-winning or benchmark record-setting Deep Learners actually use one of two *supervised* techniques: (a) recurrent LSTM (1997) trained by CTC (2006) (Sec. 5.13, 5.17, 5.21, 5.22), or (b) feedforward GPU-MPCNNs (2011, Sec. 5.19, 5.21, 5.22) based on CNNs (1979, Sec. 5.4) with MP (1992, Sec. 5.11) trained through BP (1989–2007, Sec. 5.8, 5.16).

Exceptions include two 2011 contests (Goodfellow et al., 2011; Mesnil et al., 2011; Goodfellow et al., 2012) specialised on *Transfer Learning* from one dataset to another (e.g., Caruana, 1997; Schmidhuber, 2004; Pan and Yang, 2010). However, deep GPU-MPCNNs do allow for *pure SL-based transfer* (Ciresan et al., 2012d), where pre-training on one training set greatly improves performance on quite different sets, also in more recent studies (Oquab et al., 2013; Donahue et al., 2013). In fact, deep MPCNNs pre-trained by SL can extract useful features from quite diverse off-training-set images, yielding better results than traditional, widely used features such as SIFT (Lowe, 1999, 2004) on many vision tasks (Razavian et al., 2014). To deal with changing datasets, slowly learning deep NNs were also combined with rapidly adapting “*surface*” NNs (Kak et al., 2010).

Remarkably, in the 1990s a trend went from partially unsupervised RNN stacks (Sec. 5.10) to *purely* supervised LSTM RNNs (Sec. 5.13), just like in the 2000s a trend went from partially unsupervised FNN stacks (Sec. 5.15) to *purely* supervised MPCNNs (Sec. 5.16–5.22). Nevertheless, in many applications it can still be advantageous to combine the best of both worlds—*supervised* learning and *unsupervised* pre-training (Sec. 5.10, 5.15).

5.24 近期技巧：提升SL深度神经网络（比较第5.6.2节、5.6.3节）

DBN训练（第5.15节）可以通过梯度增强和在随机梯度下降期间自动学习率调整（Cho等人，2013年；Cho，2014年）来改进， 还可以通过对RBMs（Cho等人，2012年）进行Tikhonov型（Tikhonov等人，1977年）正则化来改进。收缩自编码器（Rifai等人，2011年）会阻止隐藏单元在输入扰动时发生扰动，类似于LOCOCODE自编码器（第5.6.4节）的FMS（第5.6.3节）在权重扰动时阻止输出扰动的方式。

层次化卷积神经网络（例如，Behnke，2003b，2005）在一个<样式id='1'>神经抽象金字塔</样式>（e.g., Behnke, 2003b, 2005）中训练，用于重建被结构化噪声（Behnke, 2001）损坏的图像，从而在越来越深的层中强制执行越来越抽象的图像表示。<样式id='3'>去噪自编码器</样式>后来使用了类似的步骤（Vin- cent et al., 2008）。

<样式id='1'>Dropout</样式>（Hinton等人，2012b；Ba和Frey，2013）在训练期间从神经网络中移除单元，以改善泛化。有些人认为它是一种集成方法，可以同时训练多个数据模型（Baldi和Sadowski，2014）。在某些情况下，它也可以被视为一种训练集增强形式：实际上，越来越多的信息丰富的复杂特征被从训练数据中移除。比较循环神经网络的dropout（Pham等人，2013；Pachitariu和Sahani，2013；Pascanu等人，2013a）。一种称为<样式id='3'>快速Dropout</样式>（Wang和Manning，2013）的确定性近似可以导致更快的学习和评估，并且已针对循环神经网络进行了适配（Bayer等人，2013）。Dropout与较旧的、生物学上合理的技巧密切相关，这些技巧在训练期间向神经元或突触添加噪声（例如，Hanson，1990；Murray和Edwards，1993；Schuster，1992；Nadal和Parga，1994；Jim等人，1995；An，1996），这些技巧反过来又与寻找抗扰动低复杂度的神经网络密切相关，例如，通过FMS（第5.6.3节）。基于MDL的随机变分方法（Graves，2011）也与FMS相关。它们对循环神经网络很有用，其中经典正则化器（如权重衰减（第5.6.3节））代表对有限内存容量的偏见（例如，Pascanu等人，2013b）。比较关于变分循环自编码器的最新工作（Bayer和Osendorfer，2014）。

ReLU（整流线性单元）的激活函数 f 是 $f(x) = x$ 对于 $x > 0, f(x) = 0$ 否则——与旧概念的全波整流单元（Malik 和 Perona，1990）相比。ReLU 神经网络对 RBMs（Nair 和 Hinton，2010；Maas 等人，2013）很有用，在深度神经网络中优于 Sigmoid 激活函数（Glorot 等人，2011），并在多个领域的多个基准问题上帮助获得了最佳结果（例如，Krizhevsky 等人，2012；Dahl 等人，2013）。

具有竞争性线性单元的神经网络往往优于具有非竞争性非线性单元的神经网络，并且当训练集随时间变化时通过反向传播避免灾难性遗忘（Srivastava 等人，2013）。在这种情况下，选择学习算法可能比选择激活函数更重要（Goodfellow 等人，2014a）。*Maxout*神经网络（Goodfellow 等人，2013）结合了竞争性交互和 Dropout（见上文），在某些基准测试上取得了优异的结果。与早期用于监督学习和强化学习的具有竞争性单元的 RNN（Schmidhuber，1989b）相比。为了解决过拟合问题，而不是依赖预配置的正则化和超参数（Hertz 等人，1991；Bishop，2006），具有竞争性单元的自限定 RNN（SLIM 神经网络）（Schmidhuber，2012）原则上可以学习选择自己的运行时和自己的有效自由参数数量，从而学习自己的可计算正则化器（第 4.4 节，5.6.3），在必要时变得快速和 精简。可以惩罚特定任务的总连接长度（例如，Legenstein 和 Maass，2002；Schmidhuber，2012，2013b；Clune 等人，2013）以及未来预期在三维类脑多处理器硬件上实现的 SLIM 神经网络的通信成本。

RmsProp (Tieleman 和 Hinton，2012；Schaul 等人，2013) 可以加速一阶梯度下降方法（第 5.5 节，5.6.2 节）；比较 vario- η (Neuneier 和 Zimmermann，1996)，*Adagrad* (Duchi 等人，2011) 和 *Adadelta* (Zeiler，2012)。深度学习在神经网络中也可以通过转换隐藏单元的激活函数，使其平均输出和斜率为零来改进（Raiko 等人，2012）。许多额外的、较老的技巧（第 5.6.2 节，5.6.3 节）也应该适用于当今的深度神经网络；比较 (Orr

5.24 Recent Tricks for Improving SL Deep NNs (Compare Sec. 5.6.2, 5.6.3)

DBN training (Sec. 5.15) can be improved through gradient enhancements and automatic learning rate adjustments during stochastic gradient descent (Cho et al., 2013; Cho, 2014), and through Tikhonov-type (Tikhonov et al., 1977) regularization of RBMs (Cho et al., 2012). Contractive AEs (Rifai et al., 2011) discourage hidden unit perturbations in response to input perturbations, similar to how FMS (Sec. 5.6.3) for LOCOCODE AEs (Sec. 5.6.4) discourages output perturbations in response to weight perturbations.

Hierarchical CNNs in a *Neural Abstraction Pyramid* (e.g., Behnke, 2003b, 2005) were trained to reconstruct images corrupted by structured noise (Behnke, 2001), thus enforcing increasingly abstract image representations in deeper and deeper layers. *Denosing AEs* later used a similar procedure (Vincent et al., 2008).

Dropout (Hinton et al., 2012b; Ba and Frey, 2013) removes units from NNs during training to improve generalisation. Some view it as an ensemble method that trains multiple data models simultaneously (Baldi and Sadowski, 2014). Under certain circumstances, it could also be viewed as a form of training set augmentation: effectively, more and more informative complex features are removed from the training data. Compare dropout for RNNs (Pham et al., 2013; Pachitariu and Sahani, 2013; Pascanu et al., 2013a). A deterministic approximation coined *fast dropout* (Wang and Manning, 2013) can lead to faster learning and evaluation and was adapted for RNNs (Bayer et al., 2013). Dropout is closely related to older, biologically plausible techniques for adding noise to neurons or synapses during training (e.g., Hanson, 1990; Murray and Edwards, 1993; Schuster, 1992; Nadal and Parga, 1994; Jim et al., 1995; An, 1996), which in turn are closely related to finding perturbation-resistant low-complexity NNs, e.g., through FMS (Sec. 5.6.3). MDL-based stochastic variational methods (Graves, 2011) are also related to FMS. They are useful for RNNs, where classic regularizers such as weight decay (Sec. 5.6.3) represent a bias towards limited memory capacity (e.g., Pascanu et al., 2013b). Compare recent work on variational recurrent AEs (Bayer and Osendorfer, 2014).

The activation function f of *Rectified Linear Units* (ReLU) is $f(x) = x$ for $x > 0, f(x) = 0$ otherwise—compare the old concept of half-wave rectified units (Malik and Perona, 1990). ReLU NNs are useful for RBMs (Nair and Hinton, 2010; Maas et al., 2013), outperformed sigmoidal activation functions in deep NNs (Glorot et al., 2011), and helped to obtain best results on several benchmark problems across multiple domains (e.g., Krizhevsky et al., 2012; Dahl et al., 2013).

NNs with competing linear units tend to outperform those with non-competing nonlinear units, and avoid catastrophic forgetting through BP when training sets change over time (Srivastava et al., 2013). In this context, choosing a learning algorithm may be more important than choosing activation functions (Goodfellow et al., 2014a). *Maxout* NNs (Goodfellow et al., 2013) combine competitive interactions and dropout (see above) to achieve excellent results on certain benchmarks. Compare early RNNs with competing units for SL and RL (Schmidhuber, 1989b). To address overfitting, instead of depending on pre-wired regularizers and hyper-parameters (Hertz et al., 1991; Bishop, 2006), self-delimiting RNNs (SLIM NNs) with competing units (Schmidhuber, 2012) can in principle learn to select their own runtime and their own numbers of effective free parameters, thus learning their own computable regularisers (Sec. 4.4, 5.6.3), becoming fast and *slim* when necessary. One may penalize the task-specific total length of connections (e.g., Legenstein and Maass, 2002; Schmidhuber, 2012, 2013b; Clune et al., 2013) and communication costs of SLIM NNs implemented on the 3-dimensional brain-like multi-processor hardware to be expected in the future.

RmsProp (Tieleman and Hinton, 2012; Schaul et al., 2013) can speed up first order gradient descent methods (Sec. 5.5, 5.6.2); compare vario- η (Neuneier and Zimmermann, 1996), *Adagrad* (Duchi et al., 2011) and *Adadelta* (Zeiler, 2012). DL in NNs can also be improved by transforming hidden unit activations such that they have zero output and slope on average (Raiko et al., 2012). Many additional, older tricks (Sec. 5.6.2, 5.6.3) should also be applicable to today’s deep NNs; compare (Orr

以及 Müller, 1998; Montavon 等人, 2012)。

5.25 对神经科学的启示

很讽刺的是，人工神经网络 (ANNs) 可以帮助更好地理解生物神经网络 (BNNs) — 请参阅第 5.21 节中提到的 ISBI 2012 结果（神经元结构在电子显微镜堆栈中的分割挑战, 2012; Ciresan 等人, 2012a）。

单层视觉人工神经网络学习到的特征检测器与生物神经网络早期视觉处理阶段发现的特征检测器相似（例如，第 5.6.4 节）。同样，视觉人工神经网络深度学习到的特征检测器应该能高度预测神经科学家在生物神经网络深层会发现的内容。尽管生物神经网络的视觉皮层可能使用截然不同的学习算法，但其需要最小化的目标函数可能与视觉人工神经网络的目标函数非常相似。事实上，使用相对较深的深度信念网络（李等人, 2007b）和卷积神经网络（Yamins 等人, 2013）获得的结果似乎与灵长类大脑皮层视觉通路的研究发现相吻合，这些研究已经进行了几十年（例如，Hubel 和 Wiesel, 1968; Perrett 等人, 1982; Desimone 等人, 1984; Felleman 和 Van Essen, 1991; Perrett 等人, 1992; Kobatake 和 Tanaka, 1994; Logothetis 等人, 1995; Bichot 等人, 2005; Hung 等人, 2005; Lennie 和 Movshon, 2005; Connor 等人, 2007; Kriegeskorte 等人, 2008; DiCarlo 等人, 2012）；对比一篇面向计算机视觉的调查（Kruger 等人, 2013）。

5.26 深度学习与脉冲神经元？

许多最近的深度学习成果受益于基于 GPU 的传统深度神经网络，例如第 5.16 至 5.19 节。然而，当前的 GPU 像小烤箱一样，对能源的需求远高于生物大脑，其神经元通过短暂的脉冲高效地通信（Hodgkin 和 Huxley, 1952; FitzHugh, 1961; Nagumo 等人, 1962），并且经常保持安静。许多关于这种 脉冲神经元的计算模型已被提出和分析（例如，Gerstner 和 van Hemmen, 1992; Zipser 等人, 1993; Stemmler, 1996; Tsodyks 等人, 1996; Maex 和 Orban, 1996; Maass, 1996, 1997; Kistler 等人, 1997; Amit 和 Brunel, 1997; Tsodyks 等人, 1998; Kempter 等人, 1999; Song 等人, 2000; Stoop 等人, 2000; Brunel, 2000; Bohte 等人, 2002; Gerstner 和 Kistler, 2002; Izhikevich 等人, 2003; Seung, 2003; Deco 和 Rolls, 2005; Brette 等人, 2007; Brea 等人, 2013; Nessler 等人, 2013; Kasabov, 2014; Hoerzer 等人, 2014; Rezende 和 Gerstner, 2014）。

未来用于神经网络 (NNs) 中深度学习 (DL) 的节能硬件可能会实现此类模型的某些方面（例如，刘等人, 2001年；罗根等人, 2003年；格拉金等人, 2005年；施梅尔等人, 2006年；菲耶尔斯等人, 2008年；卡恩等人, 2008年；塞拉诺-戈塔雷多纳等人, 2009年；金等人, 2010年；因迪维里等人, 2011年；尼尔和刘, 2014年；梅罗拉等人, 2014年）。一个模拟的、事件驱动的、脉冲变体（内夫齐等人, 2014年）的受限玻尔兹曼机 (RBM) （第5.15节）由对比散度算法（希顿, 2002年）的变体进行训练。脉冲网络被进化以在小型人脸识别数据集（维索斯基等人, 2010年）上实现合理性能，并控制简单机器人（弗洛埃诺和马蒂乌西, 2001年；哈格拉斯等人, 2004年）。一个包含约25万个神经元（作为更大神经网络的一部分；埃利亚斯米斯等人, 2012年；埃利亚斯米斯, 2013年）的脉冲深度信念网络 (DBN) 在MNIST上实现了6%的错误率；比较使用神经形态事件传感器（奥康纳等人, 2013年）的深度3脉冲DBN变体的类似结果。然而，在实际应用中，当前脉冲神经元的假人网络尚无法与最佳传统深度神经网络竞争（例如，比较第5.19节的MNIST结果）。

and Müller, 1998; Montavon et al., 2012).

5.25 Consequences for Neuroscience

It is ironic that artificial NNs (ANNs) can help to better understand biological NNs (BNNs)—see the ISBI 2012 results mentioned in Sec. 5.21 (Segmentation of Neuronal Structures in EM Stacks Challenge, 2012; Ciresan et al., 2012a).

The feature detectors learned by single-layer visual ANNs are similar to those found in early visual processing stages of BNNs (e.g., Sec. 5.6.4). Likewise, the feature detectors learned in *deep* layers of visual ANNs should be highly predictive of what neuroscientists will find in *deep* layers of BNNs. While the visual cortex of BNNs may use quite different learning algorithms, its objective function to be minimised may be quite similar to the one of visual ANNs. In fact, results obtained with relatively deep artificial DBNs (Lee et al., 2007b) and CNNs (Yamins et al., 2013) seem compatible with insights about the visual pathway in the primate cerebral cortex, which has been studied for many decades (e.g., Hubel and Wiesel, 1968; Perrett et al., 1982; Desimone et al., 1984; Felleman and Van Essen, 1991; Perrett et al., 1992; Kobatake and Tanaka, 1994; Logothetis et al., 1995; Bichot et al., 2005; Hung et al., 2005; Lennie and Movshon, 2005; Connor et al., 2007; Kriegeskorte et al., 2008; DiCarlo et al., 2012); compare a computer vision-oriented survey (Kruger et al., 2013).

5.26 DL with Spiking Neurons?

Many recent DL results profit from GPU-based traditional deep NNs, e.g., Sec. 5.16–5.19. Current GPUs, however, are little ovens, much hungrier for energy than biological brains, whose neurons efficiently communicate by brief spikes (Hodgkin and Huxley, 1952; FitzHugh, 1961; Nagumo et al., 1962), and often remain quiet. Many computational models of such *spiking neurons* have been proposed and analyzed (e.g., Gerstner and van Hemmen, 1992; Zipser et al., 1993; Stemmler, 1996; Tsodyks et al., 1996; Maex and Orban, 1996; Maass, 1996, 1997; Kistler et al., 1997; Amit and Brunel, 1997; Tsodyks et al., 1998; Kempter et al., 1999; Song et al., 2000; Stoop et al., 2000; Brunel, 2000; Bohte et al., 2002; Gerstner and Kistler, 2002; Izhikevich et al., 2003; Seung, 2003; Deco and Rolls, 2005; Brette et al., 2007; Brea et al., 2013; Nessler et al., 2013; Kasabov, 2014; Hoerzer et al., 2014; Rezende and Gerstner, 2014).

Future energy-efficient hardware for DL in NNs may implement aspects of such models (e.g., Liu et al., 2001; Roggen et al., 2003; Glackin et al., 2005; Schemmel et al., 2006; Fieres et al., 2008; Khan et al., 2008; Serrano-Gotarredona et al., 2009; Jin et al., 2010; Indiveri et al., 2011; Neil and Liu, 2014; Merolla et al., 2014). A simulated, event-driven, spiking variant (Neftci et al., 2014) of an RBM (Sec. 5.15) was trained by a variant of the *Contrastive Divergence* algorithm (Hinton, 2002). Spiking nets were evolved to achieve reasonable performance on small face recognition data sets (Wysoski et al., 2010) and to control simple robots (Floreano and Mattiussi, 2001; Hagraš et al., 2004). A spiking DBN with about 250,000 neurons (as part of a larger NN; Eliasmith et al., 2012; Eliasmith, 2013) achieved 6% error rate on MNIST; compare similar results with a spiking DBN variant of depth 3 using a neuromorphic event-based sensor (O’Connor et al., 2013). In practical applications, however, current artificial networks of spiking neurons cannot yet compete with the best traditional deep NNs (e.g., compare MNIST results of Sec. 5.19).

6 深度学习在前馈神经网络和循环神经网络中的强化学习(RL)

到目前为止，我们一直专注于监督或无监督神经网络中的深度学习（DL）。此类神经网络学习感知/编码/预测/分类模式或模式序列，但它们不会学习在未知环境中以更一般意义上的强化学习（RL）（参见调查，例如，凯尔布林等人，1996年；萨顿和巴托，1998年；维リング和范·奥特洛，2012年）。在这里，我们增加了对用于强化学习的深度学习前馈神经网络（FNNs）和循环神经网络（RNNs）的讨论。这会比用于监督学习（SL）和无监督学习（UL）（第5节）的FNNs和RNNs的讨论短，反映了各个领域当前的规模。

在没有教师的情况下，仅从偶尔的实值痛苦和快乐信号中，强化学习代理必须发现如何与动态的、最初未知的环境交互以最大化其预期的累积奖励信号（第2节）。行动和可感知后果之间可能存在任意、先验未知的延迟。由于任何具有可计算描述的任务都可以在强化学习框架中表述（例如，哈特，2005年），因此该问题是计算机科学中任何问题的难度一样大。例如，对著名问题的回答，即 $P = NP$ （利文，1973b；库克，1971年）也将为一般强化学习可实现的极限设定限制。比较更具体的限制，例如（布隆代尔和齐齐克利斯，2000年；马达尼等人，2003年；弗拉西斯等人，2012年）。以下小节主要关注深度学习和强化学习之间某些明显的交集——它们不能作为一般强化学习综述。

6.1 强化学习通过神经网络世界模型生成具有深度CAPs的循环神经网络

在强化学习前馈神经网络控制器 C 与一个确定性的、可预测的环境交互的特殊情况下，一个单独的前馈神经网络称为 M 可以通过系统辨识学习成为 C 的世界模型，从先前的动作和输入预测 C 的输入（例如，Werbos，1981，1987；Munro，1987；Jordan，1988；Werbos，1989b,a；Robinson和Fallside，1989；Jordan和Rumelhart，1990；Schmidhuber，1990d；Narendra和Parthasarathy，1990；Werbos，1992；Gomi和Kawato，1993；Cochocki和Unbehauen，1993；Levin和Narendra，1995；Miller等人，1995；Ljung，1998；Prokhorov等人，2001；Ge等人，2010）。假设 M 已经学会了产生准确的预测。我们可以使用 M 来替代环境。然后 M 和 C 形成一个循环神经网络，其中 M 的输出成为 C 的输入，而 C 的输出（动作）又成为 M 的输入。现在可以使用循环神经网络的反向传播（第5.5.1节）来实现期望的输入事件，例如高实值奖励信号：当 M 的权重保持固定时，关于 C 权重的梯度信息会通过 M 反向传播到 C ，并通过 M 等。在某种程度上，该方法也适用于概率性或不确定环境，只要 M 的基于 C 的梯度估计与 M 的“真实”梯度内积趋于为正即可。

通常来说，这种方法为 C 带来了深度 CAPs，这与基于动态规划的传统强化学习（第 6.2 节）不同。几十年前，该方法被用于学习倒车模型卡车（Nguyen and Widrow，1989）。一个强化学习主动视觉系统用它来学习fovea的顺序移动（saccades），以在视觉场景中检测目标（Schmidhuber and Huber，1991），从而学习控制选择性注意力。对比没有神经网络的强化学习注意力学习（Whitehead，1992）。

为了允许在<样式 id='l'>部分可观察世界</样式>（第 6.3 节）中记住先前事件，这种技术的最通用变体使用循环神经网络而不是前馈神经网络来实现 M 和 C （Schmidhuber，1990d，1991c；Feldkamp and Puskorius，1998）。这可能不仅为 C 带来深度 CAPs，也为 M 带来深度 CAPs。

M 也可以通过规划未来动作序列来优化预期奖励（Schmidhuber，1990d）。事实上，2004年RoboCup世界锦标赛快速联赛的冠军（Egorova等人，2004）训练了神经网络来预测对具有4个电机的4个不同轮子的快速机器人施加转向信号的效果。在比赛过程中，使用此类神经网络模型通过快速提前规划来优化动作序列，以实现期望的子目标。该方法还用于创建自愈机器人，这些机器人能够补偿故障电机，其效果不再与神经网络模型的预测相匹配（Gloye等人，2005；Schmidhuber，2007）。

6 DL in FNNs and RNNs for Reinforcement Learning (RL)

So far we have focused on Deep Learning (DL) in supervised or unsupervised NNs. Such NNs learn to perceive / encode / predict / classify patterns or pattern sequences, but they do not learn to act in the more general sense of *Reinforcement Learning* (RL) in unknown environments (see surveys, e.g., Kaelbling et al., 1996; Sutton and Barto, 1998; Wiering and van Otterlo, 2012). Here we add a discussion of DL FNNs and RNNs for RL. It will be shorter than the discussion of FNNs and RNNs for SL and UL (Sec. 5), reflecting the current size of the various fields.

Without a teacher, solely from occasional real-valued pain and pleasure signals, RL agents must discover how to interact with a dynamic, initially unknown environment to maximize their expected cumulative reward signals (Sec. 2). There may be arbitrary, a priori unknown delays between actions and perceivable consequences. The problem is as hard as any problem of computer science, since any task with a computable description can be formulated in the RL framework (e.g., Hutter, 2005). For example, an answer to the famous question of whether $P = NP$ (Levin, 1973b; Cook, 1971) would also set limits for what is achievable by general RL. Compare more specific limitations, e.g., (Blondel and Tsitsiklis, 2000; Madani et al., 2003; Vlassis et al., 2012). The following subsections mostly focus on certain obvious intersections between DL and RL—they cannot serve as a general RL survey.

6.1 RL Through NN World Models Yields RNNs With Deep CAPs

In the special case of an RL FNN controller C interacting with a *deterministic, predictable* environment, a separate FNN called M can learn to become C ’s *world model* through *system identification*, predicting C ’s inputs from previous actions and inputs (e.g., Werbos，1981，1987；Munro，1987；Jordan，1988；Werbos，1989b,a；Robinson and Fallside，1989；Jordan and Rumelhart，1990；Schmidhuber，1990d；Narendra and Parthasarathy，1990；Werbos，1992；Gomi and Kawato，1993；Cochocki and Unbehauen，1993；Levin and Narendra，1995；Miller et al.，1995；Ljung，1998；Prokhorov et al.，2001；Ge et al.，2010). Assume M has learned to produce accurate predictions. We can use M to substitute the environment. Then M and C form an RNN where M ’s outputs become inputs of C ，whose outputs (actions) in turn become inputs of M . Now BP for RNNs (Sec. 5.5.1) can be used to achieve *desired input events* such as high real-valued reward signals: While M ’s weights remain fixed, gradient information for C ’s weights is propagated back through M down into C and back through M etc. To a certain extent, the approach is also applicable in probabilistic or uncertain environments, as long as the inner products of M ’s C -based gradient estimates and M ’s “true” gradients tend to be positive.

In general, this approach implies deep CAPs for C ，unlike in DP-based traditional RL (Sec. 6.2). Decades ago, the method was used to learn to back up a model truck (Nguyen and Widrow，1989). An RL active vision system used it to learn sequential shifts (saccades) of a fovea, to detect targets in visual scenes (Schmidhuber and Huber，1991), thus learning to control selective attention. Compare RL-based attention learning without NNs (Whitehead，1992).

To allow for memories of previous events in *partially observable worlds* (Sec. 6.3), the most general variant of this technique uses RNNs instead of FNNs to implement both M and C （Schmidhuber，1990d，1991c；Feldkamp and Puskorius，1998). This may cause deep CAPs not only for C but also for M .

M can also be used to optimize expected reward by *planning* future action sequences (Schmidhuber，1990d). In fact, the winners of the 2004 RoboCup World Championship in the fast league (Egorova et al., 2004) trained NNs to predict the effects of steering signals on fast robots with 4 motors for 4 different wheels. During play, such NN models were used to achieve desirable subgoals, by optimizing action sequences through quickly planning ahead. The approach also was used to create *self-healing* robots able to compensate for faulty motors whose effects do not longer match the predictions of the NN models (Gloye et al., 2005; Schmidhuber, 2007).

通常 M 不会预先给出。然后一个关键问题是：应该 C 进行哪些实验才能快速提升 M ？《趣味与创造性的形式理论》（*Schmidhuber, 2006a, 2013b*）（无监督学习）形式化了这种好奇和探索行为背后的驱动力和价值函数：对 M 的学习进展 的衡量成为 C 的内在奖励（Schmidhuber, 1991a）；参见（Singh 等人，2005；Oudeyer 等人，2013）。这促使 C 创建动作序列（实验），使得 M 能快速取得进展。

6.2 传统RLand马尔可夫决策过程（MDPs）

强化学习（RL）的经典方法（Samuel, 1959；Bertsekas和Tsitsiklis, 1996）做出了简化假设，即马尔可夫决策过程(MDPs)：强化学习智能体的当前输入包含了计算最优下一个输出事件或决策所需的所有信息。这允许通过使用动态规划(DP)技巧（Bellman, 1957）大幅降低强化学习神经网络中的CAP深度（第3节，6.1节）。后者通常在概率框架下解释（例如，Sutton和Barto, 1998），但其基本思想已经可以在确定性设置中传达。为简化起见，使用第2节的符号，让输入事件 x_t 编码环境的整个当前状态，包括实值奖励 r_t （无需引入额外的向量值符号，因为实值可以编码任意实值向量）。原始的强化学习目标（找到最大化一个回合所有奖励总和的权重）被替换为一组等效的替代目标，这些目标由定义在输入事件上的实值值函数 V 设定。考虑任意两个后续输入事件 x_t, x_k 。递归定义 $V(x_t) = r_t + V(x_k)$ ，其中 $V(x_k) = r_k$ 如果 x_k 是最后一个输入事件。现在寻找最大化所有输入事件 V 的权重，通过引发适当的输出事件或动作。

由于马尔可夫假设，一个前馈神经网络就足以实现将输入映射到输出事件的策略。相关的CAPs不会比这个前馈神经网络更深。 V 本身通常由一个独立的前馈神经网络（通常也会产生较短的CAPs）建模，仅从局部信息 $r_t, V(x_k)$ 中学习到近似 $V(x_t)$ 。

许多传统的强化学习变体存在（例如，Barto等人，1983年；Watkins, 1989年；Watkins和Dayan, 1992年；Moore和Atkeson, 1993年；Schwartz, 1993年；Rummery和Niranjan, 1994年；Singh, 1994年；Baird, 1995年；Kaelbling等人，1995年；Peng和Williams, 1996年；Mahadevan, 1996年；Tsitsiklis和van Roy, 1996年；Bradtke等人，1996年；Santamaría等人，1997年；Prokhorov和Wunsch, 1997年；Sutton和Barto, 1998年；Wiering和Schmidhuber, 1998b年；Baird和Moore, 1999年；Meuleau等人，1999年；Morimoto和Doya, 2000年；Bertsekas, 2001年；Brafman和Tennenholtz, 2002年；Abounadi等人，2002年；Lagoudakis和Parr, 2003年；Sutton等人，2008年；Maei和Sutton, 2010年；van Hasselt, 2012年）。大多数都在概率框架下进行，并评估输入和输出（动作）事件对（而不仅仅是输入事件）。为了便于某些数学推导，一些方法会折扣延迟的奖励，但这种对原始强化学习问题的扭曲是有问题的。

或许最著名的强化学习神经网络是世界级的强化学习西洋双陆棋玩家（Tesauro, 1994年），通过自我对战达到了人类世界冠军的水平。它的非线性、相对较浅的前馈神经网络将大量但有限的离散棋盘状态映射到值。最近，一种相当深的GPU-卷积神经网络被用于传统的强化学习框架中，直接从84x84像素60Hz视频输入中玩几个Atari 2600电脑游戏（Mnih等人，2013年），使用经验回放（Lin, 1993年），扩展了之前关于神经拟合Q学习(NFQ)（Riedmiller, 2005年）的研究。使用（慢速）蒙特卡洛树规划来训练相对较快的深度神经网络可以取得更好的结果（Guo等人，2014年）。比较基于RBM的强化学习（Sallans和Hinton, 2004年）与高维输入（Elfwing等人，2010年），早期的强化学习Atari玩家（Grüttner等人，2010年），以及一个早期基于原始视频的强化学习神经网络用于电脑游戏（Koutník等人，2013年），该网络由间接策略搜索(第6.7节)训练。

Typically M is not given in advance. Then an essential question is: which experiments should C conduct to quickly improve M ? The *Formal Theory of Fun and Creativity* (e.g., Schmidhuber, 2006a, 2013b) formalizes driving forces and value functions behind such curious and exploratory behavior: A measure of the *learning progress* of M becomes the intrinsic reward of C (Schmidhuber, 1991a); compare (Singh et al., 2005; Oudeyer et al., 2013). This motivates C to create action sequences (experiments) such that M makes quick progress.

6.2 Deep FNNs for Traditional RL and Markov Decision Processes (MDPs)

The classical approach to RL (Samuel, 1959; Bertsekas and Tsitsiklis, 1996) makes the simplifying assumption of *Markov Decision Processes* (MDPs): the current input of the RL agent conveys all information necessary to compute an optimal next output event or decision. This allows for greatly reducing CAP depth in RL NNs (Sec. 3, 6.1) by using the *Dynamic Programming* (DP) trick (Bellman, 1957). The latter is often explained in a probabilistic framework (e.g., Sutton and Barto, 1998), but its basic idea can already be conveyed in a deterministic setting. For simplicity, using the notation of Sec. 2, let input events x_t encode the entire current state of the environment, including a real-valued reward r_t (no need to introduce additional vector-valued notation, since real values can encode arbitrary vectors of real values). The original RL goal (find weights that maximize the sum of all rewards of an episode) is replaced by an equivalent set of alternative goals set by a real-valued value function V defined on input events. Consider any two subsequent input events x_t, x_k . Recursively define $V(x_t) = r_t + V(x_k)$, where $V(x_k) = r_k$ if x_k is the *last* input event. Now search for weights that maximize the V of all input events, by causing appropriate output events or actions.

Due to the Markov assumption, an FNN suffices to implement the policy that maps input to output events. Relevant CAPs are not deeper than this FNN. V itself is often modeled by a *separate FNN* (also yielding typically short CAPs) learning to approximate $V(x_t)$ only from *local* information $r_t, V(x_k)$.

Many variants of traditional RL exist (e.g., Barto et al., 1983; Watkins, 1989; Watkins and Dayan, 1992; Moore and Atkeson, 1993; Schwartz, 1993; Rummery and Niranjan, 1994; Singh, 1994; Baird, 1995; Kaelbling et al., 1995; Peng and Williams, 1996; Mahadevan, 1996; Tsitsiklis and van Roy, 1996; Bradtke et al., 1996; Santamaría et al., 1997; Prokhorov and Wunsch, 1997; Sutton and Barto, 1998; Wiering and Schmidhuber, 1998b; Baird and Moore, 1999; Meuleau et al., 1999; Morimoto and Doya, 2000; Bertsekas, 2001; Brafman and Tennenholtz, 2002; Abounadi et al., 2002; Lagoudakis and Parr, 2003; Sutton et al., 2008; Maei and Sutton, 2010; van Hasselt, 2012). Most are formulated in a probabilistic framework, and evaluate pairs of input and output (action) events (instead of input events only). To facilitate certain mathematical derivations, some discount delayed rewards, but such distortions of the original RL problem are problematic.

Perhaps the most well-known RL NN is the world-class RL backgammon player (Tesauro, 1994), which achieved the level of human world champions by playing against itself. Its nonlinear, rather shallow FNN maps a large but finite number of discrete board states to values. More recently, a rather deep GPU-CNN was used in a traditional RL framework to play several Atari 2600 computer games directly from 84x84 pixel 60 Hz video input (Mnih et al., 2013), using experience replay (Lin, 1993), extending previous work on *Neural Fitted Q-Learning* (NFQ) (Riedmiller, 2005). Even better results are achieved by using (slow) Monte Carlo tree planning to train comparatively fast deep NNs (Guo et al., 2014). Compare RBM-based RL (Sallans and Hinton, 2004) with high-dimensional inputs (Elfwing et al., 2010), earlier RL Atari players (Grüttner et al., 2010), and an earlier, raw video-based RL NN for computer games (Koutník et al., 2013) trained by *Indirect Policy Search* (Sec. 6.7).

6.3 用于部分可观察马尔可夫决策过程（POMDP）的深度RLRNN

<样式id='1'>马尔可夫假设</样式>(第6.2节)通常是现实的。我们无法直接感知我们背后是什么，更不用说整个宇宙的当前状态了。然而，先前事件的记忆可以帮助处理<样式id='3'>部分可观察马尔可夫决策问题</样式>(POMDPs)（例如，Schmidhuber, 1990d, 1991c; Ring, 1991年, 1993年, 1994年; Williams, 1992a; Lin, 1993年; Teller, 1994年; Kaelbling等人, 1995年; Littman等人, 1995年; Boutilier和Poole, 1996年; Jaakkola等人, 1995年; McCallum, 1996年; Kimura等人, 1997年; Wiering和Schmidhuber, 1996年, 1998a; Otsuka等人, 2010年）。在MDP框架内（第6.2节）实现记忆的一种简单方法是将一个可能巨大的状态空间简单地考虑为所有可能的观察历史及其前缀的集合。一种更现实的方法是使用函数逼近器，如循环神经网络，它们将到目前为止看到的整个历史作为函数产生紧凑的状态特征。一般来说，POMDP强化学习通常使用深度学习循环神经网络来学习要记忆哪些事件以及要忽略哪些事件。三种基本替代方案是：

1. 使用RNN作为值函数，将任意事件历史映射到值（例如，Schmidhuber, 1990b, 1991c; Lin, 1993; Bakker, 2002）。例如，深度LSTM RNN被用于这种方式为RL机器人（Bakker等人, 2003年）。
2. 使用一个RNN控制器与第二个RNN作为预测世界模型相结合，以获得一个具有深度CAPs的联合RNN——参见第6.1节。
3. 使用RNN通过直接搜索（第6.6节）或间接搜索（第6.7节）在权重空间中进行RL。

然而，一般来说，POMDP可能意味着CAP深度大大增加。

6.4 由深度UL在前馈神经网络和循环神经网络中促进的RL

强化学习机器可以通过无监督学习进行输入预处理（例如，Jodogne 和 Piater, 2007）。特别是，一个无监督学习的神经网络可以学习紧凑地编码环境输入，例如图像或视频，例如第 5.7、5.10、5.15 节。这些紧凑的编码（而不是高维原始数据）可以被输入到强化学习机器中，其任务因此可能变得容易得多（Legenstein 等人, 2010; Cuccu 等人, 2011），就像监督学习可能通过无监督学习受益一样，例如第 5.7、5.10、5.15 节。例如，NFQ (Riedmiller, 2005) 被应用于现实世界的控制任务（Lange 和 Riedmiller, 2010; Riedmiller 等人, 2012），其中纯粹的视觉输入通过深度自动编码器被紧凑地编码（第 5.7、5.15 节）。结合基于慢特征分析（Wiskott 和 Sejnowski, 2002; Kompella 等人, 2012）的无监督学习的强化学习使一个真实的人形机器人能够从原始的高维视频流中学习技能（Luciw 等人, 2013）。为了处理涉及高维输入的 POMDP（第 6.3 节），使用了基于 RBM 的强化学习（Otsuka, 2010），并采用 RAAM（Pollack, 1988）（第 5.7 节）作为深度无监督序列编码器用于强化学习（Gisslen 等人, 2011）。某些类型的强化学习和无监督学习也被结合在具有脉冲神经元的生物合理型循环神经网络中（第 5.26 节）（例如，Yin 等人, 2012; Klampfl 和 Maass, 2013; Rezende 和 Gerstner, 2014）。

6.5 深度分层强化学习（分层强化学习）与基于前馈神经网络和循环神经网络的子目标学习

多可学习抽象层次（Fu, 1977; Lenat and Brown, 1984; Ring, 1994; Bengio 等人, 2013; Deng 和 Yu, 2014）对强化学习和监督学习都同样重要。基于神经网络的层次化强化学习（HRL）的研究自20世纪90年代初就已开始。特别是基于梯度的子目标发现，使用前馈神经网络或循环神经网络将强化学习任务分解为子任务，以便强化学习子模块（Schmidhuber, 1991b; Schmidhuber 和 Wahnsiedler, 1992）。许多替代的 HRL 技术已被提出（例如，Ring, 1991, 1994; Jameson, 1991; Tenenbergeretal., 1993; Weiss, 1994; Moore 和 Atkeson, 1995; Precup 等人, 1998; Dietterich, 2000b; Menache 等人, 2002; Doya 等人, 2002; Ghavamzadeh 和 Mahadevan, 2003; Barto 和 Mahadevan, 2003; Samejima 等人, 2003; Bakker 和 Schmidhuber, 2004; Whiteson 等人, 2005; Simsek 和 Barto, 2008）。

6.3 Deep RL RNNs for Partially Observable MDPs (POMDPs)

The *Markov assumption* (Sec. 6.2) is often unrealistic. We cannot directly perceive what is behind our back, let alone the current state of the entire universe. However, memories of previous events can help to deal with *partially observable Markov decision problems* (POMDPs) (e.g., Schmidhuber, 1990d, 1991c; Ring, 1991, 1993, 1994; Williams, 1992a; Lin, 1993; Teller, 1994; Kaelbling et al., 1995; Littman et al., 1995; Boutilier and Poole, 1996; Jaakkola et al., 1995; McCallum, 1996; Kimura et al., 1997; Wiering and Schmidhuber, 1996, 1998a; Otsuka et al., 2010). A naive way of implementing memories without leaving the MDP framework (Sec. 6.2) would be to simply consider a possibly huge state space, namely, the set of all possible observation histories and their prefixes. A more realistic way is to use function approximators such as RNNs that produce compact state features as a function of the entire history seen so far. Generally speaking, POMDP RL often uses DL RNNs to learn which events to memorize and which to ignore. Three basic alternatives are:

1. Use an RNN as a value function mapping arbitrary event histories to values (e.g., Schmidhuber, 1990b, 1991c; Lin, 1993; Bakker, 2002). For example, deep LSTM RNNs were used in this way for RL robots (Bakker et al., 2003).
2. Use an RNN controller in conjunction with a second RNN as predictive world model, to obtain a combined RNN with deep CAPs—see Sec. 6.1.
3. Use an RNN for RL by *Direct Search* (Sec. 6.6) or *Indirect Search* (Sec. 6.7) in weight space.

In general, however, POMDPs may imply greatly increased CAP depth.

6.4 RL Facilitated by Deep UL in FNNs and RNNs

RL machines may profit from UL for input preprocessing (e.g., Jodogne and Piater, 2007). In particular, an UL NN can learn to compactly encode environmental inputs such as images or videos, e.g., Sec. 5.7, 5.10, 5.15. The compact codes (instead of the high-dimensional raw data) can be fed into an RL machine, whose job thus may become much easier (Legenstein et al., 2010; Cuccu et al., 2011), just like SL may profit from UL, e.g., Sec. 5.7, 5.10, 5.15. For example, NFQ (Riedmiller, 2005) was applied to real-world control tasks (Lange and Riedmiller, 2010; Riedmiller et al., 2012) where purely visual inputs were compactly encoded by deep autoencoders (Sec. 5.7, 5.15). RL combined with UL based on *Slow Feature Analysis* (Wiskott and Sejnowski, 2002; Kompella et al., 2012) enabled a real humanoid robot to learn skills from raw high-dimensional video streams (Luciw et al., 2013). To deal with POMDPs (Sec. 6.3) involving high-dimensional inputs, RBM-based RL was used (Otsuka, 2010), and a RAAM (Pollack, 1988) (Sec. 5.7) was employed as a deep unsupervised sequence encoder for RL (Gisslen et al., 2011). Certain types of RL and UL also were combined in biologically plausible RNNs with spiking neurons (Sec. 5.26) (e.g., Yin et al., 2012; Klampfl and Maass, 2013; Rezende and Gerstner, 2014).

6.5 Deep Hierarchical RL (HRL) and Subgoal Learning with FNNs and RNNs

Multiple learnable levels of abstraction (Fu, 1977; Lenat and Brown, 1984; Ring, 1994; Bengio et al., 2013; Deng and Yu, 2014) seem as important for RL as for SL. Work on NN-based *Hierarchical RL* (HRL) has been published since the early 1990s. In particular, gradient-based *subgoal discovery* with FNNs or RNNs decomposes RL tasks into subtasks for RL submodules (Schmidhuber, 1991b; Schmidhuber and Wahnsiedler, 1992). Numerous alternative HRL techniques have been proposed (e.g., Ring, 1991, 1994; Jameson, 1991; Tenenberget al., 1993; Weiss, 1994; Moore and Atkeson, 1995; Precup et al., 1998; Dietterich, 2000b; Menache et al., 2002; Doya et al., 2002;

Ghavamzadeh 和 Mahadevan, 2003; Barto 和 Mahadevan, 2003; Samejima 等人, 2003; Bakker 和 Schmidhuber, 2004; Whiteson 等人, 2005; Simsek 和 Barto, 2008)。虽然像封建强化学习 (Dayan 和 Hinton, 1993) 和选项 (Sutton 等人, 1999b; Barto 等人, 2004; Singh 等人, 2005) 这样的 HRL 框架不直接解决自动子目标发现的问题, 但*HQ-Learning* (Wiering 和 Schmidhuber, 1998a) 自动将部分可观察马尔可夫决策过程 (Sec. 6.3) 分解为可由无记忆策略学习的、可由反应式子代理解决的一系列更简单的子任务。最近的 HRL 将潜在的深度神经网络强化学习子模块组织成自组织的、二维的运动控制图 (Ring 等人, 2011), 灵感来自神经生理学发现 (Graziano, 2009)。

6.6 直接神经网络搜索 / 策略梯度 / 进化

与第6.8节的方法相比, 它们并非那么普适, 但既实用又比大多数传统强化学习算法 (第6.2节) 更通用, 这些是直接策略搜索 (DS)方法。无需价值函数或马尔可夫假设 (第6.2节、6.3节), 前馈神经网络或循环神经网络的权重直接在给定的强化学习问题上进行评估。连续试验的结果会指导进一步搜索以获得更好的权重。与基于反向传播 (第5.5节、6.3节、6.1节) 的强化学习不同, CAP深度 (第3节、5.9节) 并非关键问题。DS可以在不回溯可修改参数的深层因果链的情况下解决信用分配问题——它既不关心这些参数的存在, 也不试图利用它们。

对于神经网络而言, DS方法的一个重要类别是策略梯度方法 (Williams, 1986、1988、1992a; Sutton等人, 1999a; Baxter和Bartlett, 2001; Aberdeen, 2003; Ghavamzadeh和Mahadevan, 2003; Kohl和Stone, 2004; Wierstra等人, 2008; Rückstieß等人, 2008; Peters和Schaal, 2008b,a; Sehnke等人, 2010; Gruttner等人, 2010; Wierstra等人, 2010; Peters, 2010; Grondman等人, 2012; Heess等人, 2012)。通过重复神经网络评估来估计总奖励相对于策略 (神经网络权重) 的梯度 (然后加以利用)。

强化学习神经网络也可以通过<样式 id='1'>进化算法</样式>(Rechenberg, 1971; Schwefel, 1974; Holland, 1975; Fogel 等人, 1966; Goldberg, 1989)在一系列试验中进化。这里, 多个策略由一个神经网络种群表示, 该种群通过突变和/或重复重组种群中最适应的个体 (例如, Montana 和 Davis, 1989; Fogel 等人, 1990; Maniezzo, 1994; Happel 和 Murre, 1994; Nolfi 等人, 1994b) 得到改进。比较<样式 id='3'>遗传编程</样式>(GP) (Cramer, 1985) (另见 Smith, 1980), 它可用于进化可变大小的计算机程序 (Dickmanns 等人, 1987; Koza, 1992), 以及<样式 id='5'>笛卡尔遗传编程</样式>(Miller 和 Thomson, 2000; Miller 和 Harding, 2009)用于进化图状程序, 包括神经网络 (Khan 等人, 2010) 及其拓扑结构 (Turner 和 Miller, 2013)。相关方法包括<样式 id='7'>基于概率分布的进化算法</样式>(Baluja, 1994; Saravanan 和 Fogel, 1995; Śaustowicz 和 Schmidhuber, 1997; Larraanaga 和 Lozano, 2001), <样式 id='9'>协方差矩阵估计进化策略</样式>(CMA-ES) (Hansen 和 Ostermeier, 2001; Hansen 等人, 2003; Igel, 2003; Heidrich-Meisner 和 Igel, 2009), 以及<样式 id='11'>增强拓扑神经进化</样式>(NEAT) (Stanley 和 Miikkulainen, 2002)。混合方法结合了传统的基于神经网络的强化学习 (第 6.2 节) 和进化算法 (例如, Whiteson 和 Stone, 2006)。

由于循环神经网络 (RNNs) 是一般计算机, RNN的进化在意义上类似于遗传编程 (GP), 因为它可以进化通用程序。然而, 与传统GP学习的序列程序不同, RNNs可以自然且高效地混合序列和并行信息处理, 正如第1节中已经提到的。已经提出了许多RNN进化器 (例如, Miller等人, 1989年; Wieland, 1991年; Cliff等人, 1993年; Yao, 1993年; Nolfi等人, 1994a; Sims, 1994年; Yamauchi和Beer, 1994年; Miglino等人, 1995年; Moriarty, 1997年; Pasemann等人, 1999年; Juang, 2004年; Whiteson, 2012年)。一种特别有效的方法族 协同进化 神经元, 将它们组合成网络, 并选择那些参与表现最佳网络的神经元进行繁殖 (Moriarty和Miikkulainen, 1996年; Gomez, 2003年; Gomez和Miikkulainen, 2003年)。这有助于解决深度部分可观察马尔可夫决策过程 (POMDPs) (Gomez和Schmidhuber, 2005年)。协同突触神经进化 (CoSyNE) 在突触层面上做类似的事情

Ghavamzadeh 和 Mahadevan, 2003; Barto 和 Mahadevan, 2003; Samejima et al., 2003; Bakker 和 Schmidhuber, 2004; Whiteson et al., 2005; Simsek 和 Barto, 2008)。While HRL frameworks such as *Feudal RL* (Dayan and Hinton, 1993) and *options* (Sutton et al., 1999b; Barto et al., 2004; Singh et al., 2005) do not directly address the problem of automatic subgoal discovery, *HQ-Learning* (Wiering and Schmidhuber, 1998a) automatically decomposes POMDPs (Sec. 6.3) into sequences of simpler subtasks that can be solved by memoryless policies learnable by reactive sub-agents. Recent HRL organizes potentially deep NN-based RL sub-modules into self-organizing, 2-dimensional motor control maps (Ring et al., 2011) inspired by neurophysiological findings (Graziano, 2009).

6.6 Deep RL by Direct NN Search / Policy Gradients / Evolution

Not quite as universal as the methods of Sec. 6.8, yet both practical and more general than most traditional RL algorithms (Sec. 6.2), are methods for *Direct Policy Search* (DS). Without a need for value functions or Markovian assumptions (Sec. 6.2, 6.3), the weights of an FNN or RNN are directly evaluated on the given RL problem. The results of successive trials inform further search for better weights. Unlike with RL supported by BP (Sec. 5.5, 6.3, 6.1), CAP depth (Sec. 3, 5.9) is not a crucial issue. DS may solve the credit assignment problem without backtracking through deep causal chains of modifiable parameters—it neither cares for their existence, nor tries to exploit them.

An important class of DS methods for NNs are *Policy Gradient* methods (Williams, 1986, 1988, 1992a; Sutton et al., 1999a; Baxter and Bartlett, 2001; Aberdeen, 2003; Ghavamzadeh and Mahadevan, 2003; Kohl and Stone, 2004; Wierstra et al., 2008; Rückstieß et al., 2008; Peters and Schaal, 2008b,a; Sehnke et al., 2010; Grüttner et al., 2010; Wierstra et al., 2010; Peters, 2010; Grondman et al., 2012; Heess et al., 2012). Gradients of the total reward with respect to policies (NN weights) are estimated (and then exploited) through repeated NN evaluations.

RL NNs can also be evolved through *Evolutionary Algorithms* (EAs) (Rechenberg, 1971; Schwefel, 1974; Holland, 1975; Fogel et al., 1966; Goldberg, 1989) in a series of trials. Here several policies are represented by a population of NNs improved through mutations and/or repeated recombinations of the population’s fittest individuals (e.g., Montana and Davis, 1989; Fogel et al., 1990; Maniezzo, 1994; Happel and Murre, 1994; Nolfi et al., 1994b). Compare *Genetic Programming* (GP) (Cramer, 1985) (see also Smith, 1980) which can be used to evolve computer programs of variable size (Dickmanns et al., 1987; Koza, 1992), and *Cartesian GP* (Miller and Thomson, 2000; Miller and Harding, 2009) for evolving graph-like programs, including NNs (Khan et al., 2010) and their topology (Turner and Miller, 2013). Related methods include *probability distribution-based EAs* (Baluja, 1994; Saravanan and Fogel, 1995; Śaustowicz and Schmidhuber, 1997; Larraanaga and Lozano, 2001), *Covariance Matrix Estimation Evolution Strategies* (CMA-ES) (Hansen and Ostermeier, 2001; Hansen et al., 2003; Igel, 2003; Heidrich-Meisner and Igel, 2009), and *NeuroEvolution of Augmenting Topologies* (NEAT) (Stanley and Miikkulainen, 2002). Hybrid methods combine traditional NN-based RL (Sec. 6.2) and EAs (e.g., Whiteson and Stone, 2006).

Since RNNs are general computers, RNN evolution is like GP in the sense that it can evolve general programs. Unlike sequential programs learned by traditional GP, however, RNNs can mix sequential and parallel information processing in a natural and efficient way, as already mentioned in Sec. 1. Many RNN evolvers have been proposed (e.g., Miller et al., 1989; Wieland, 1991; Cliff et al., 1993; Yao, 1993; Nolfi et al., 1994a; Sims, 1994; Yamauchi and Beer, 1994; Miglino et al., 1995; Moriarty, 1997; Pasemann et al., 1999; Juang, 2004; Whiteson, 2012). One particularly effective family of methods *coevolves* neurons, combining them into networks, and selecting those neurons for reproduction that participated in the best-performing networks (Moriarty and Miikkulainen, 1996; Gomez, 2003; Gomez and Miikkulainen, 2003). This can help to solve deep POMDPs (Gomez and Schmidhuber, 2005). *Co-Synaptic Neuro-Evolution* (CoSyNE) does something similar on the level of

突触或权重 (Gomez 等人, 2008) ; 这种优势在困难的非线性 POMDP 基准测试中得到了展示。

自然进化策略 (NES) (Wierstra 等人, 2008; Glasmachers 等人, 2010; Sun 等人, 2009,2013) 通过 自然梯度 (Amari, 1998) 的概念将策略梯度方法和进化方法联系起来。RNN 进化也可能通过 *Evolino*(Schmidhuber 等人, 2007) (第 5.9 节) 帮助改进深度 RNN 的监督学习。

6.7 通过间接策略搜索 / 压缩 NN 搜索的深度强化学习

一些 DS 方法 (第 6.6 节) 可以进化具有数百或数千个权重的神经网络, 但不能数百万个。如何搜索大型和深度神经网络? 到目前为止提到的多数监督学习和强化学习方法都在某种程度上搜索权重空间 w_{io} 。一些方法通过共享 w_i (例如在卷积神经网络 (第 5.4、5.8、5.16、5.21 节) 中, 或在用于监督学习 (第 5.5、5.13、5.17 节) 和强化学习 (第 6.1、6.3、6.6 节) 的循环神经网络中) 中重复使用, 从而减少了搜索空间。

然而, 可能通过间接搜索权重空间来利用解决方案空间中的附加规律性/可压缩性。与其直接进化大型神经网络 (第6.6节), 人们有时可以通过进化神经网络的紧凑编码来大大缩小搜索空间, 例如, 通过*Lindenmayer*系统(Lindenmayer, 1968; Jacob等人, 1994)、图重写(Kitano, 1990)、细胞编码 (Gruau等人, 1996)、*HyperNEAT* (D’ Ambrosio和Stanley, 2007; Stanley等人, 2009; Clune等人, 2011; van den Berg和Whiteson, 2013) (扩展NEAT; 第6.6节), 及其扩展 (例如, Risiand Stanley, 2012)。这有助于避免过拟合 (比较第5.6.3、5.24节), 并与正则化和MDL (第4.4节) 的主题密切相关。

一种通用方法 (Schmidhuber, 1997) 用于监督学习和强化学习, 旨在通过用通用编程语言 (Gödel, 1931; Church, 1936; Turing, 1936; Post, 1936) 编写的程序来紧凑地编码大型神经网络的权重 (Schmidhuber, 1997)。通常, 系统地搜索此类程序的空间, 并倾向于短而快的程序 (Levin, 1973b; Schmidhuber, 1997, 2004), 比直接搜索可能的神经网络权重矩阵的巨大空间更高效。先前用于编码神经网络的通用语言类似于汇编语言 (Schmidhuber, 1997)。最近的工作使用基于常用变换系数 (傅里叶变换、小波变换等) 的更实用的语言。特别是, RNN权重矩阵可以通过编码它们通过离散余弦变换 (DCT) 的系数来像图像一样压缩 (Koutník 等人, 2010, 2013)。基于 DCT 的紧凑描述可以通过 NES 或 CoSyNE (第6.6节) 进行进化。一个具有超过一百万个权重的 RNN (在没有教师的情况下) 学习了在 TORCS 驾驶游戏中驱动模拟汽车 (Loiacono 等人, 2009, 2011), 基于高维视频类似的视觉输入流 (Koutník 等人, 2013)。RNN 从零开始学习了控制和视觉处理, 而没有 UL 的帮助。(当然, UL 可能有助于生成更紧凑的图像代码 (第6.4节, 4.2节) 以输入到更小的 RNN 中, 以减少整体计算工作量。)

6.8 通用强化学习

通用学习算法可以在终身学习环境中以开放方式和特定环境的方式自我改进 (Schmidhuber, 1987; Schmidhuber等人, 1997b,a; Schaul和Schmidhuber, 2010)。最通用的 RL 类型仅受限于理论计算机科学创始人发现的 可计算性基本限制 (Gödel, 1931; Church, 1936; Turing, 1936; Post, 1936)。值得注意的是, 存在针对无限问题深度的通用问题解决器或 通用 *RL* 机器的蓝图, 在多种理论意义上是时间最优的 (Hutter, 2005, 2002; Schmidhuber, 2002, 2006b)。特别是, *Gödel* 机器可以在通用计算机 (如 RNNs) 上实现, 并以可证明的时间最优方式改进其软件的任何部分 (包括学习算法本身)。

synapses or weights (Gomez et al., 2008); benefits of this were shown on difficult nonlinear POMDP benchmarks.

Natural Evolution Strategies (NES) (Wierstra et al., 2008; Glasmachers et al., 2010; Sun et al., 2009, 2013) link policy gradient methods and evolutionary approaches through the concept of *Natural Gradients* (Amari, 1998). RNN evolution may also help to improve SL for deep RNNs through *Evolino* (Schmidhuber et al., 2007) (Sec. 5.9).

6.7 Deep RL by Indirect Policy Search / Compressed NN Search

Some DS methods (Sec. 6.6) can evolve NNs with hundreds or thousands of weights, but not millions. How to search for large and deep NNs? Most SL and RL methods mentioned so far somehow search the space of weights w_i . Some profit from a reduction of the search space through shared w_i that get reused over and over again, e.g., in CNNs (Sec. 5.4, 5.8, 5.16, 5.21), or in RNNs for SL (Sec. 5.5, 5.13, 5.17) and RL (Sec. 6.1, 6.3, 6.6).

It may be possible, however, to exploit *additional* regularities/compressibilities in the space of solutions, through *indirect search in weight space*. Instead of evolving large NNs directly (Sec. 6.6), one can sometimes greatly reduce the search space by evolving *compact encodings* of NNs, e.g., through *Lindenmeyer Systems* (Lindenmayer, 1968; Jacob et al., 1994), *graph rewriting* (Kitano, 1990), *Cellular Encoding* (Gruau et al., 1996), *HyperNEAT* (D’Ambrosio and Stanley, 2007; Stanley et al., 2009; Clune et al., 2011; van den Berg and Whiteson, 2013) (extending NEAT; Sec. 6.6), and extensions thereof (e.g., Risi and Stanley, 2012). This helps to avoid overfitting (compare Sec. 5.6.3, 5.24) and is closely related to the topics of regularisation and MDL (Sec. 4.4).

A general approach (Schmidhuber, 1997) for both SL and RL seeks to compactly encode weights of large NNs (Schmidhuber, 1997) through programs written in a *universal programming language* (Gödel, 1931; Church, 1936; Turing, 1936; Post, 1936). Often it is much more efficient to systematically search the space of such programs with a bias towards short and fast programs (Levin, 1973b; Schmidhuber, 1997, 2004), instead of directly searching the huge space of possible NN weight matrices. A previous universal language for encoding NNs was assembler-like (Schmidhuber, 1997). More recent work uses more practical languages based on coefficients of popular transforms (Fourier, wavelet, etc). In particular, RNN weight matrices may be compressed like images, by encoding them through the coefficients of a *discrete cosine transform* (DCT) (Koutník et al., 2010, 2013). Compact DCT-based descriptions can be evolved through NES or CoSyNE (Sec. 6.6). An RNN with over a million weights learned (without a teacher) to drive a simulated car in the TORCS driving game (Loiacono et al., 2009, 2011), based on a high-dimensional video-like visual input stream (Koutník et al., 2013). The RNN learned both control and visual processing from scratch, without being aided by UL. (Of course, UL might help to generate more compact image codes (Sec. 6.4, 4.2) to be fed into a smaller RNN, to reduce the overall computational effort.)

6.8 Universal RL

General purpose learning algorithms may improve themselves in open-ended fashion and environment-specific ways in a lifelong learning context (Schmidhuber, 1987; Schmidhuber et al., 1997b,a; Schaul and Schmidhuber, 2010). The most general type of RL is constrained only by the fundamental limitations of computability identified by the founders of theoretical computer science (Gödel, 1931; Church, 1936; Turing, 1936; Post, 1936). Remarkably, there exist blueprints of *universal problem solvers* or *universal RL machines* for unlimited problem depth that are time-optimal in various theoretical senses (Hutter, 2005, 2002; Schmidhuber, 2002, 2006b). In particular, the *Gödel Machine* can be implemented on general computers such as RNNs and may improve any part of its software (including the learning algorithm itself) in a way that is provably time-optimal in a certain

在某种意义上（Schmidhuber, 2006b）。它可以由一个渐近最优元方法（Hut-ter, 2002）（也适用于RNNs）初始化，该方法将以解决任何明确定义问题的最快未知方式尽可能快地解决它，除了一个随着问题规模增长而变得微不足道的加性常数开销。请注意，大多数问题都是大问题；只有少数是小问题。AI和DL研究人员仍在从事相关工作，因为许多人感兴趣的问题非常小，值得尝试通过不太通用的方法（包括启发式方法）来降低开销。在这里，我不会进一步讨论通用RL方法，这些方法超出了通常所说的DL范畴。

7 结论与展望

深度学习 (DL) 在 神经网络 (NNs) 中的相关性与 监督学习 (SL) (Sec. 5)、无监督学习 (UL) (Sec. 5) 以及 强化学习 (RL) (Sec. 6) 都有关联。通过缓解深度 信用分配路径 (CAPs, Sec. 3, 5.9) 的问题，UL (Sec. 5.6.4) 不仅能够促进序列 (Sec. 5.10) 和平稳模式 (Sec. 5.7, 5.15) 的 SL，还能促进 RL (Sec. 6.4, 4.2)。动态规划 (DP, Sec. 4.1) 对深度 SL (Sec. 5.5) 和传统使用深度神经网络的 RL (Sec. 6.2) 都很重要。寻找计算解决方案、抗扰动 (Sec. 5.6.3, 5.15, 5.24)、低复杂度的神经网络，这些网络可以用少量信息描述 (Sec. 4.4)，可以减少过拟合，并改进深度 SL & UL (Sec. 5.6.3, 5.6.4) 以及 RL (Sec. 6.7)，即使在部分可观察环境中 (Sec. 6.3) 也适用。深度 SL、UL、RL 常常会创建越来越抽象的平稳数据 (Sec. 5.3, 5.7, 5.15)、序列数据 (Sec. 5.10) 或 RL 策略 (Sec. 6.5) 的层次结构。虽然 UL 可以促进 SL，但前馈神经网络 (FNNs) (Sec. 5.5, 5.8, 5.16, 5.18) 和循环神经网络 (RNNs) (Sec. 5.5, 5.13) 的纯 SL 不仅赢得了早期的竞赛 (Sec. 5.12, 5.14)，还赢得了最近的大部分竞赛 (Sec. 5.17–5.22)。特别是 FNNs 中的深度学习受益于 GPU 实现 (Sec. 5.16–5.19)。特别是，基于 GPU 的 (Sec. 5.19) 最大池化 (Sec. 5.11) 卷积神经网络 (Sec. 5.4, 5.8, 5.16) 不仅在模式识别 (Sec. 5.19–5.22) 中赢得了竞赛，还在图像分割 (Sec. 5.21) 和目标检测 (Sec. 5.21, 5.22) 中赢得了竞赛。

与这些系统不同，人类通过依次将注意力引导至可用数据的 relevant 部分，来 主动感知模式。不久的将来，深度 NNs 也会这样做，扩展自 1990 年以来关于通过 RL 学习选择性注意力的 NNs 的研究，这些研究涉及 (a) 运动行为，如注视控制（第 6.1 节），以及 (b) 内部行为，这些行为控制着 RNNs 内部注意力光斑，从而通过外部和内部反馈（例如，第 2、5.21、6.6、6.7 节）关闭通用的感觉运动回路。

许多未来的深度神经网络也会考虑到激活神经元以及它们之间传递信号需要消耗能量。大脑似乎至少通过两种方式在解决问题时最小化这种计算成本：(1) 在某一时刻，只有所有神经元中的一小部分处于激活状态，因为通过赢者通吃机制进行的局部竞争会关闭许多相邻神经元，只有获胜者才能通过输出连接激活其他神经元（比较SLIM神经网络；第5.24节）。(2) 在紧凑的3D空间内，大量神经元通过许多短程连接和少量远程连接进行稀疏连接（就像传统超级计算机中的微芯片一样）。通常相邻的神经元被分配执行单个任务，从而降低了通信成本。物理学似乎表明，任何高效的计算硬件在未来也必须像大脑一样，遵循这两个约束条件。然而，目前最成功的深度循环神经网络却并非如此。与某些脉冲神经网络（第5.26节）不同，它们通常会稍微激活所有单元，并且倾向于强连接，忽略了3D硬件的自然约束。通过采用(1)和(2)，并通过在程序（权重）空间中进行直接搜索来最小化不可微分的能量和通信成本（例如，第6.6、6.7节），应该可以改进它们。这些更类似大脑的循环神经网络将把相邻的循环神经网络部分分配给相关的行为，而将较远的循环神经网络部分分配给不太相关的行为，从而以一种比传统前馈神经网络中的自组织映射（第5.6.4节）更通用的方式实现自模块化。它们还将作为能量最小化的副产品实现奥卡姆剃刀（第4.4、5.6.3节），通过寻找需要少量激活神经元和少量、主要是短连接的简单（高度泛化）问题解决方案。

sense (Schmidhuber, 2006b). It can be initialized by an *asymptotically optimal* meta-method (Hut-ter, 2002) (also applicable to RNNs) which will solve any well-defined problem as quickly as the unknown fastest way of solving it, save for an additive constant overhead that becomes negligible as problem size grows. Note that most problems are large; only few are small. AI and DL researchers are still in business because many are interested in problems so small that it is worth trying to reduce the overhead through less general methods, including heuristics. Here I won’t further discuss universal RL methods, which go beyond what is usually called DL.

7 Conclusion and Outlook

Deep Learning (DL) in *Neural Networks* (NNs) is relevant for *Supervised Learning* (SL) (Sec. 5), *Unsupervised Learning* (UL) (Sec. 5), and *Reinforcement Learning* (RL) (Sec. 6). By alleviating problems with deep *Credit Assignment Paths* (CAPs, Sec. 3, 5.9), UL (Sec. 5.6.4) can not only facilitate SL of sequences (Sec. 5.10) and stationary patterns (Sec. 5.7, 5.15), but also RL (Sec. 6.4, 4.2). *Dynamic Programming* (DP, Sec. 4.1) is important for both deep SL (Sec. 5.5) and traditional RL with deep NNs (Sec. 6.2). A search for solution-computing, perturbation-resistant (Sec. 5.6.3, 5.15, 5.24), low-complexity NNs describable by few bits of information (Sec. 4.4) can reduce overfitting and improve deep SL & UL (Sec. 5.6.3, 5.6.4) as well as RL (Sec. 6.7), also in the case of partially observable environments (Sec. 6.3). Deep SL, UL, RL often create hierarchies of more and more abstract representations of stationary data (Sec. 5.3, 5.7, 5.15), sequential data (Sec. 5.10), or RL policies (Sec. 6.5). While UL can facilitate SL, pure SL for feedforward NNs (FNNs) (Sec. 5.5, 5.8, 5.16, 5.18) and recurrent NNs (RNNs) (Sec. 5.5, 5.13) did not only win early contests (Sec. 5.12, 5.14) but also most of the recent ones (Sec. 5.17–5.22). Especially DL in FNNs profited from GPU implementations (Sec. 5.16–5.19). In particular, GPU-based (Sec. 5.19) Max-Pooling (Sec. 5.11) Convolutional NNs (Sec. 5.4, 5.8, 5.16) won competitions not only in pattern recognition (Sec. 5.19–5.22) but also image segmentation (Sec. 5.21) and object detection (Sec. 5.21, 5.22).

Unlike these systems, humans *learn to actively perceive* patterns by sequentially directing attention to relevant parts of the available data. Near future deep NNs will do so, too, extending previous work since 1990 on NNs that learn selective attention through RL of (a) *motor actions* such as saccade control (Sec. 6.1) and (b) *internal actions* controlling spotlights of attention within RNNs, thus closing the general sensorimotor loop through both external and internal feedback (e.g., Sec. 2, 5.21, 6.6, 6.7).

Many future deep NNs will also take into account that it costs energy to activate neurons, and to send signals between them. Brains seem to minimize such computational costs during problem solving in at least two ways: (1) At a given time, only a small fraction of all neurons is active because local competition through winner-take-all mechanisms shuts down many neighbouring neurons, and only winners can activate other neurons through outgoing connections (compare SLIM NNs; Sec. 5.24). (2) Numerous neurons are sparsely connected in a compact 3D volume by many short-range and few long-range connections (much like microchips in traditional supercomputers). Often neighbouring neurons are allocated to solve a single task, thus reducing communication costs. Physics seems to dictate that any efficient computational hardware will in the future also have to be brain-like in keeping with these two constraints. The most successful current deep RNNs, however, are not. Unlike certain spiking NNs (Sec. 5.26), they usually activate all units at least slightly, and tend to be strongly connected, ignoring natural constraints of 3D hardware. It should be possible to improve them by adopting (1) and (2), and by minimizing non-differentiable energy and communication costs through direct search in program (weight) space (e.g., Sec. 6.6, 6.7). These more brain-like RNNs will allocate neighboring RNN parts to related behaviors, and distant RNN parts to less related ones, thus self-modularizing in a way more general than that of traditional self-organizing maps in FNNs (Sec. 5.6.4). They will also implement Occam’s razor (Sec. 4.4, 5.6.3) as a by-product of energy min-

通过找到需要少量激活神经元和少量、主要是短连接的简单（高度泛化）问题解决方案。

更遥远的未来可能属于通用学习算法，它们以可证明的最优方式自我改进（第6.8节），但这些算法目前还不实用或具有商业意义。

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imization, by finding simple (highly generalizing) problem solutions that require few active neurons and few, mostly short connections.

The more distant future may belong to general purpose learning algorithms that improve themselves in provably optimal ways (Sec. 6.8), but these are not yet practical or commercially relevant.

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